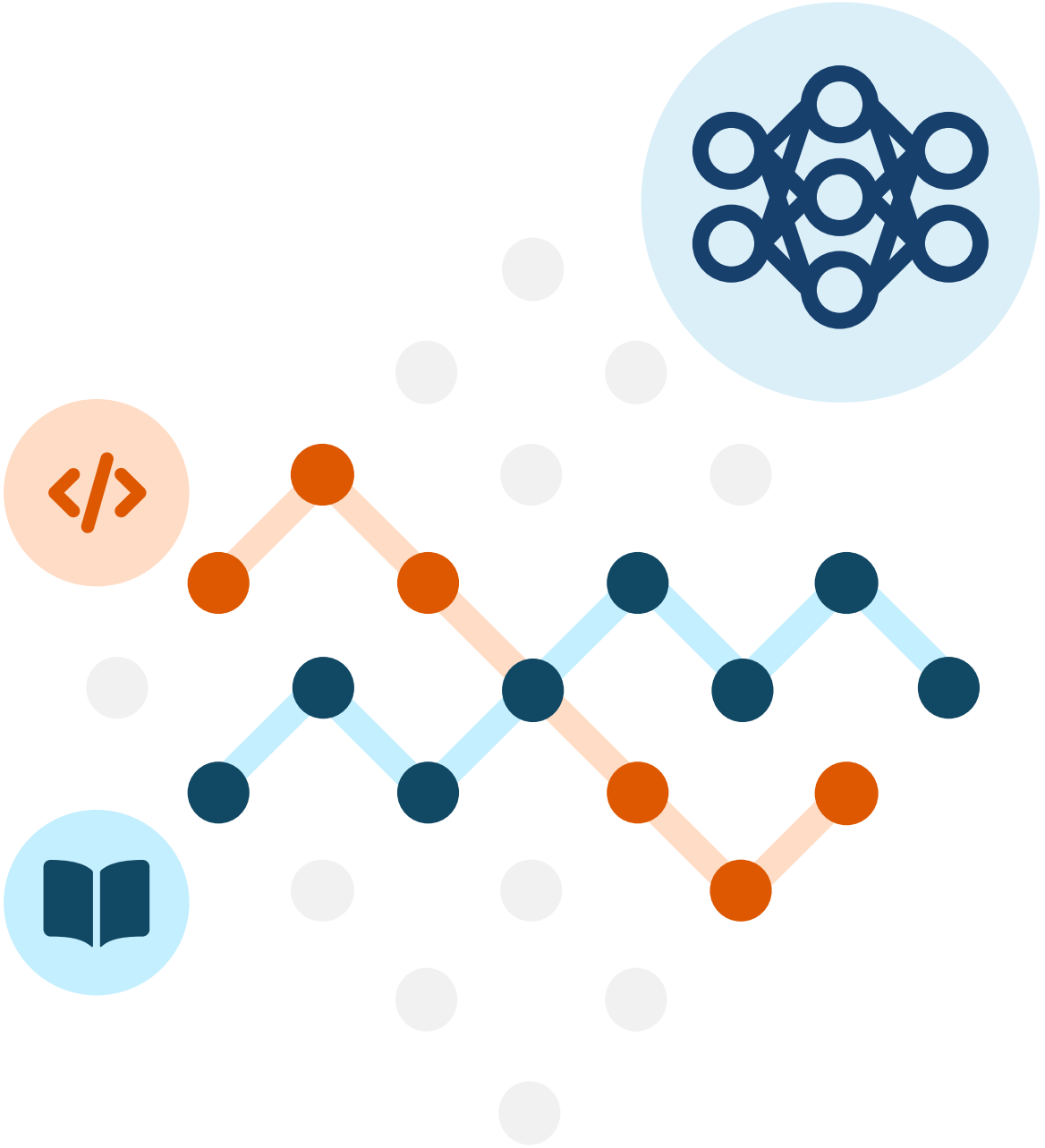
ML in PL CONFERENCE 2026

Security in Machine Learning

Adam Dziedzic












SprintML

S - ecure

P - rivate

R - obust

In - terpretable

T – rustworthy

Machine Learning



Tutorial on Security in Machine Learning



Louis Kerner
Research Talk:
Watermarking



Michel Meintz
Research Talk:
Radioactivity



Bihe Zhao
Research Talk:
Radioactivity



Maitri Shah
Coding Task:
Dataset Inference

Schedule

14:00 

Overview: Security in ML

Adam Dziedzic

14:30 

Research talks from our SprintML lab

Louis Kerner, Michel Meintz, Bihe Zhao

15:15 

Discussion tables

15:45 

Coding challenge

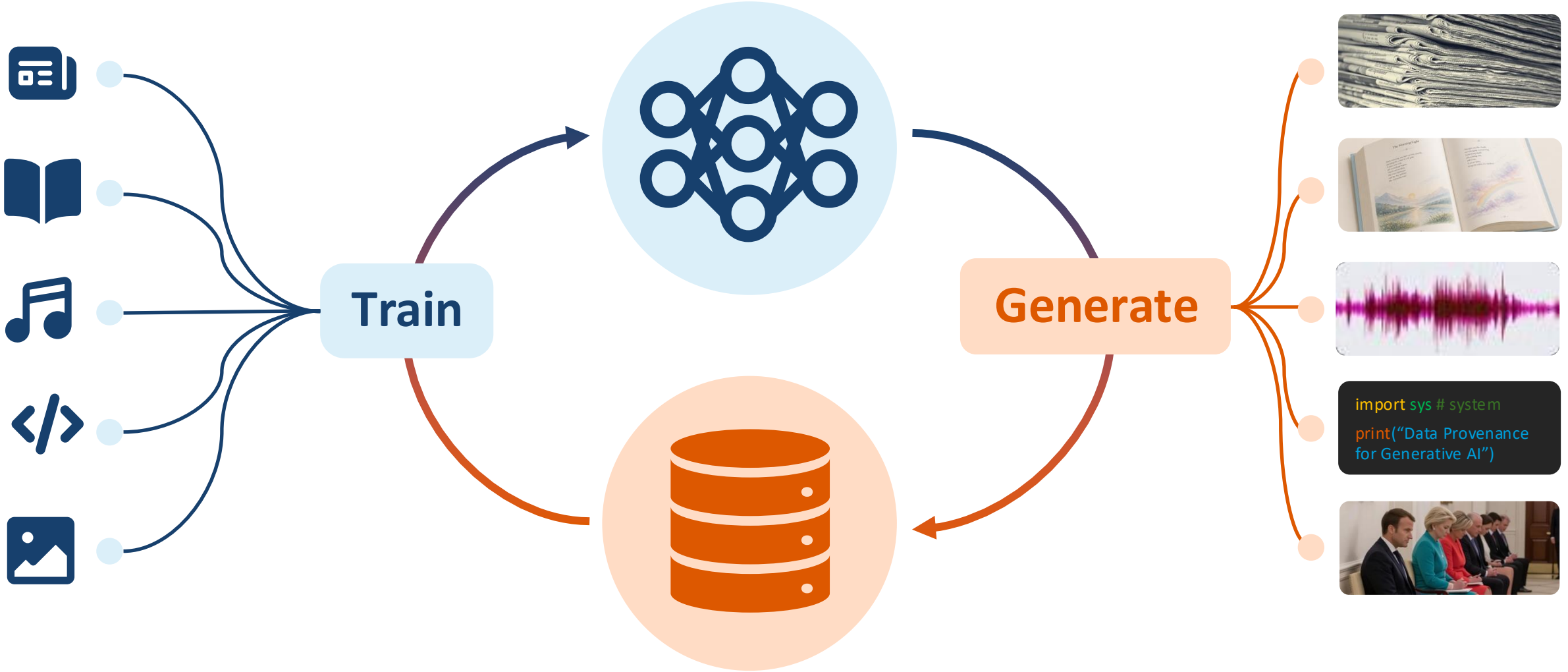
Membership inference on Kaggle · Maitri Shah

17:30 

Summary

Our Large Data Leads to Impressive Capabilities

Models: GPT Gemini ••• LLaMA



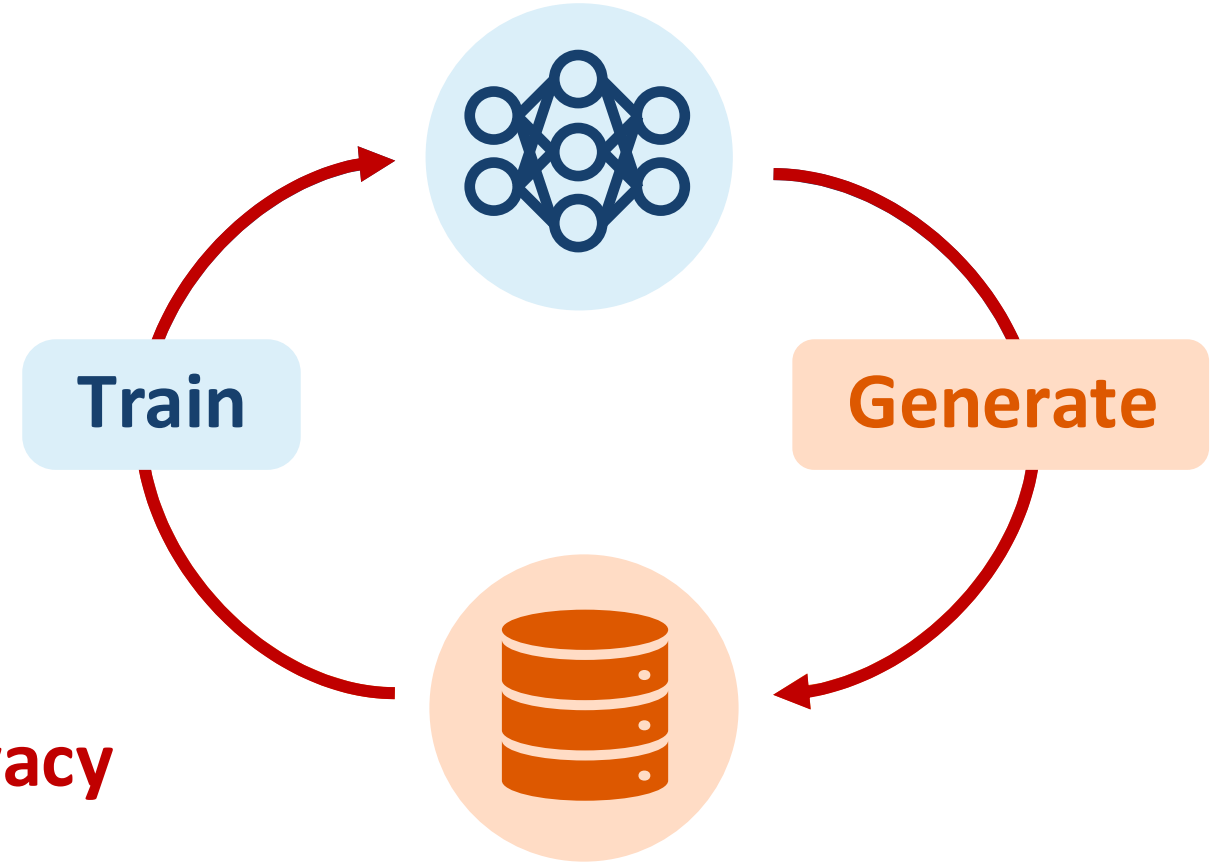
Models **Memorize Training Data**



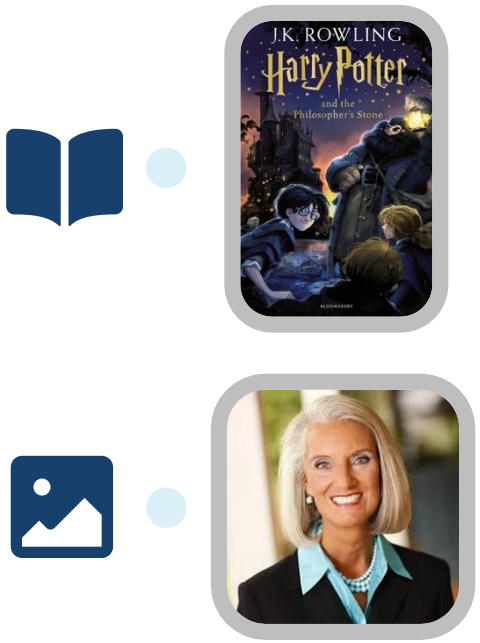


Copyright & Privacy Violations

> 100 lawsuits, 100 bn €



Models **Produce Misinformation**



Copyright & Privacy Violations

Train

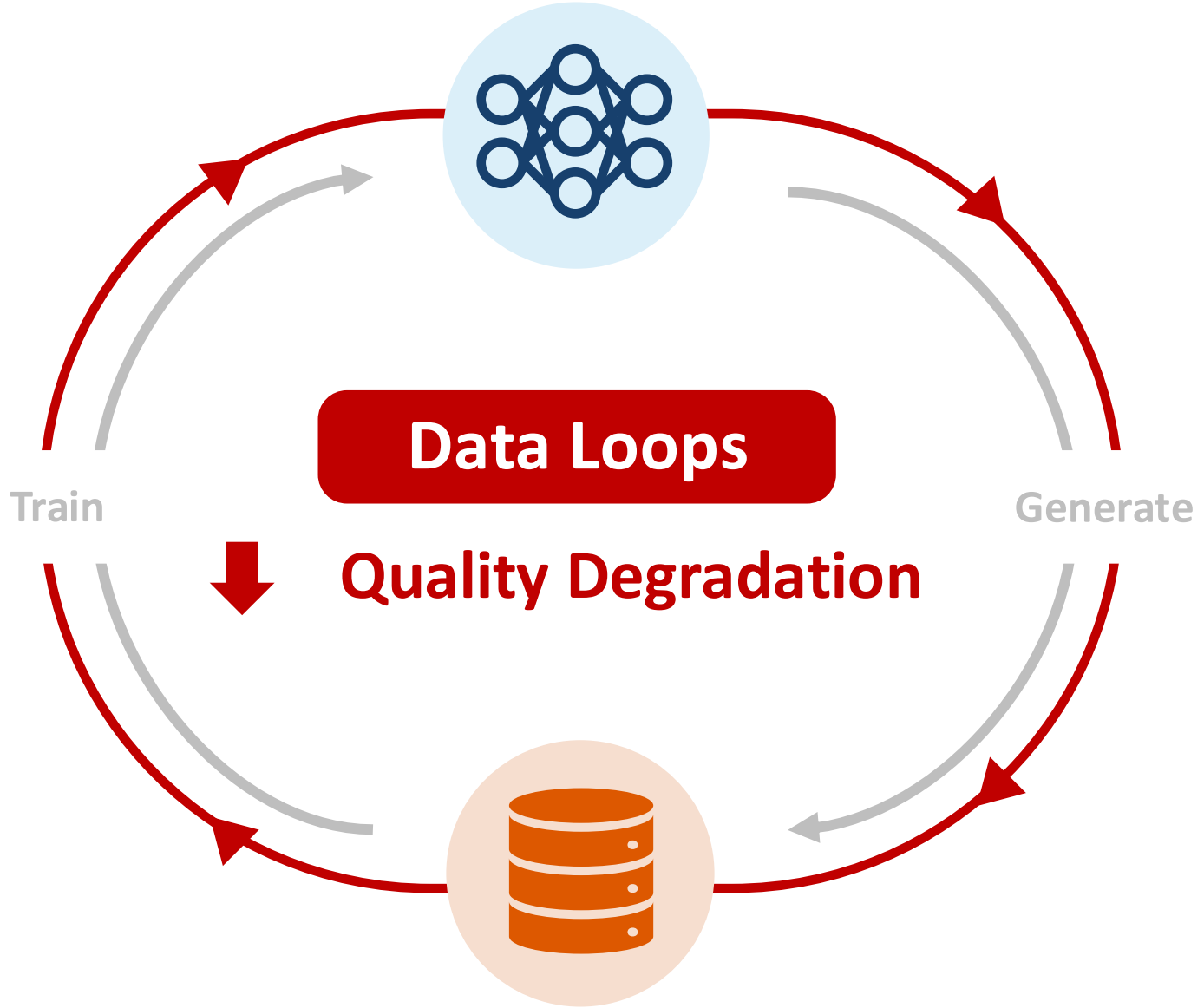
Generate



Fake Data

Models **Create Data Loops**

Copyright & Privacy Violations

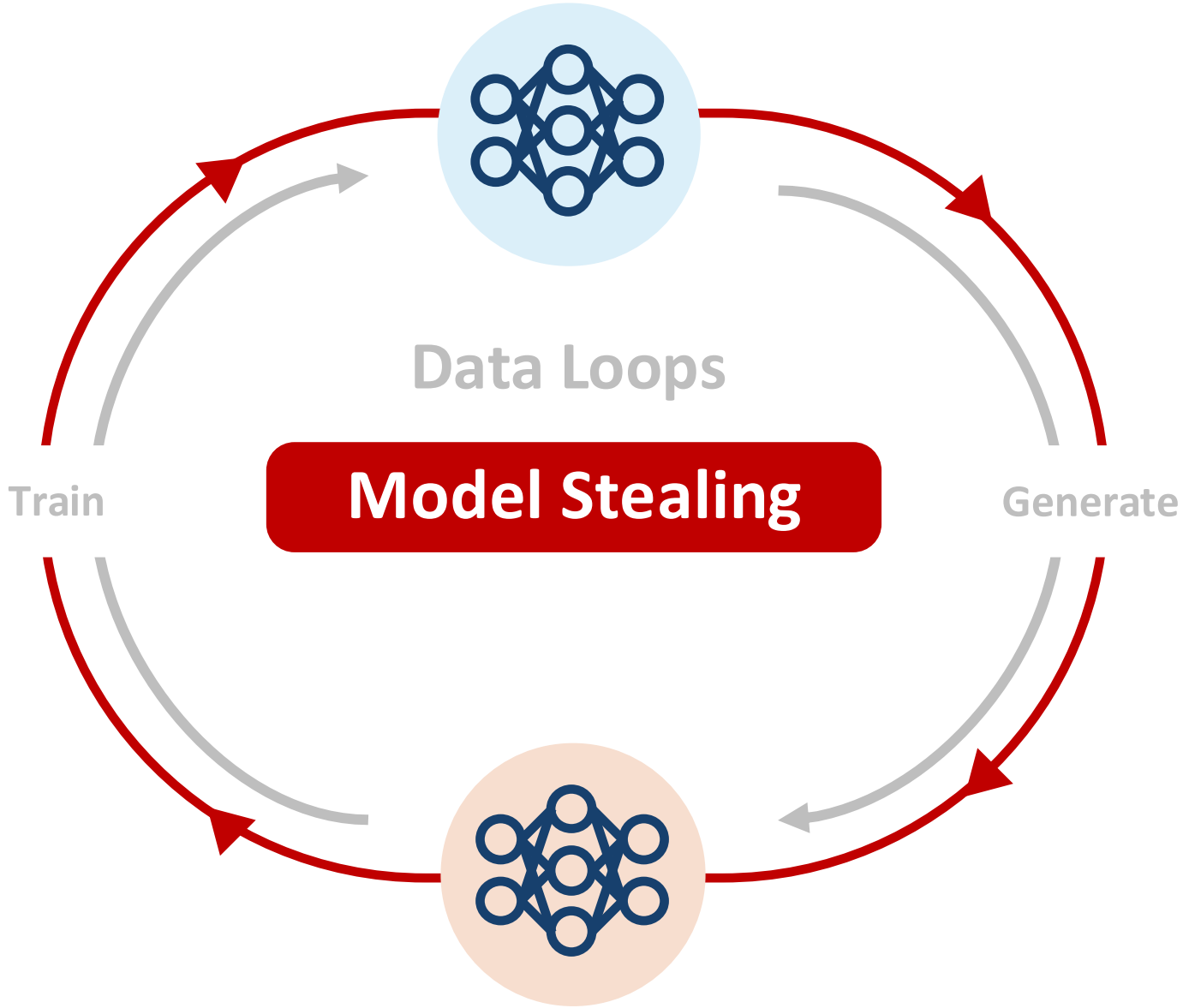


Fake Data

Models are **Stolen**

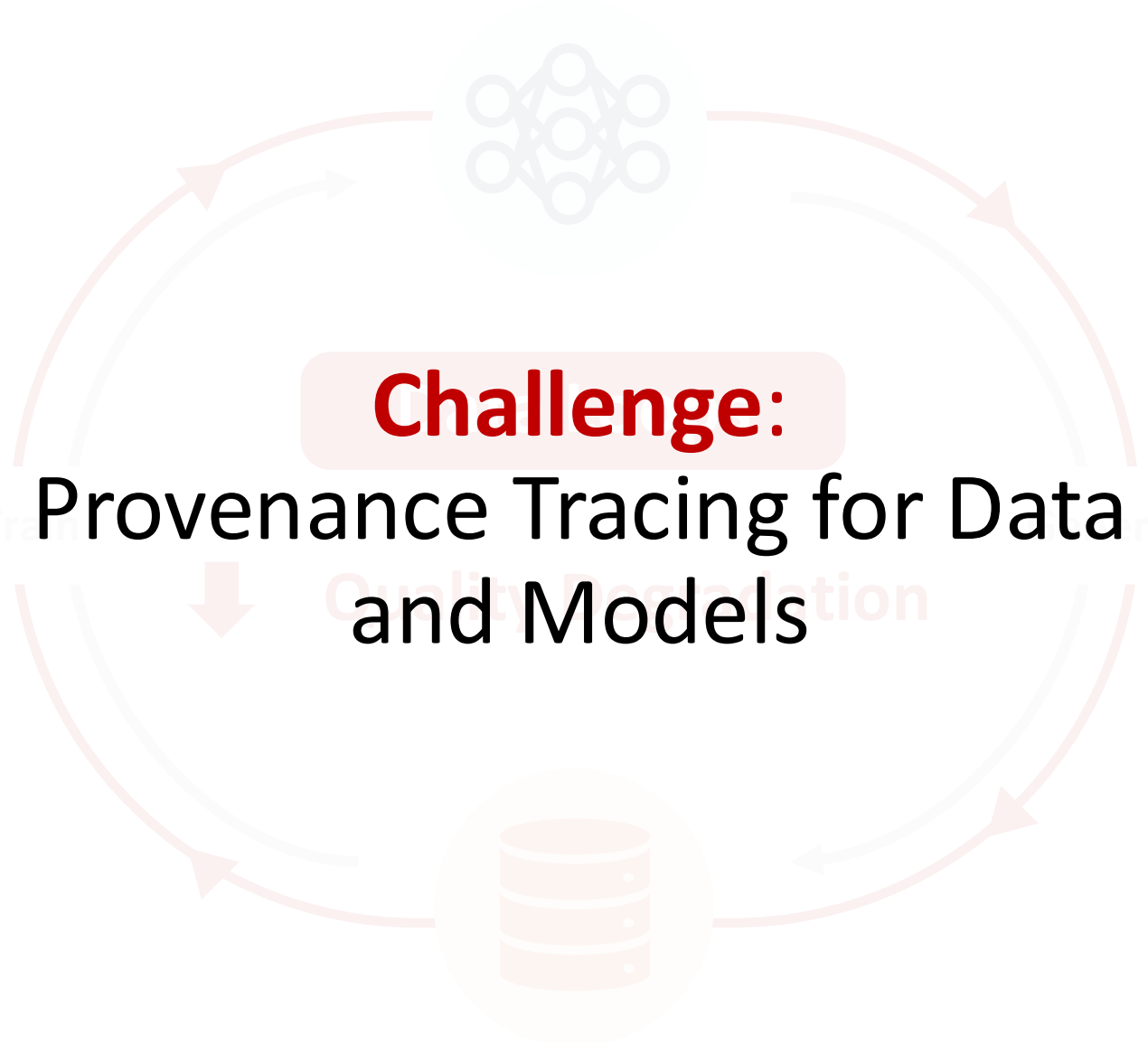
OpenAI alleges DeepSeek stole its intellectual property
Anthropic accuses Alibaba of extracting its AI capabilities
...

Copyright & Privacy Violations



Fake Data

Profound Risks in Machine Learning

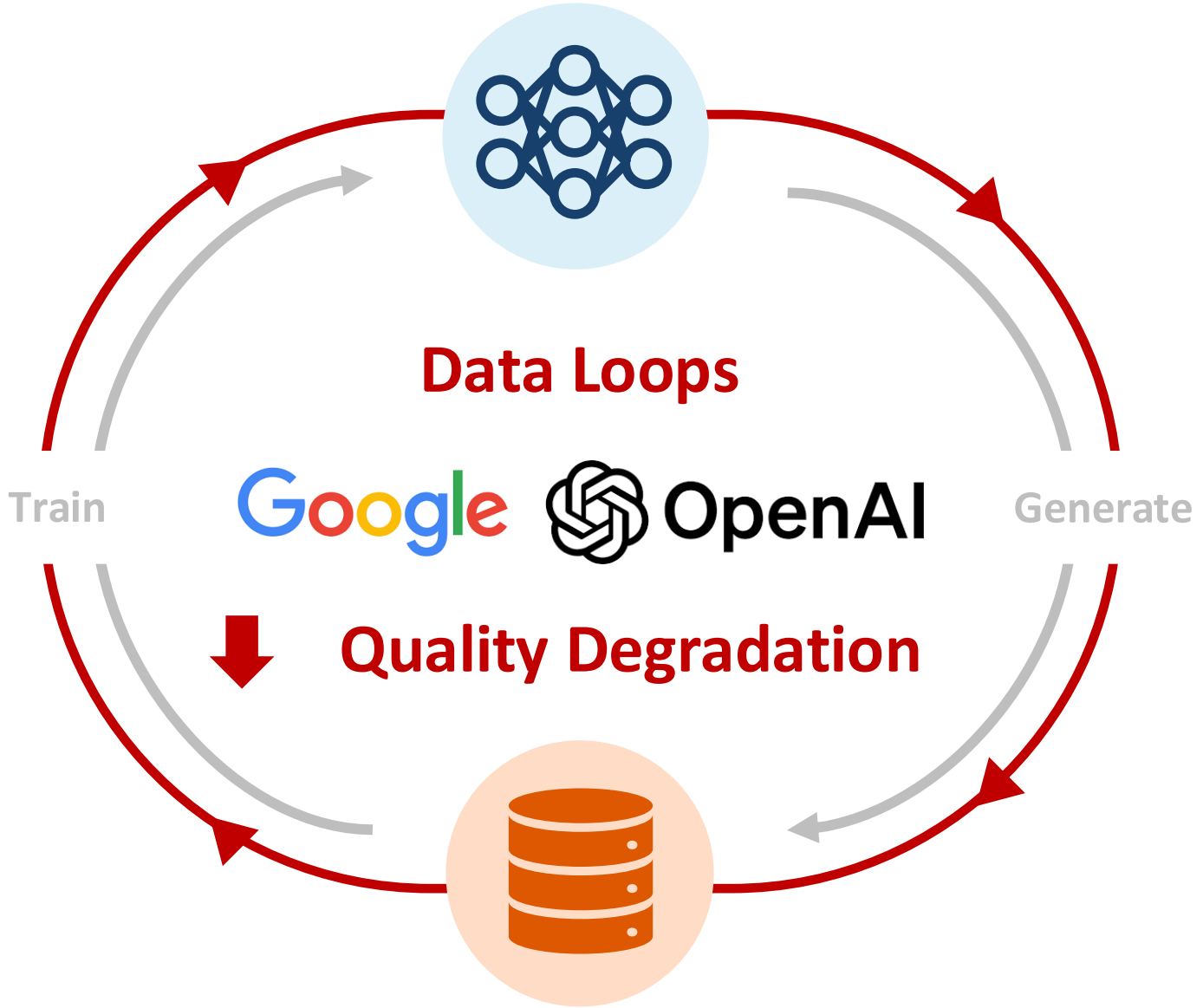


Challenge:
Provenance Tracing for Data and Models

We Need Data Provenance Now!

Copyright & Privacy Violations

EU AI Act:
Article 53



2026

Fake Data

EU AI Act:
Article 50

Tutorial **Roadmap**

How can we **protect models and data**?

0

Model Stealing

1

Data & Model Identification

2

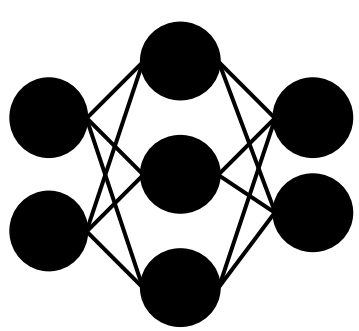
Memorization

3

Watermarking

Hands-on task: Dataset Inference challenge on a live Kaggle leaderboard

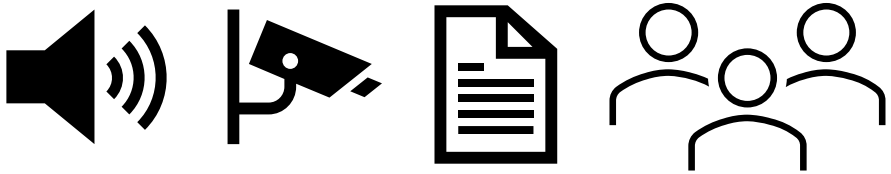
APIs: Dominant Paradigm for Using Models



Machine Learning API

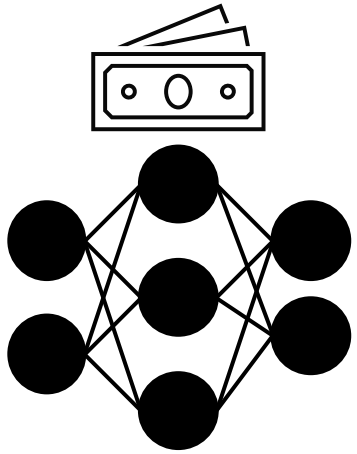
Query

Answer



High Cost of Training ML Models

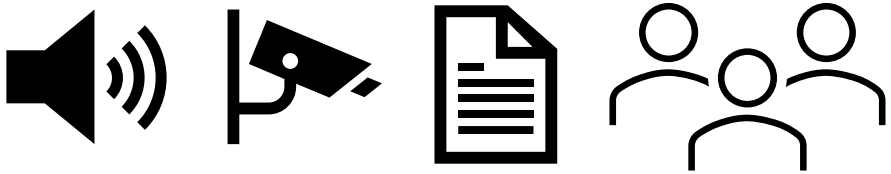
Collect and Clean Data



Machine Learning API

Query

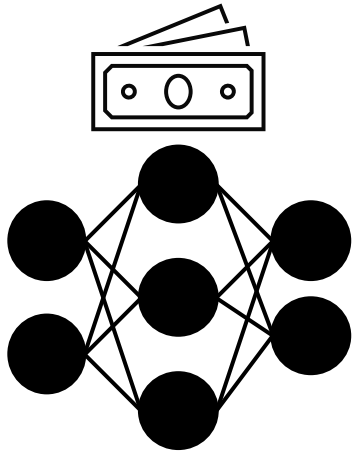
Answer



High Cost of Training ML Models

Collect and Clean Data

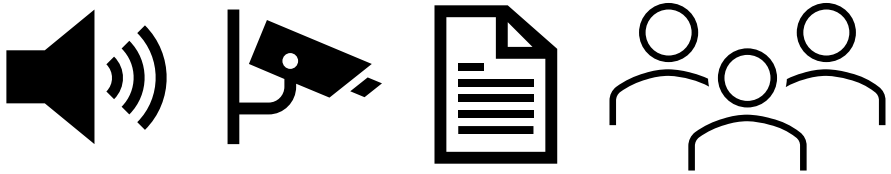
Label Data



Machine Learning API

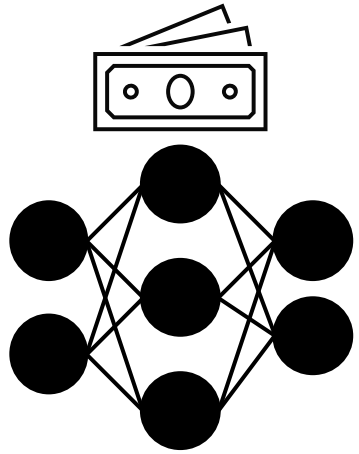
Query

Answer



High Cost of Training ML Models

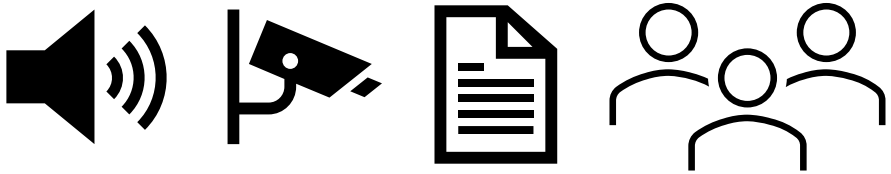
- Collect and Clean Data
- Label Data
- Tune Parameters



Machine Learning API

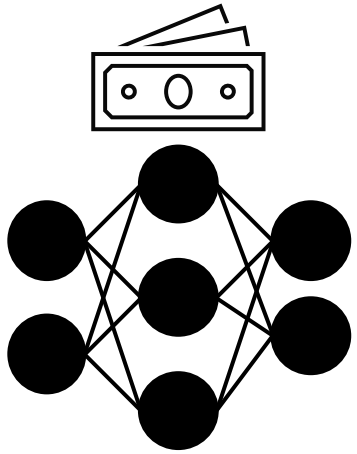
Query

Answer



High Cost of Training ML Models

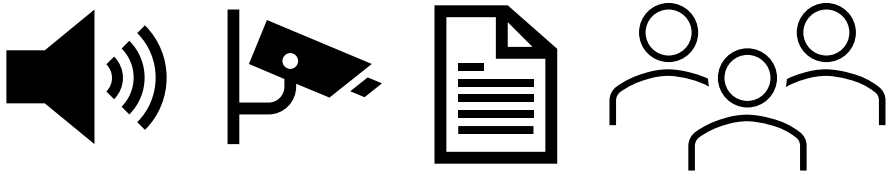
- Collect and Clean Data
- Label Data
- Tune Parameters
- Run on GPU/TPU/CPU



Machine Learning API

Query

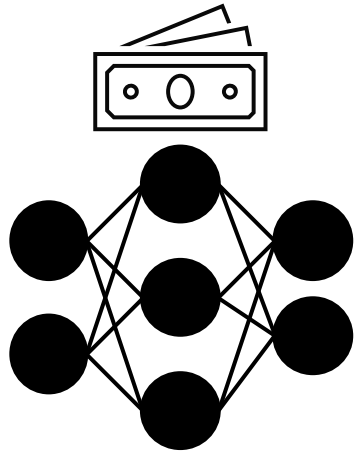
Answer



High Cost of Training ML Models

- Collect and Clean Data
- Label Data
- Tune Parameters
- Run on GPU/TPU/CPU

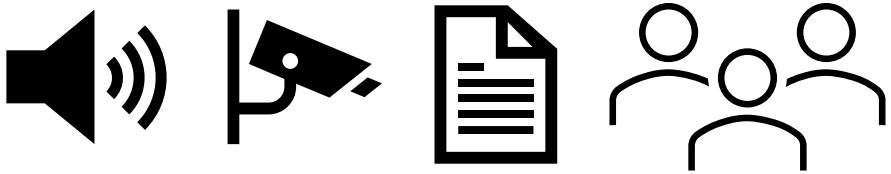
$12M GPT-3



Machine Learning API

Query

Answer



High Cost of Training ML Models

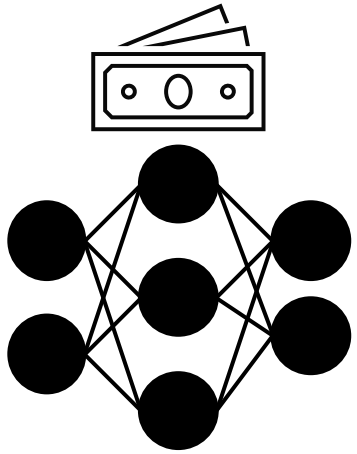
Collect and Clean Data

Label Data

Tune Parameters

Run on GPU/TPU/CPU

$12M GPT-3



Machine Learning API

Query

Answer

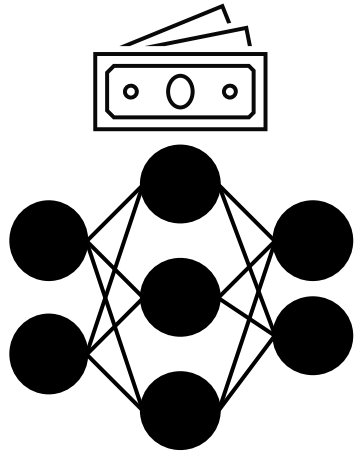


Model Stealing - ranked among the most severe attacks against ML

[Shankar et al. SWP 2020]

The Threat of Stealing the Models is Real

$12M GPT-3



Machine Learning API

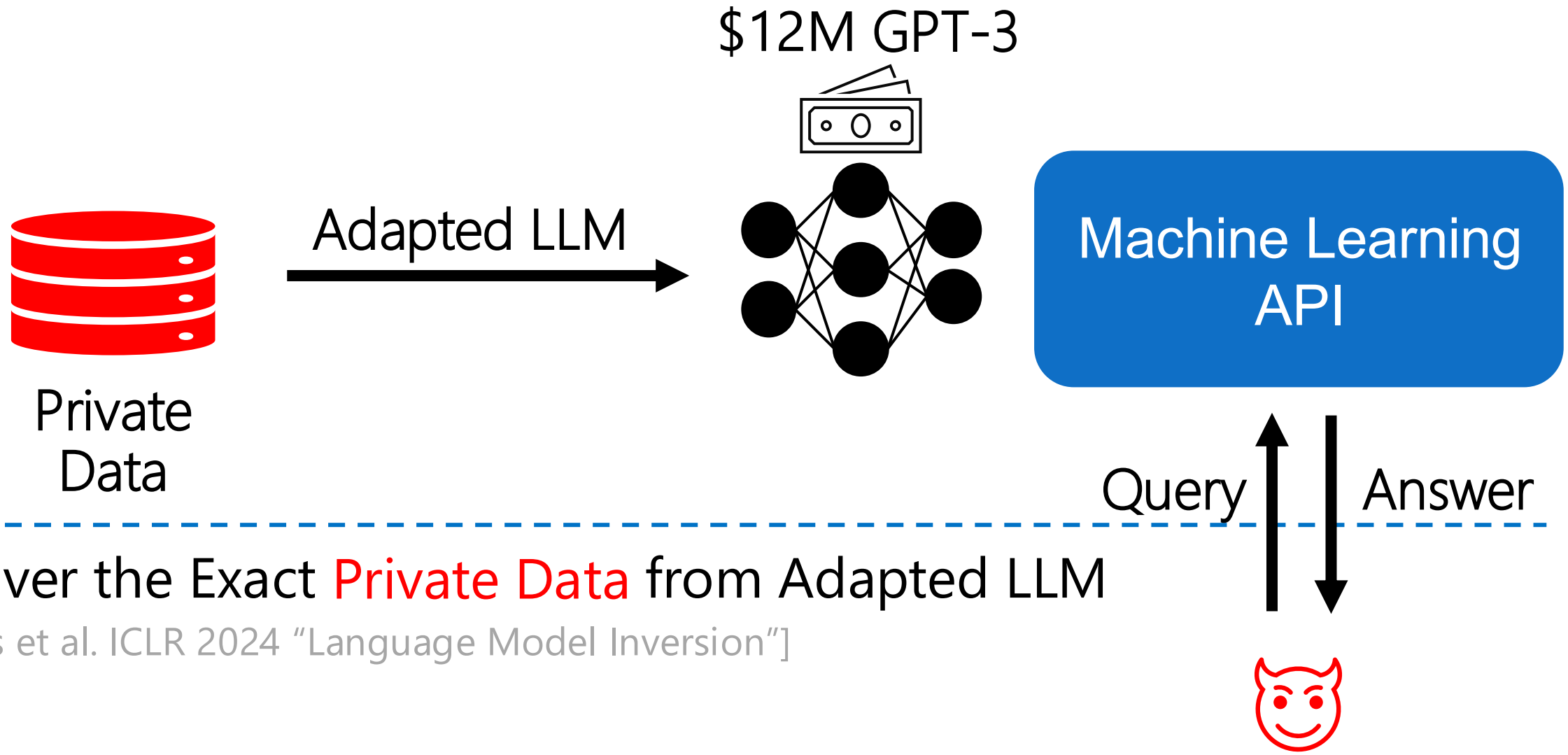
Query

Answer

Steal *parameters* of the last layer of GPT-3 for < **$20**

[Carlini et al. ICML 2024, "Stealing Part of a Production Language Model"]

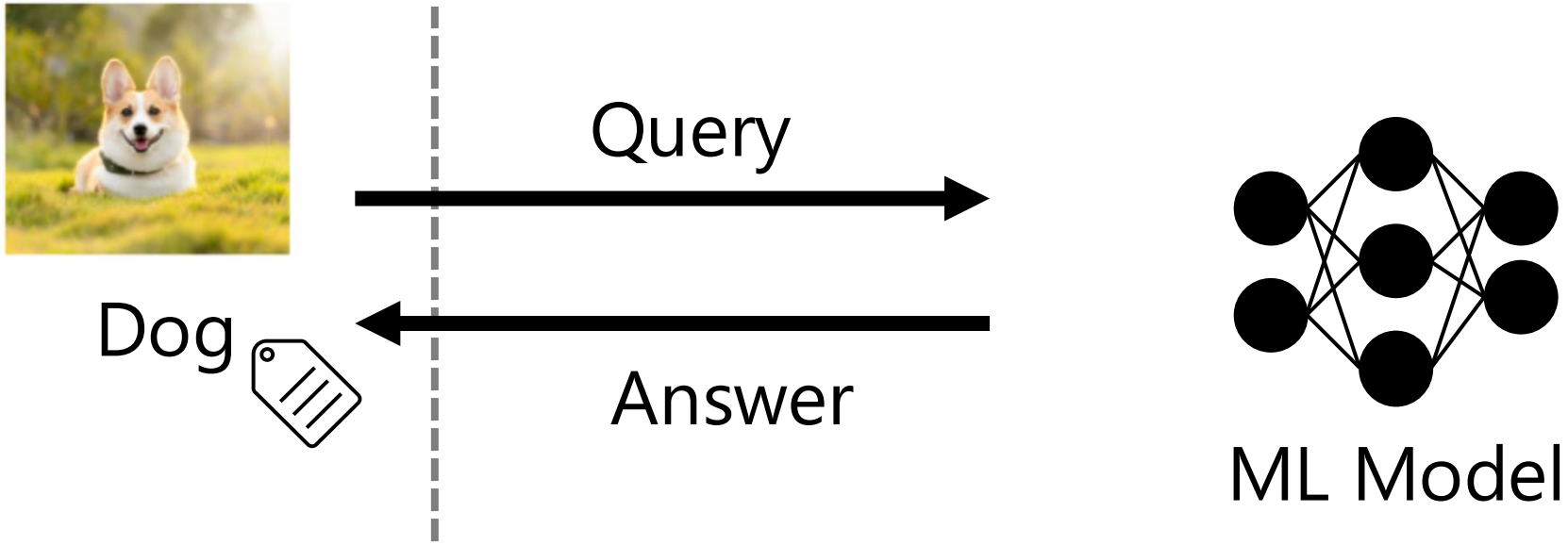
The Threat of Data Leakage is Real



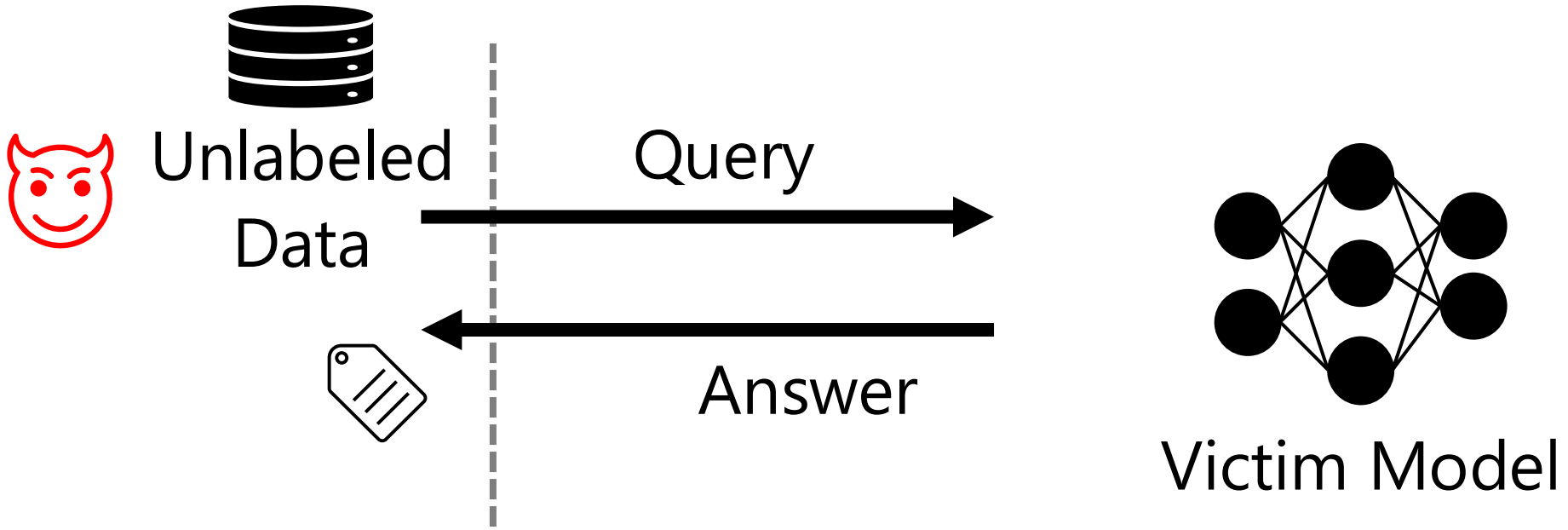
Recover the Exact Private Data from Adapted LLM

[Morris et al. ICLR 2024 "Language Model Inversion"]

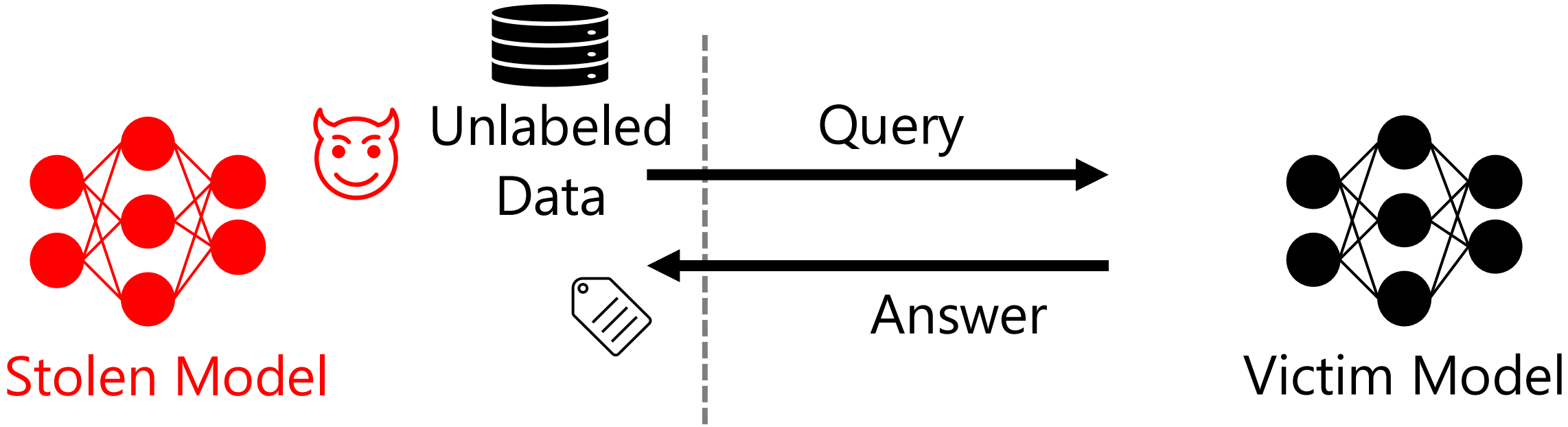
Machine Learning **APIs**



Stealing Models from **APIs**

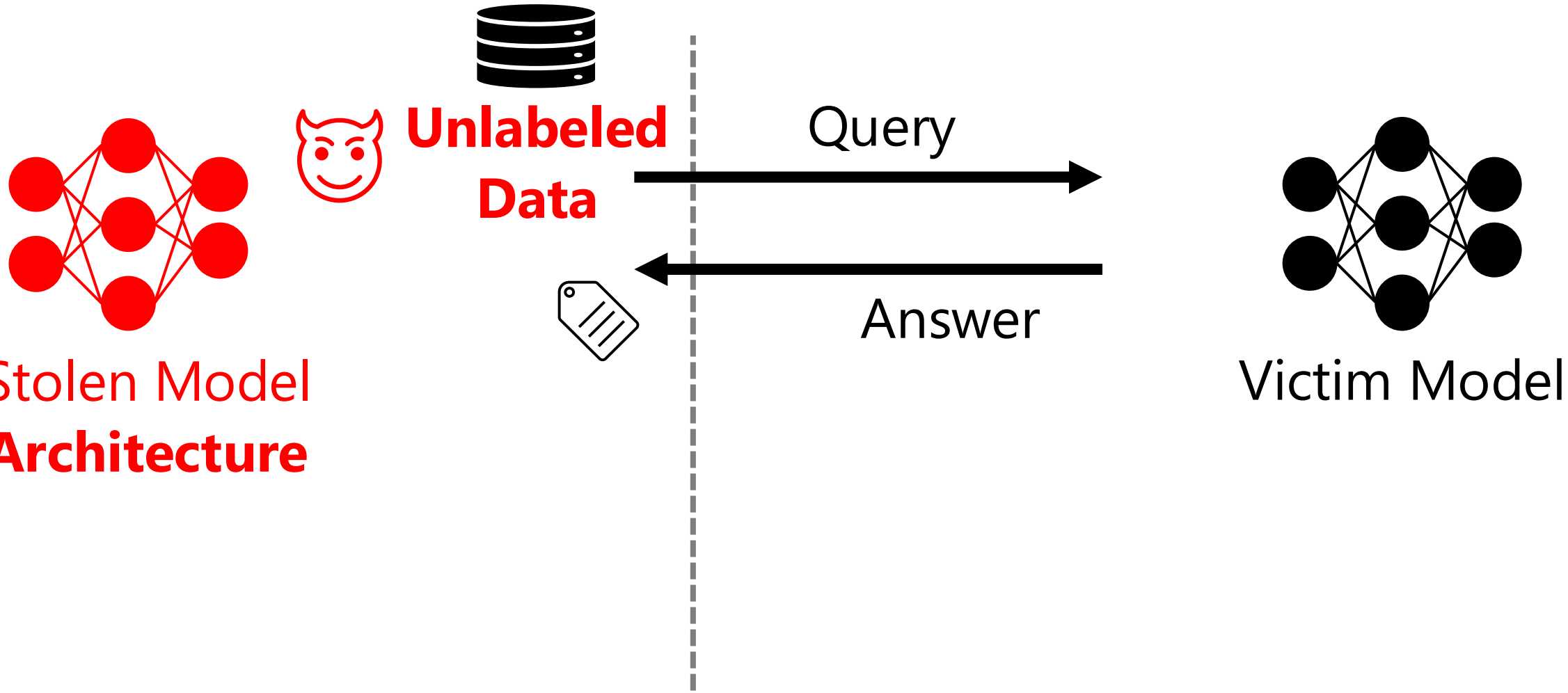


Stealing Models from **APIs**

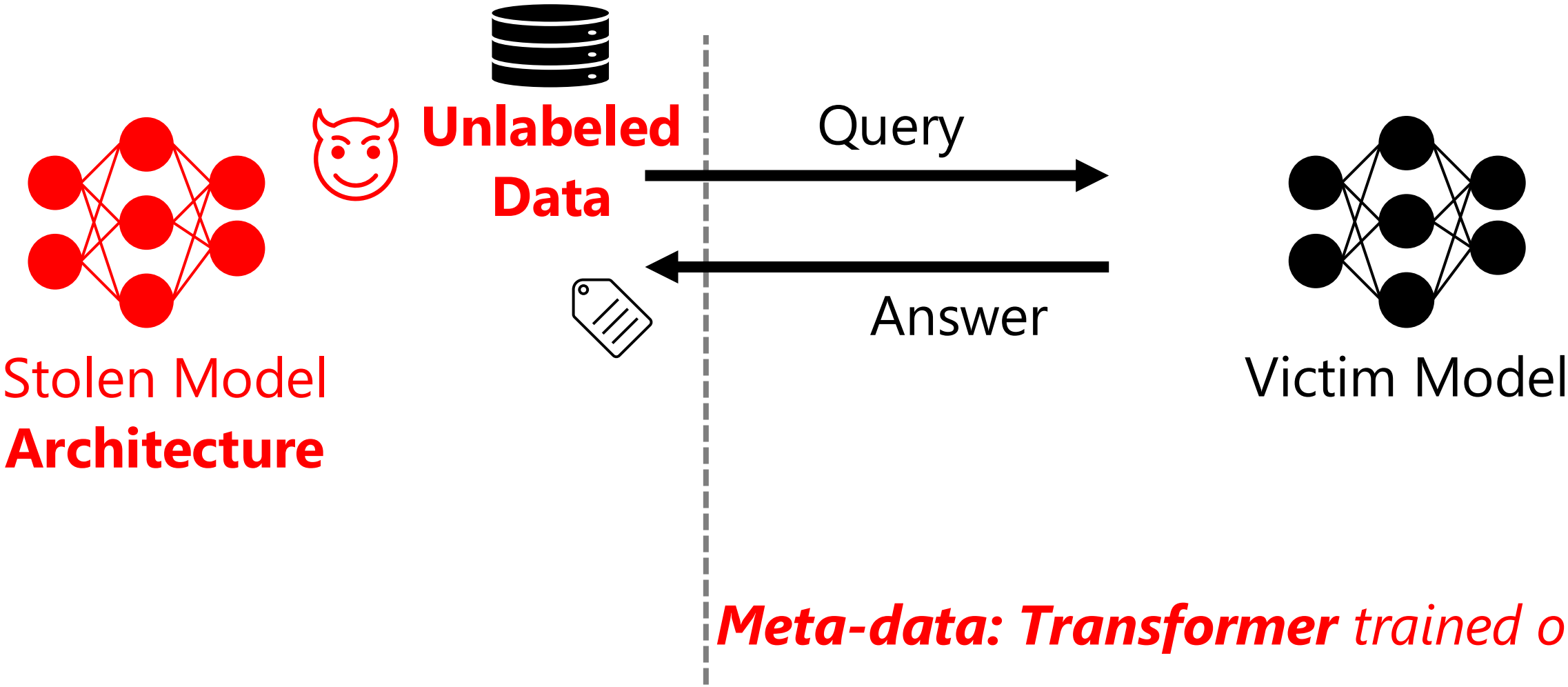


Assumptions for Stealing Models from **APIs**

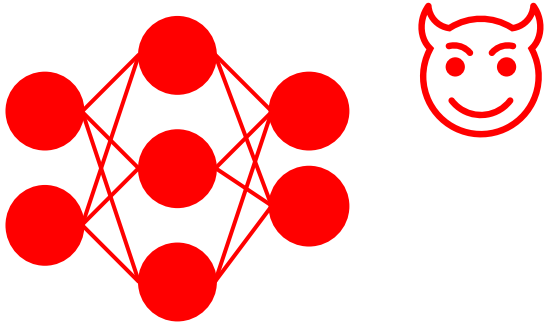
[Orekondy et al. CVPR 2019]



Assumptions for Stealing Models from **APIs**



Defenses Against Model Stealing



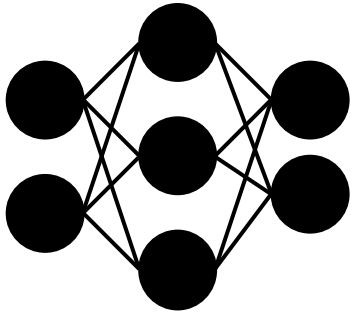
Stolen Model



Active Defenses



Ownership Resolution

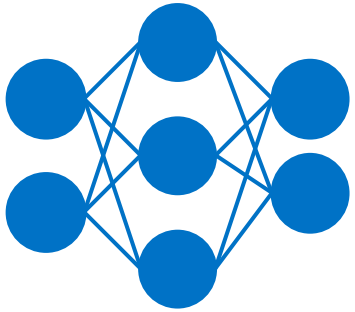


Victim Model

Active Defenses Against Model Stealing

Attacker

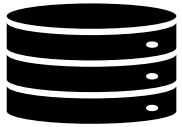
Defender



Victim Model

Adam Dziedzic, Muhammad Ahmad Kaleem, Yu Shen Lu, Nicolas Papernot *"Increasing the Cost of Model Extraction with Calibrated Proof of Work"* [ICLR 2022 SPOTLIGHT]

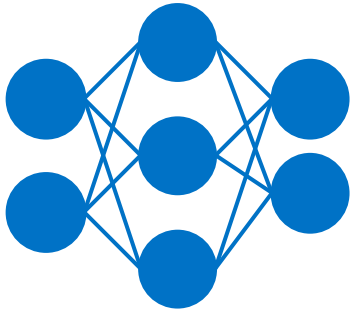
Estimate Victim Model Information Leakage



Unlabeled Data

Send Queries

Estimate Information Leakage

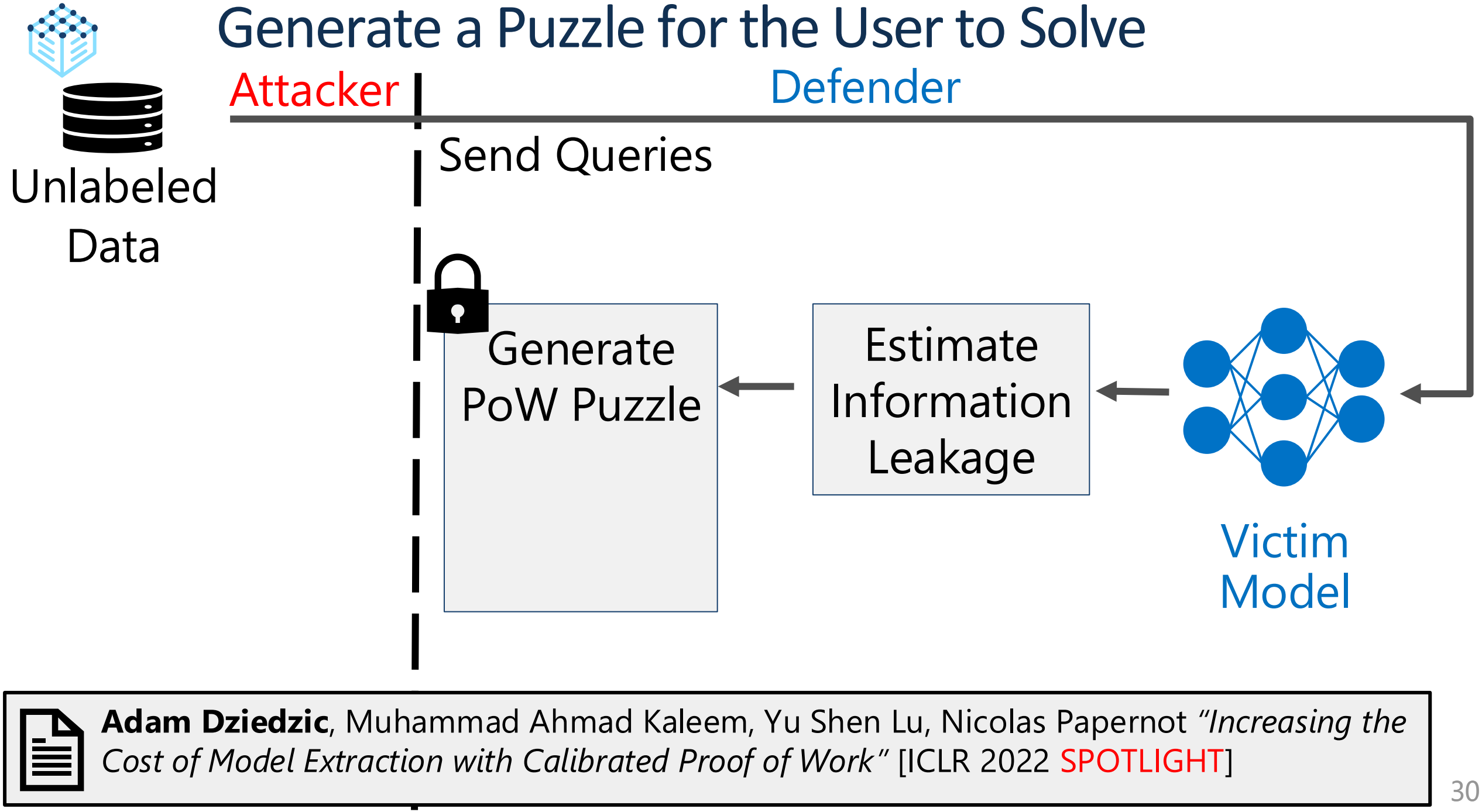


Victim Model



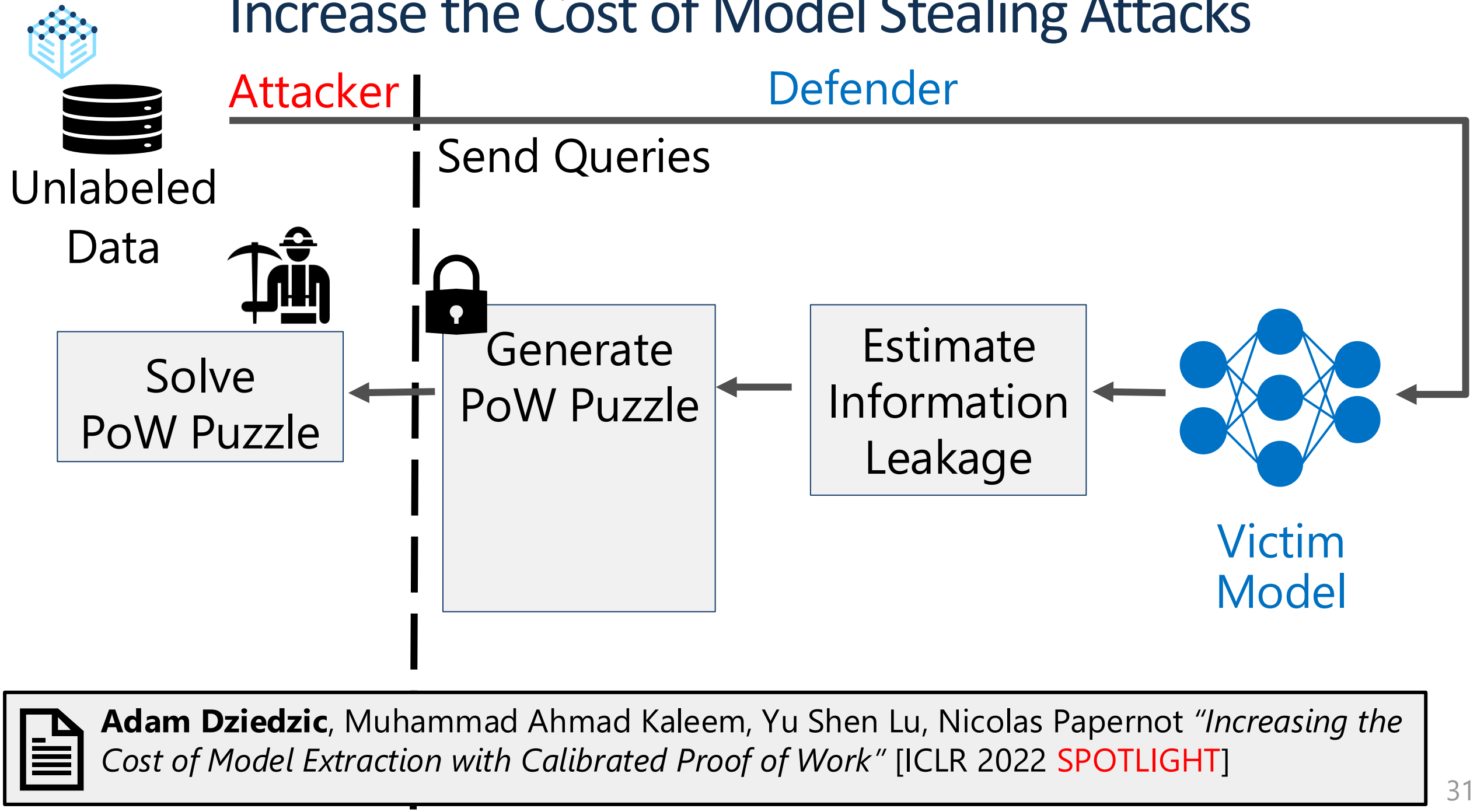
Adam Dziedzic, Muhammad Ahmad Kaleem, Yu Shen Lu, Nicolas Papernot *"Increasing the Cost of Model Extraction with Calibrated Proof of Work"* [ICLR 2022 SPOTLIGHT]

Generate a Puzzle for the User to Solve



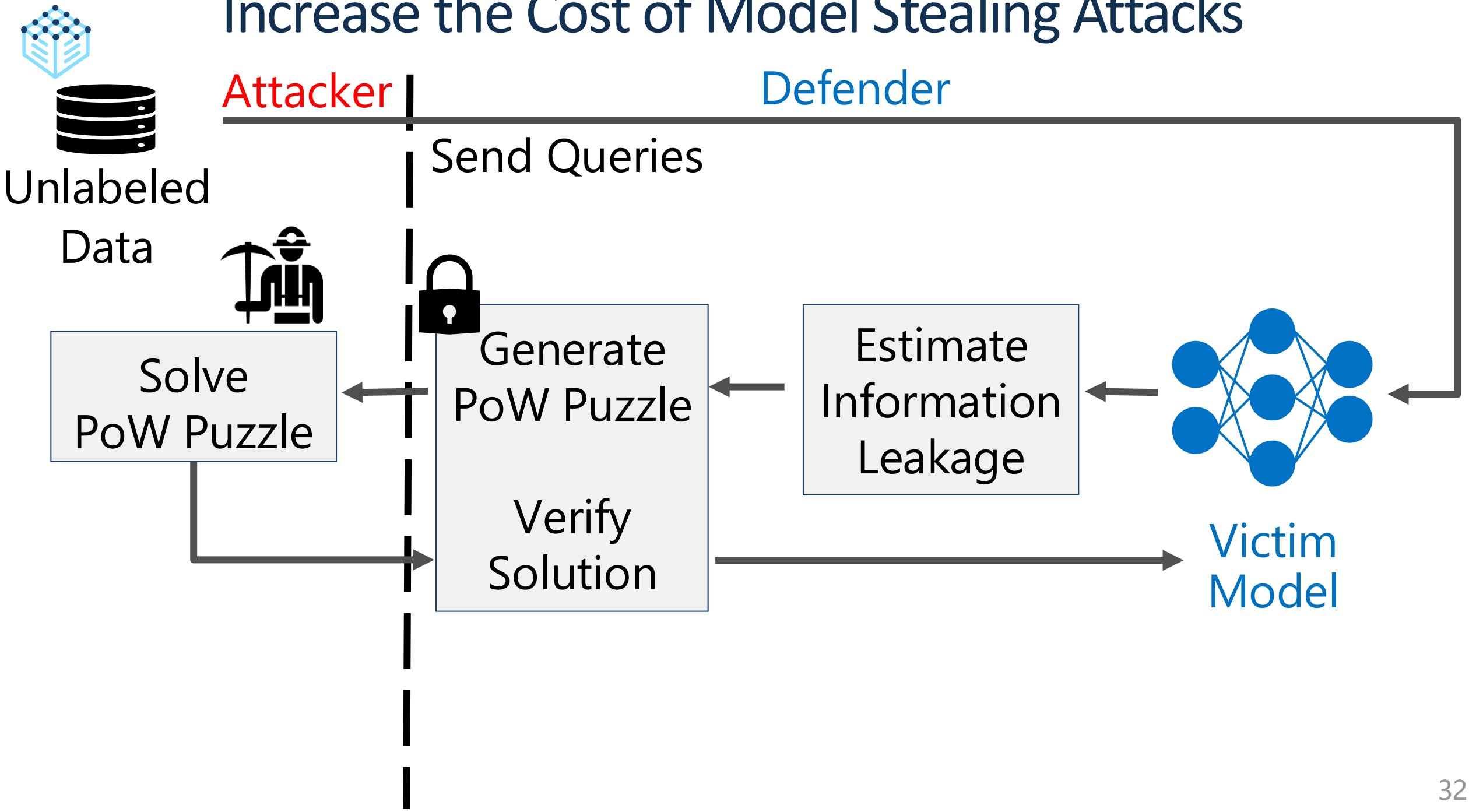
Adam Dziedzic, Muhammad Ahmad Kaleem, Yu Shen Lu, Nicolas Papernot *"Increasing the Cost of Model Extraction with Calibrated Proof of Work"* [ICLR 2022 SPOTLIGHT]

Increase the Cost of Model Stealing Attacks



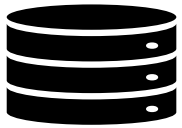
Adam Dziedzic, Muhammad Ahmad Kaleem, Yu Shen Lu, Nicolas Papernot *"Increasing the Cost of Model Extraction with Calibrated Proof of Work"* [ICLR 2022 SPOTLIGHT]

Increase the Cost of Model Stealing Attacks



Increase the Cost of Model Stealing Attacks





Unlabeled Data

Attacker

Defender

Send Queries



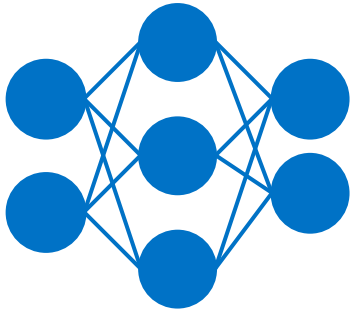


Solve PoW Puzzle

Generate PoW Puzzle

Verify Solution

Estimate Information Leakage



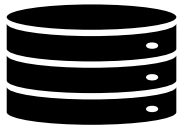
Victim Model

Return Output



Increase the Cost of Model Stealing Attacks





Unlabeled Data

Attacker

Defender

Send Queries



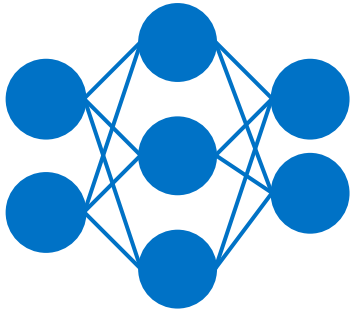


Solve PoW Puzzle

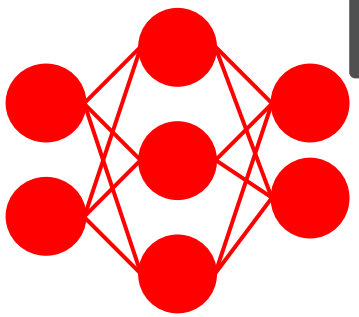
Generate PoW Puzzle

Verify Solution

Estimate Information Leakage



Victim Model



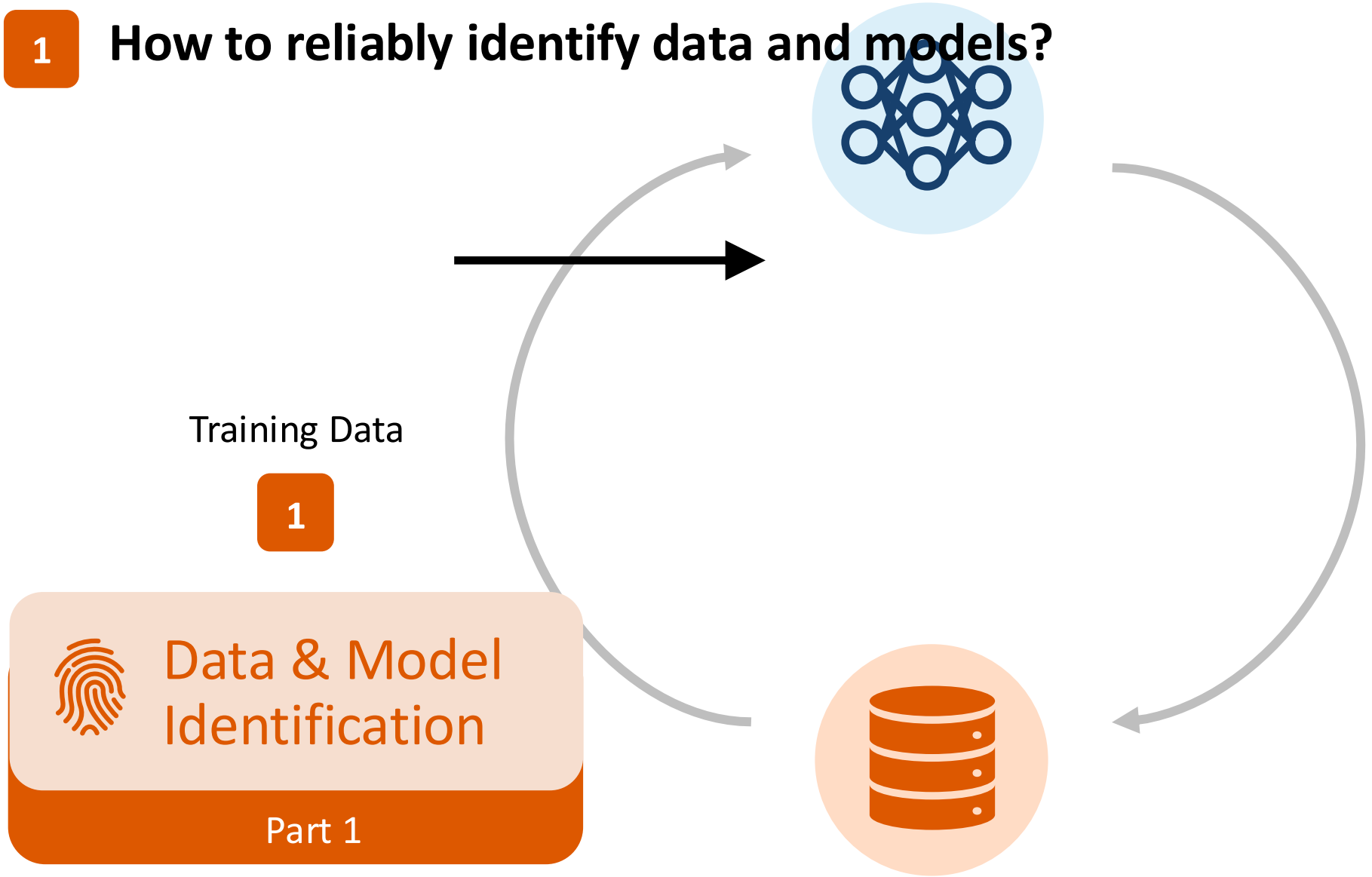
Stolen Model

Return Output



Unified Provenance Tracing for Data & Models

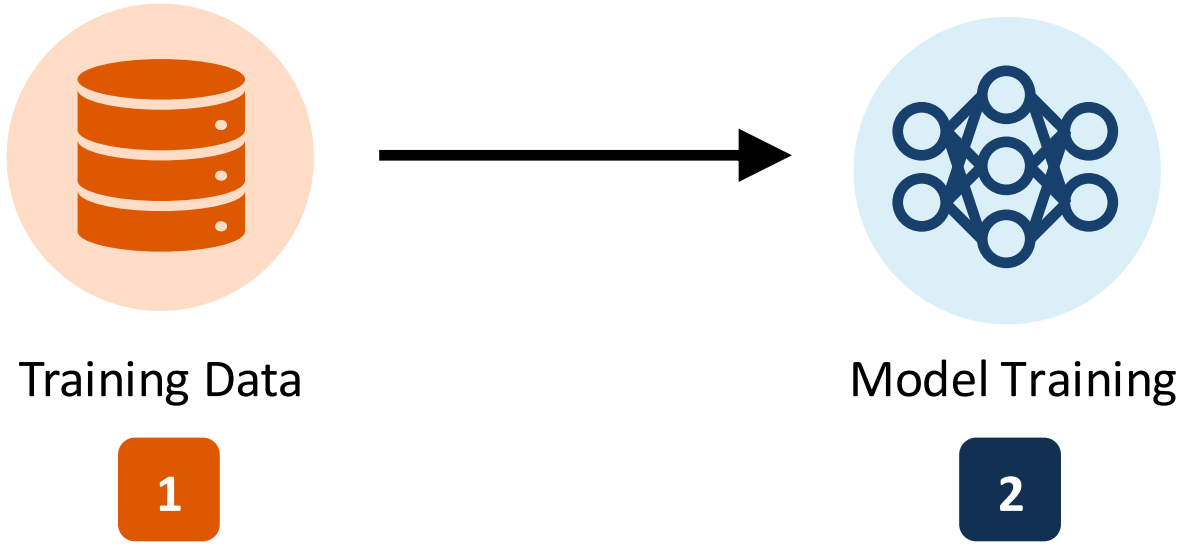
1 **How to reliably identify data and models?**



Unified Provenance Tracing for Data & Models

2

How to learn patterns instead of memorizing individual samples?



Data & Model Identification

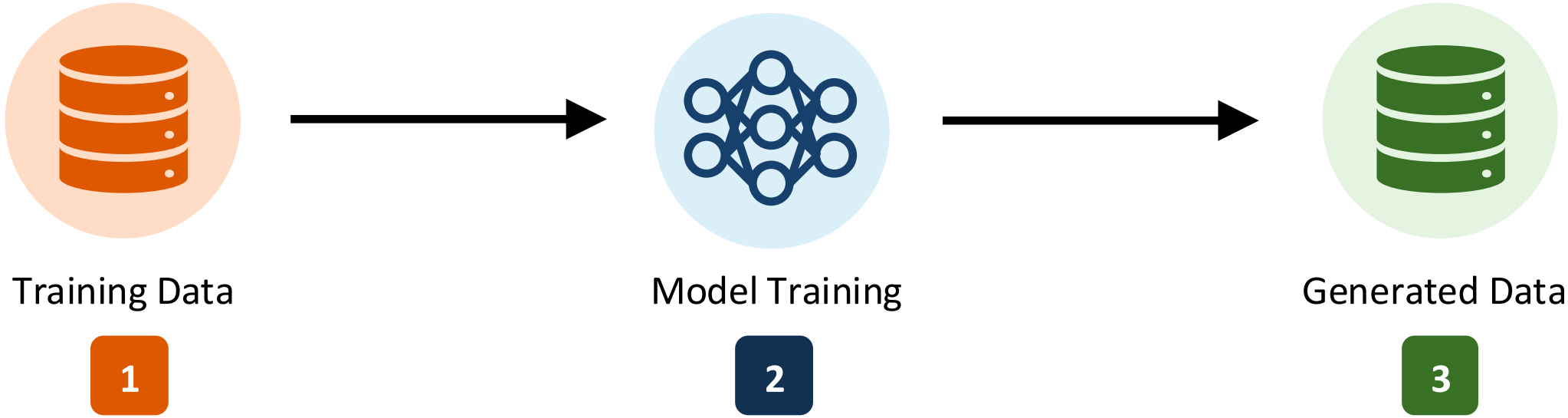
Part 1

Verbatim Data Memorization

Part 2

Our Solution: **Unified Provenance Tracing**

3 **How to persistently mark and identify generated data?**



Data & Model Identification

Part 1

Verbatim Data Memorization

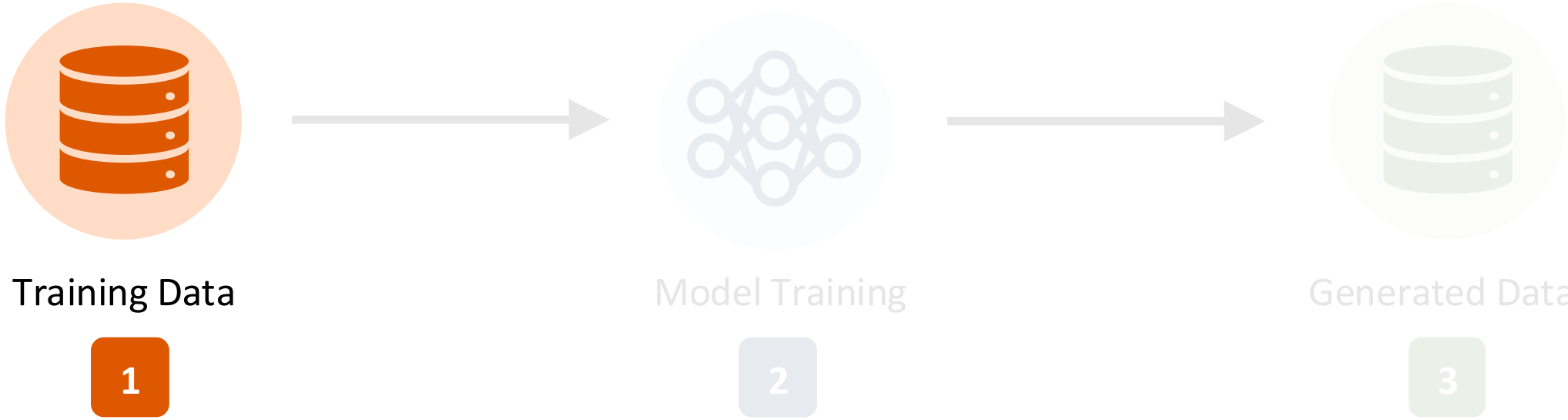
Part 2

Persistent Data Marking

Part 3

Our Solution: **Unified Provenance Tracing**

1 How to reliably identify data and models?



Data & Model Identification

Part 1

Controlled Data Memorization

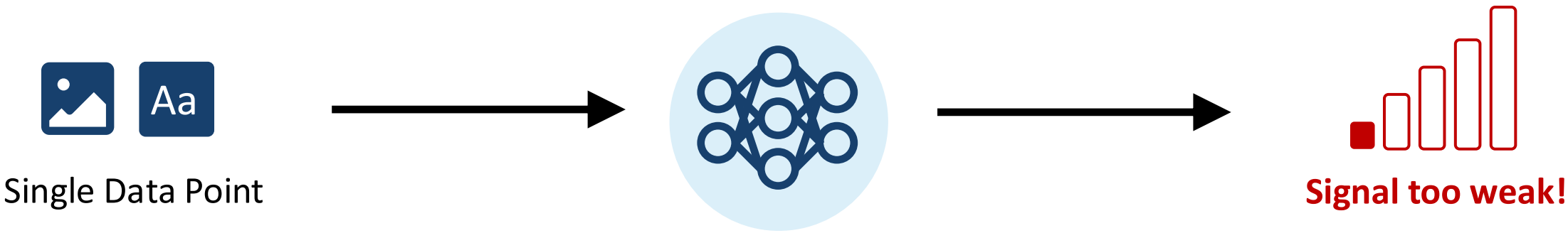
Part 2

Persistent Data Marking

Part 3

Limited **Membership Inference Methods**

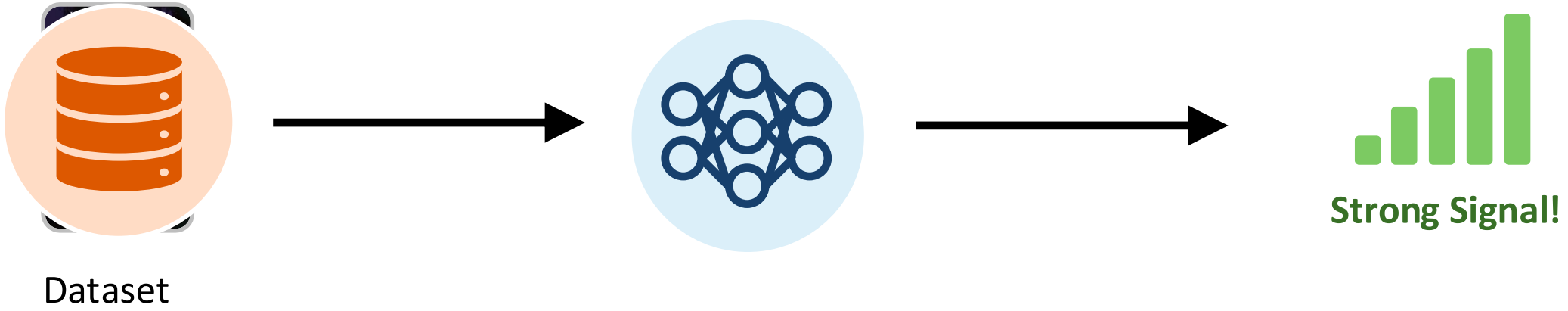
Was this **data point** used to train a model?



Our breakthrough: 1 Signals from single samples are too weak to identify them as training data

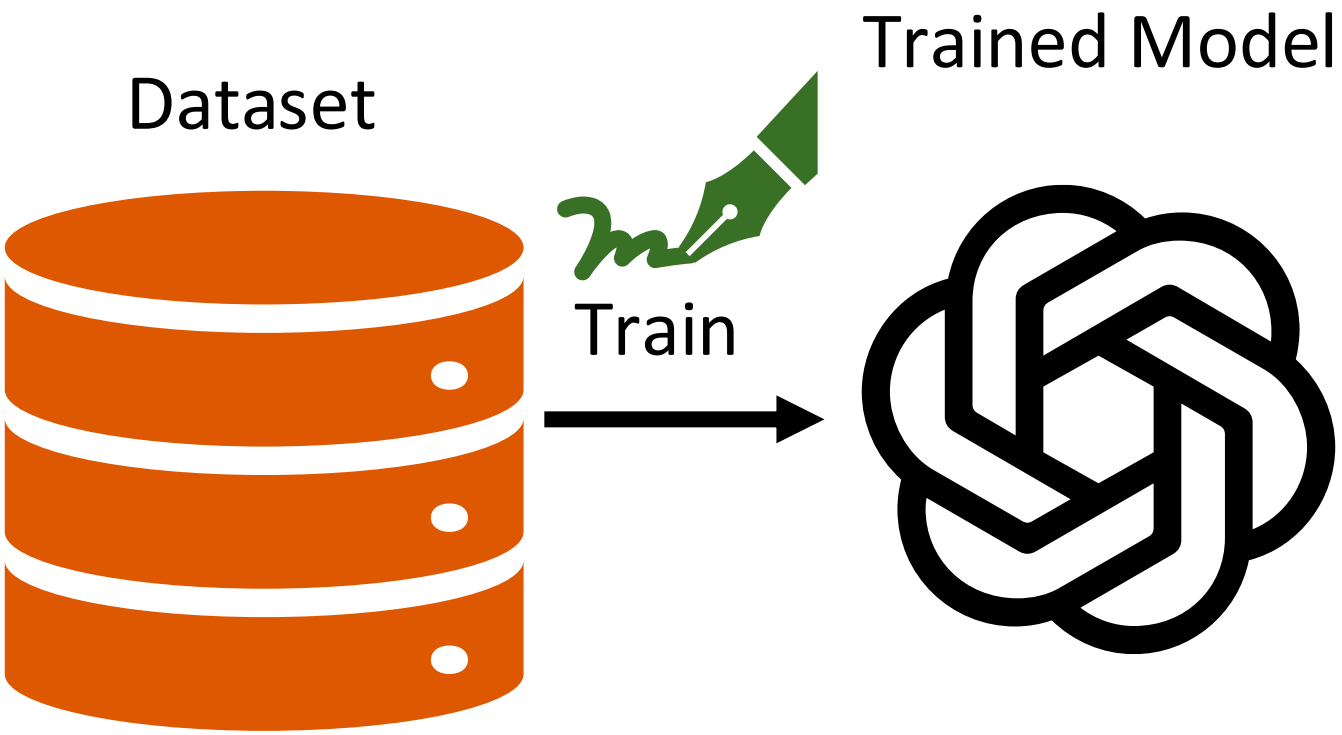
Dataset Inference: Large Language Models (LLMs)

Was this **dataset** used to train a model?



Our breakthrough: 2 Extract and aggregate a large number of signals per sample and across a dataset

Training Dataset is a Signature of the Model



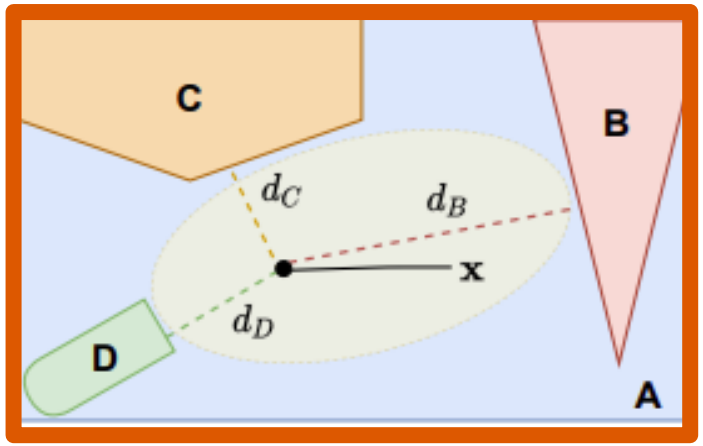
Training Dataset is a Signature of the Model

Dataset

Trained Model

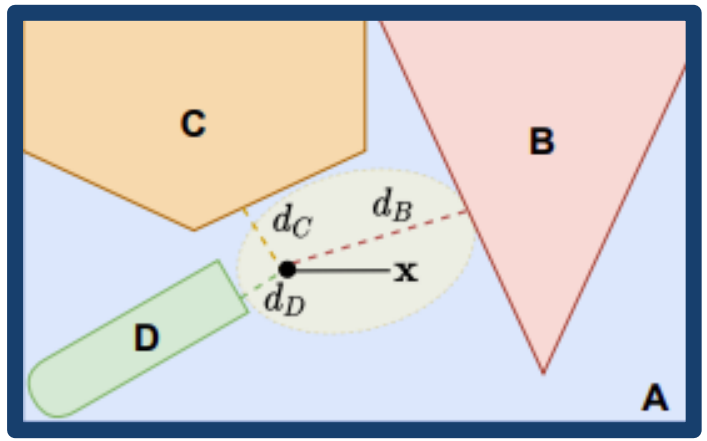
Train

Test

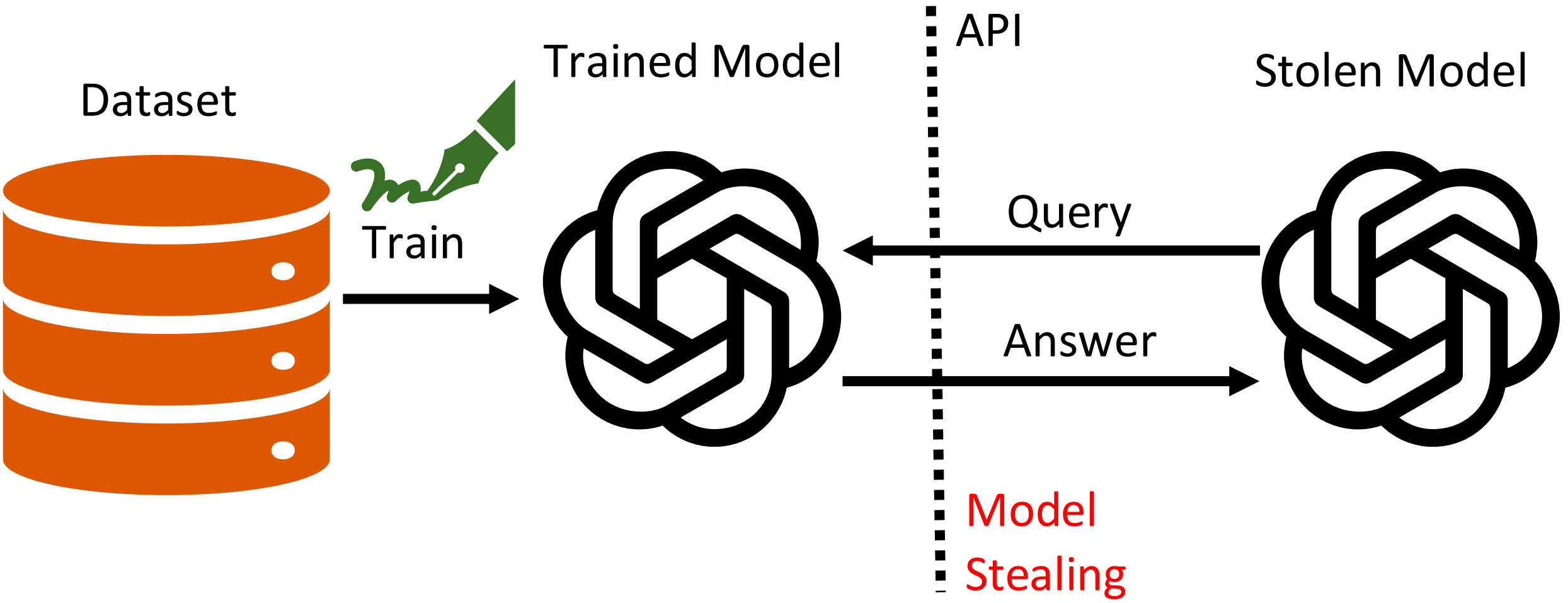


Train Set

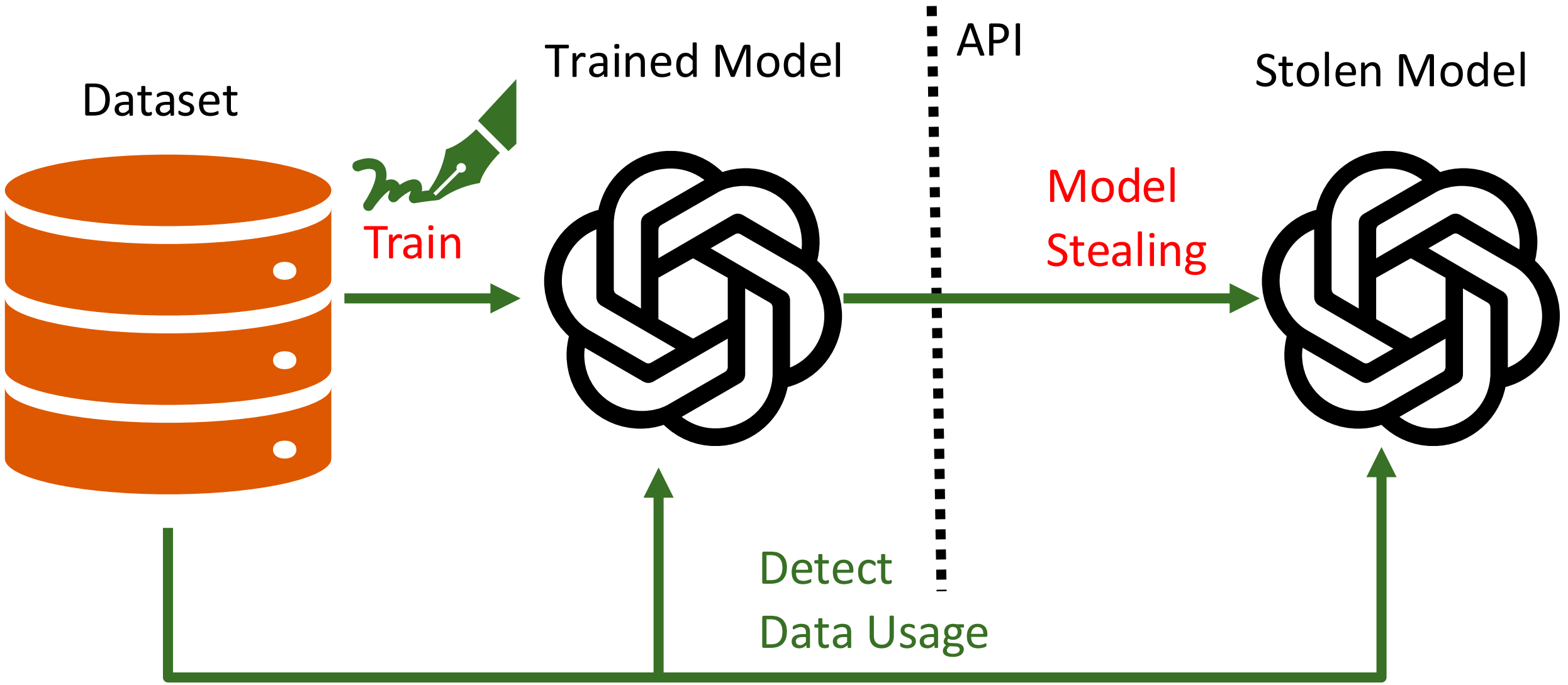
Test Set



Dataset Inference to Identify Stolen Models

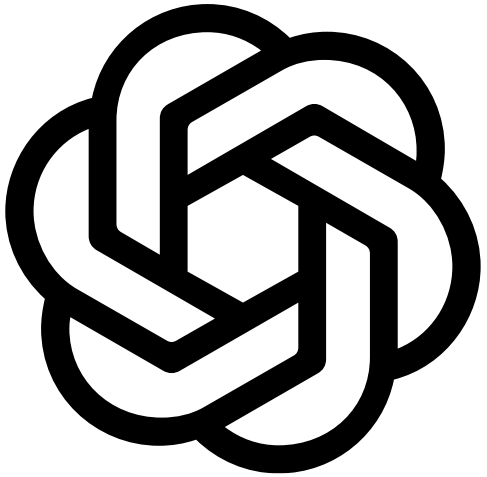


Dataset Inference: Identify Data & Models

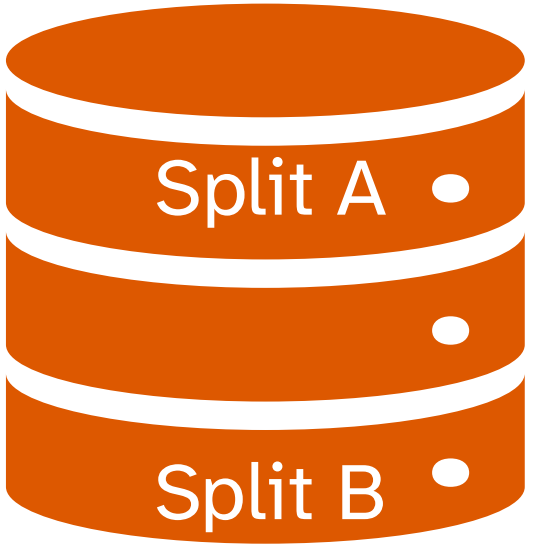


Dataset Inference Procedure: Stage 1/4

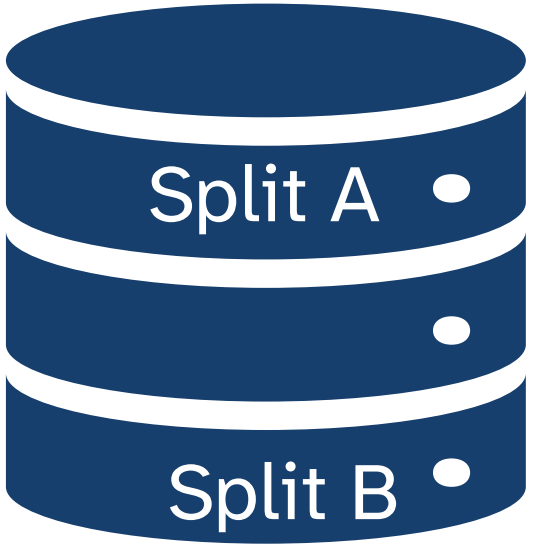
Suspect Model



Suspect Set



Held-out Set

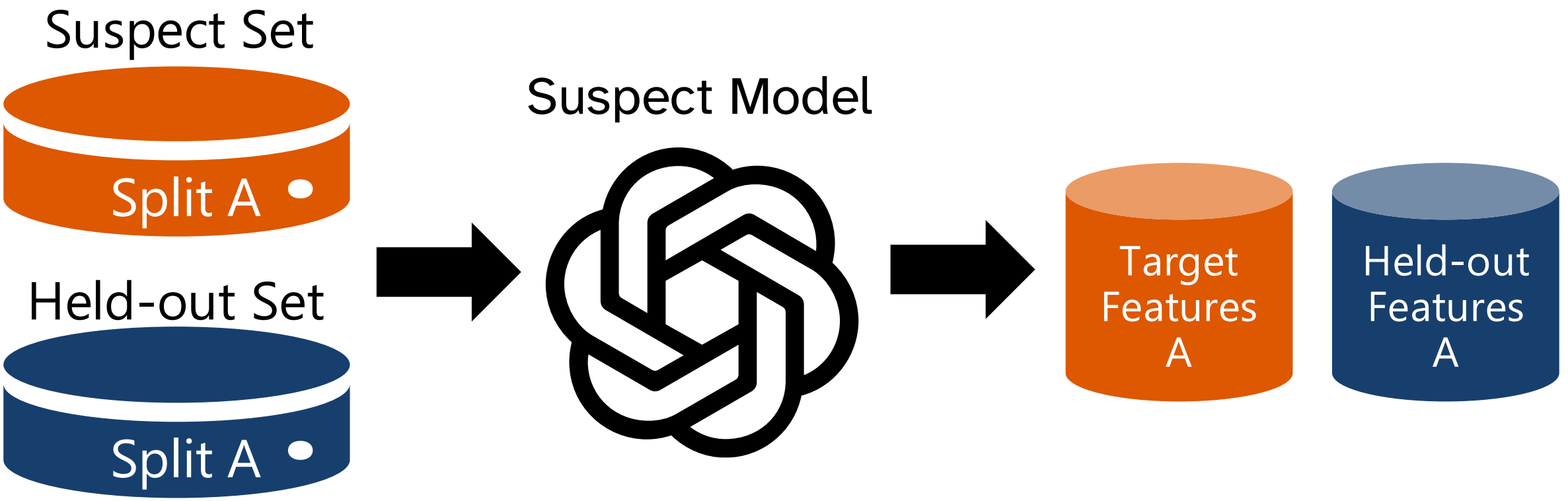


Test: Was the Suspect LLM trained on the Suspect Set?

Assumption: Suspect and Held-out Sets are IID

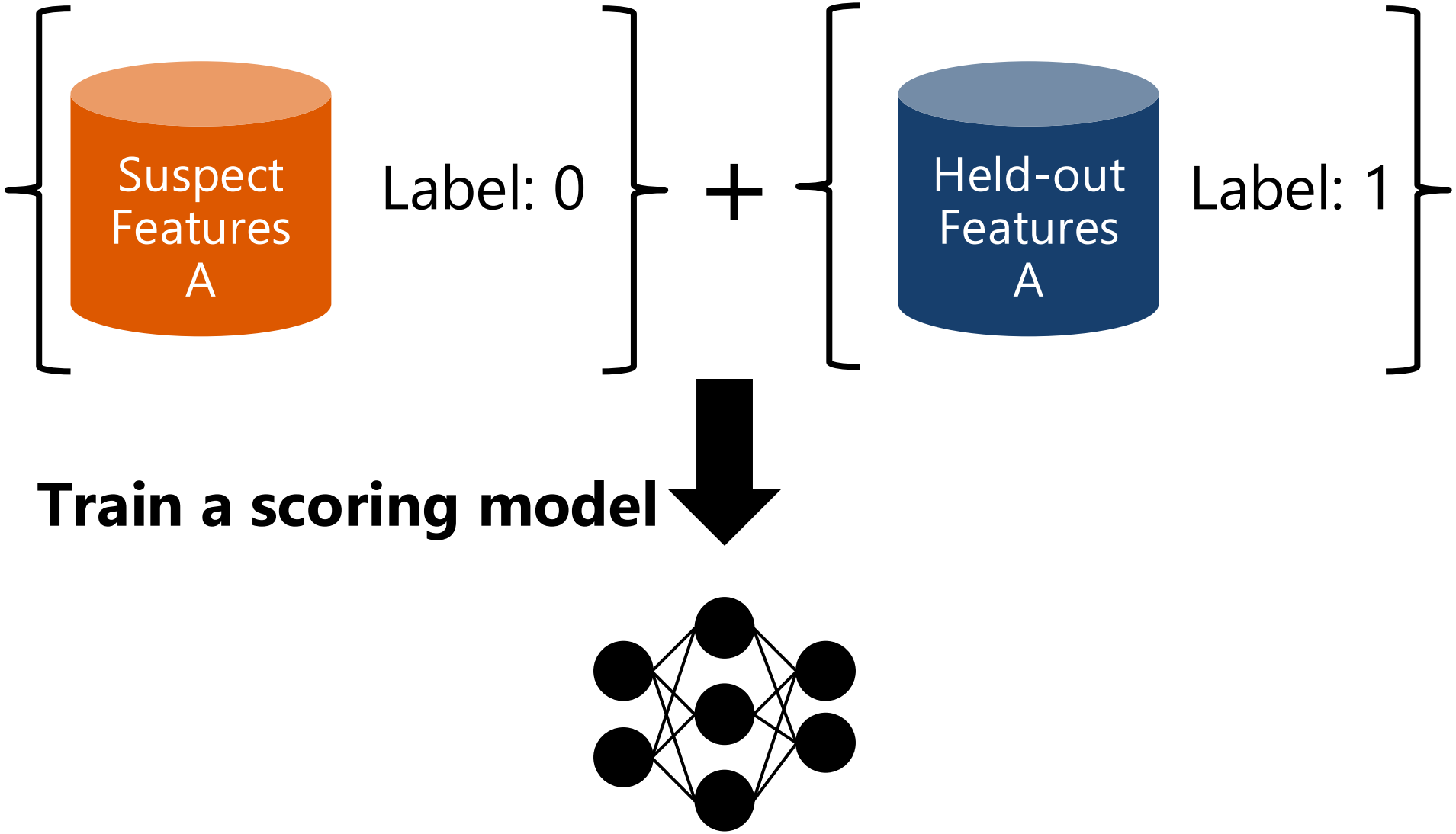
"LLM Dataset Inference: Did you train on my dataset?" (...) NeurIPS'24

Extract Membership Features: DI Stage 2/4



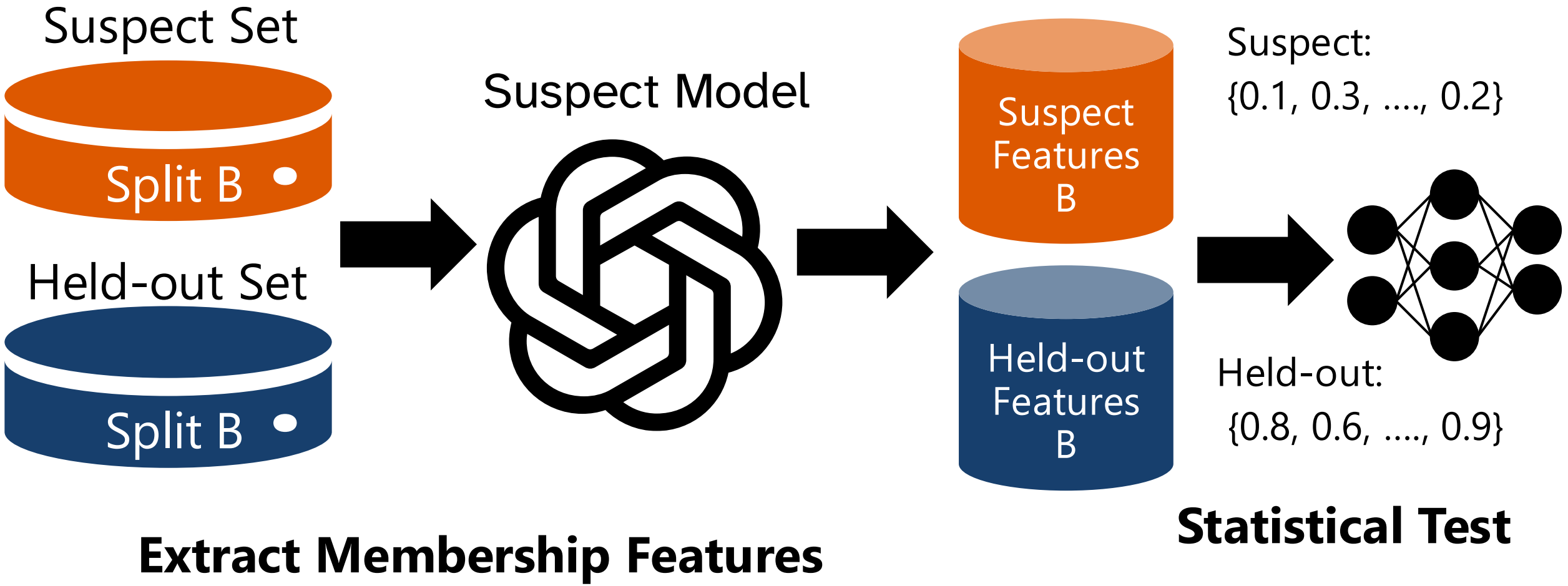
Extract Membership Features

Learn Feature Membership: DI Stage 3/4

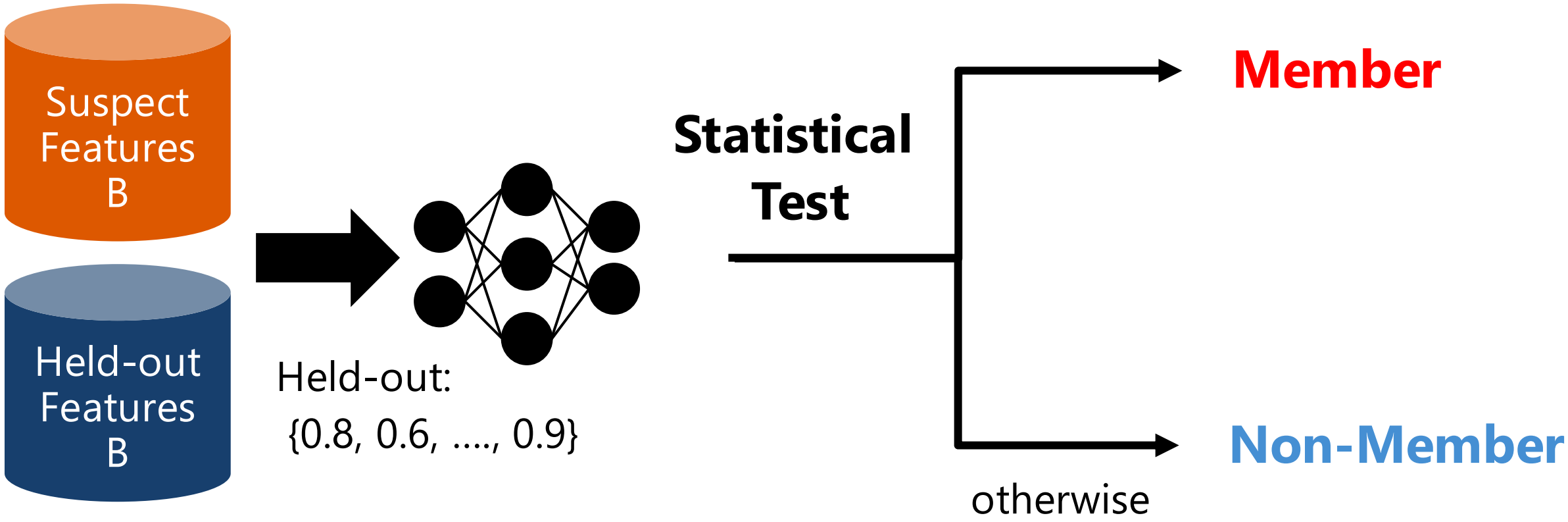


Scores: Suspect: {0.1, 0.3, …., 0.2} Held-out: {0.8, 0.6, …., 0.9}

Perform Dataset Inference: Stage 4/4

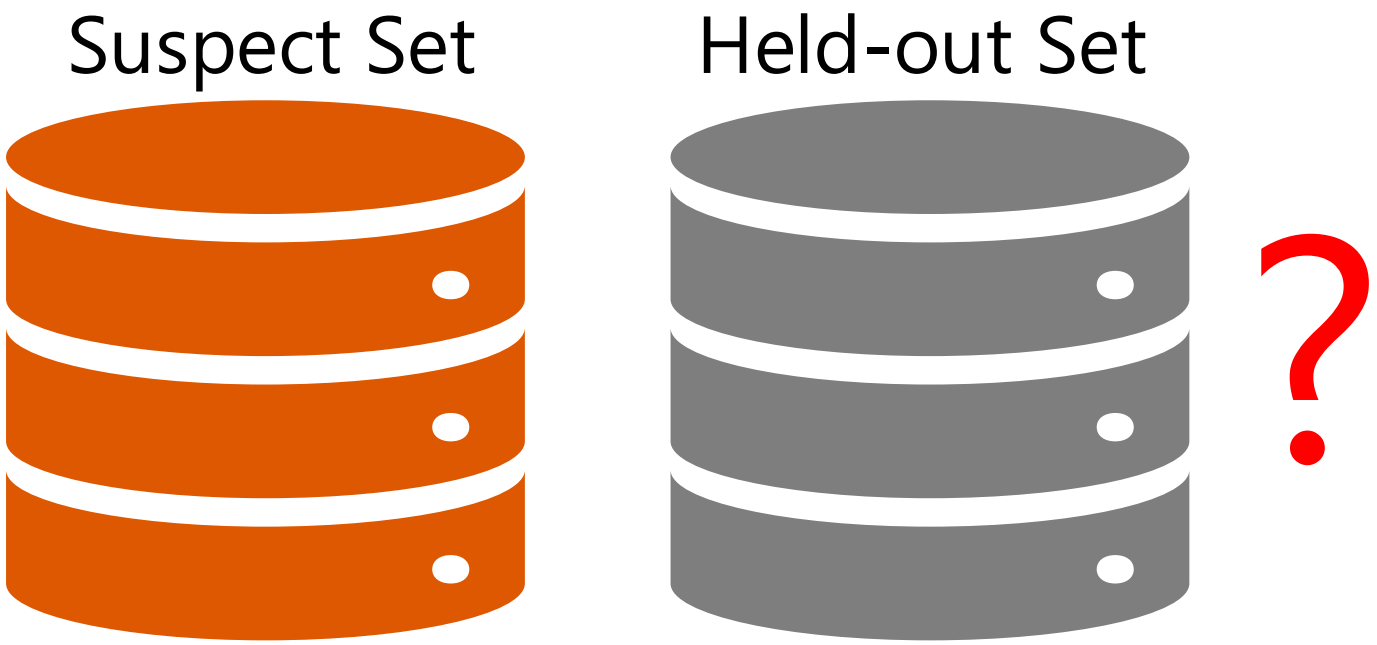


Perform Dataset Inference: Stage 4/4



Null Hypothesis: Suspect set was not used to train the suspect model

How to Provide the Held-out Set for DI?



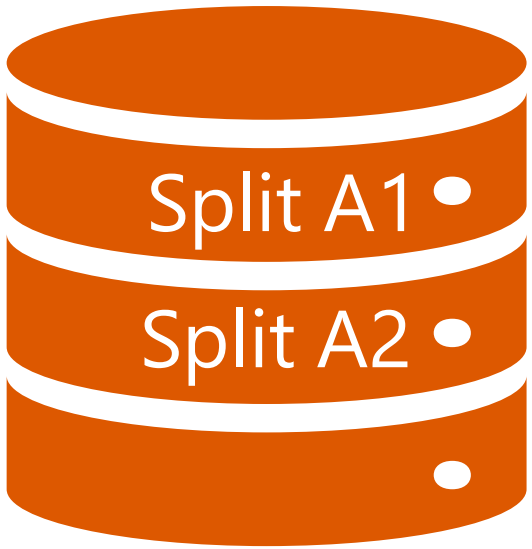
Assumption: Access to a Held-out Set from the same distribution as the Suspect Set

"Unlocking Post-hoc Dataset Inference with Synthetic Data" (...) ICML'25

Generate Held-out Set from the Suspect Set

Segment Text:

Mr. and Mrs. Dursley, of number four Private Drive, //



Suspect Set

Train Generator

were proud to say that they were perfectly normal.

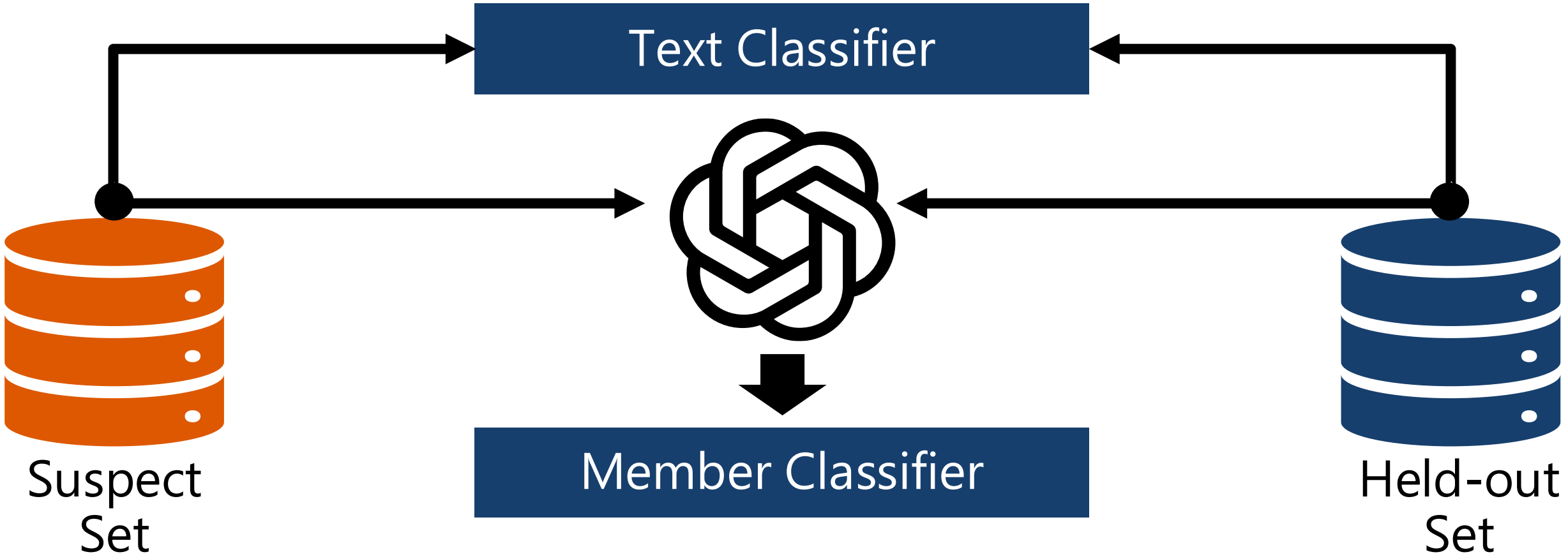
They were the last people you'd expect to be, //



Held-out Generation

involved in anything strange or mysterious.

Distribution Shift vs Membership Signal



$\mathcal{H}_1$(Alternative Hypothesis):
$\mathbb{E}$[Member(Held) – Member(Suspect)] ≫ $\mathbb{E}$[Text(Held)-Text(Suspect)]

Natural Identifiers (NIDs) for Post-hoc Audits

NIDs are structured random **identifiers**, e.g., URLs or crypto hashes

NID from RefinedWeb Dolma based on MD5 cryptographic hash

For a future birthday party – fairy party favors. But I want to figure out a different fairy, not Disney…
34d42a69a258fa51222a2e94b4563007.jpg 300×300 pixels



GID: **9659875b92ba8fa639ba476aedbb73b9**

"Natural Identifiers for Privacy and Data Audits in LLMs" (…) ICLR'26

From NIDs to Generated Identifiers (GIDs)

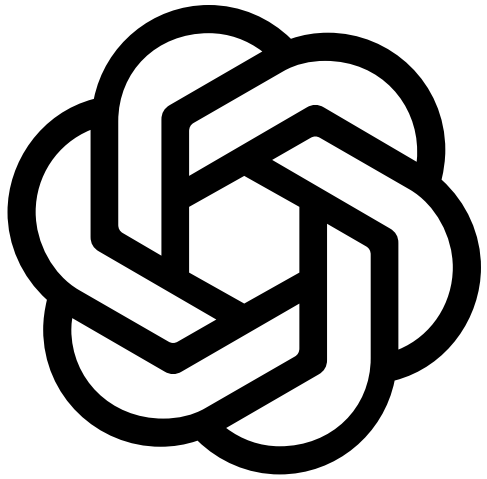
Natura Identifiers (NIDs) are structured random **IDs** naturally occurring in common training datasets. **Generated Identifiers (GIDs)** are generated **IDs** from the exact same distribution as **NIDs**.

NID from RefinedWeb Dolma based on MD5 cryptographic hash
For a future birthday party – fairy party favors. But I want to figure out a different fairy, not Disney… **34d42a69a258fa51222a2e94b4563007**.jpg 300×300 pixels

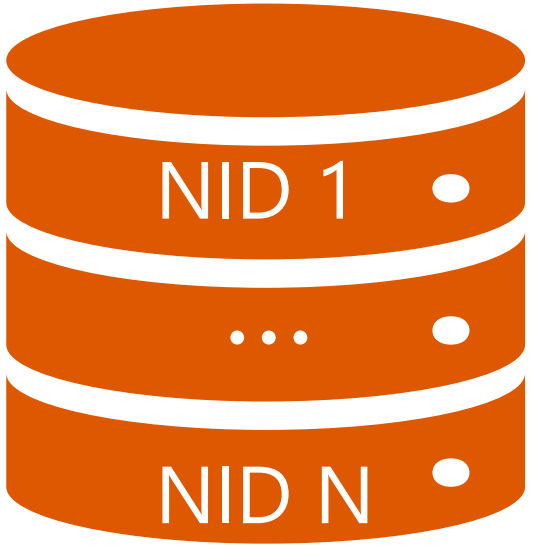
GID from RefinedWeb Dolma based on MD5 cryptographic hash
For a future birthday party – fairy party favors. But I want to figure out a different fairy, not Disney… **9659875b92ba8fa639ba476aedbb73b9**.jpg 300×300 pixels

Natural Identifiers (NIDs) for Post-hoc Audits

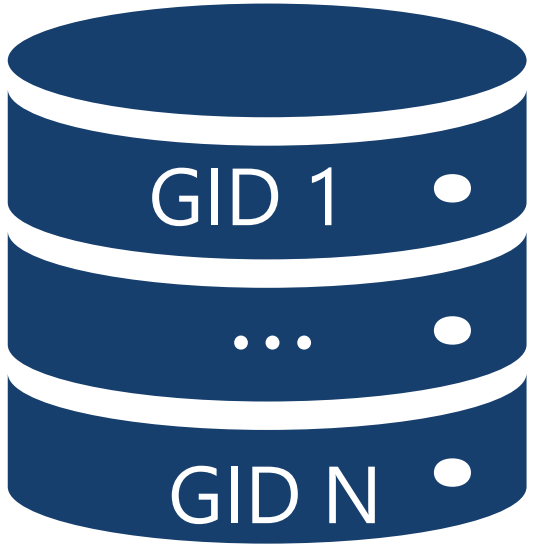
NIDs are structured random **identifiers**, e.g., URLs or crypto hashes



Suspect Model



Suspect Set



Held-out Set

"Natural Identifiers for Privacy and Data Audits in LLMs" (...) ICLR'26

Hands-on task: **Dataset Inference Challenge**

1 **Analyze a trained image classifier**

Use its outputs to decide which candidate image data subsets were used during training.

2 **Start from a scaffolded baseline**

Develop stronger ways of combining evidence across multiple examples using dataset inference.

3 **Submit dataset membership scores**

Compete on the live Kaggle leaderboard with your dataset-membership scores.

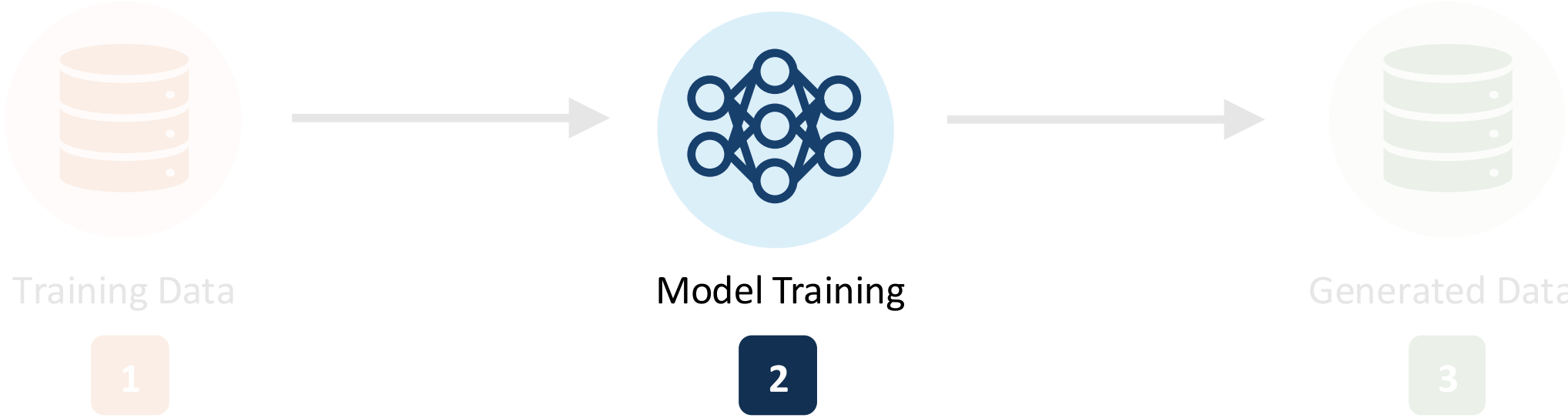
Kaggle competition

huggingface.co/datasets/SprintML/ml-in-pl-dataset-inference

Our Solution: **Unified Provenance Tracing**

2

How do generative models memorize individual training samples?



Data & Model Identification

Part 1

Verbatim Data Memorization

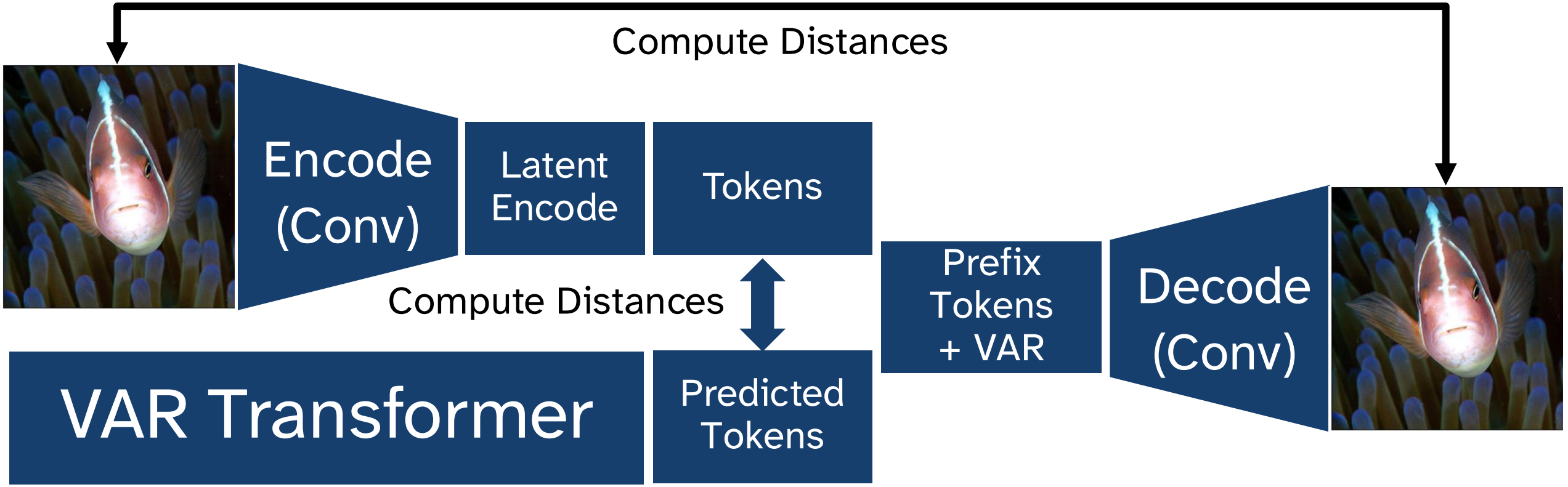
Part 2

Persistent Data Marking

Part 3

Memorization: Image Autoregressive Models

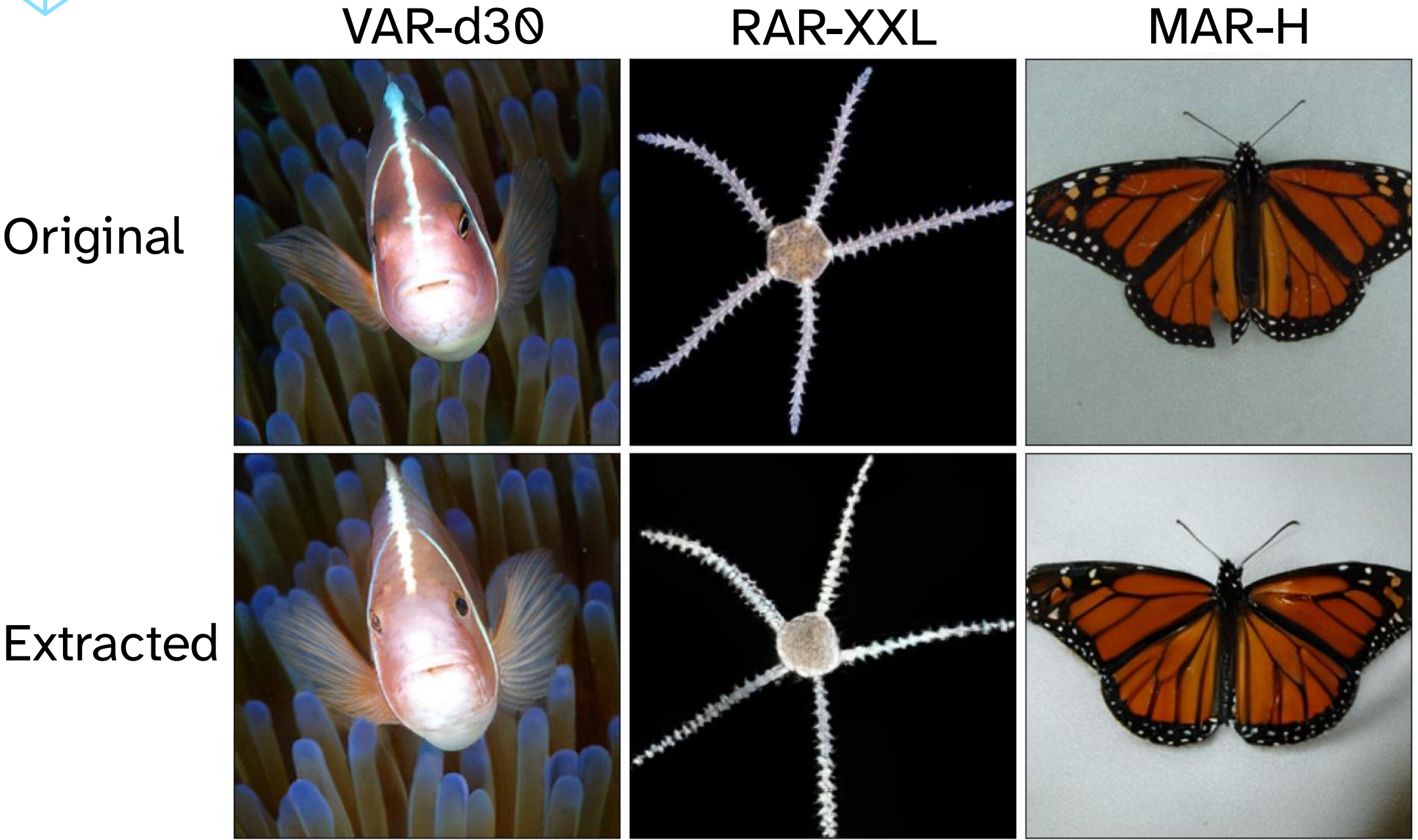
We extract individual memorized training samples from Image Autoregressive Models



Model	VAR-d30	MAR-H	RAR-XXL
# of Extracted Memorized Images	698	5	36

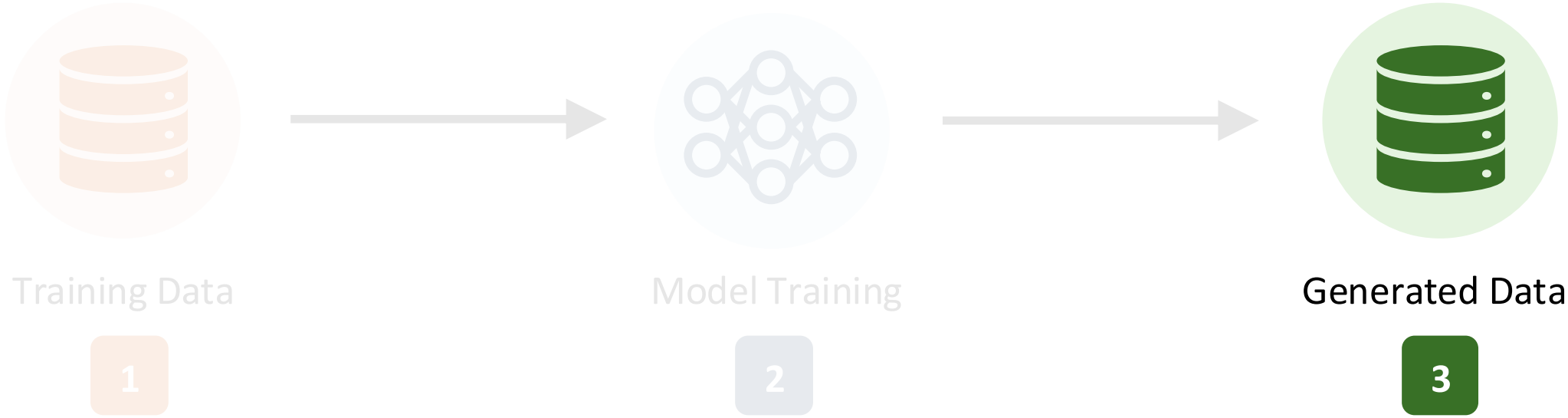
"Privacy Attacks on Image Autoregressive Models" (...) ICML'25

Verbatim Memorized Images in IARs



Unified Provenance for Data & Models

3 **How to persistently mark and identify generated data?**



Data & Model Identification

Part 1

Verbatim Data Memorization

Part 2

Persistent Data Marking

Part 3

Is the Image AI Generated or Human Created?



Detector

Hard to predict!

Is the Image AI Generated or Human Created?



Detector

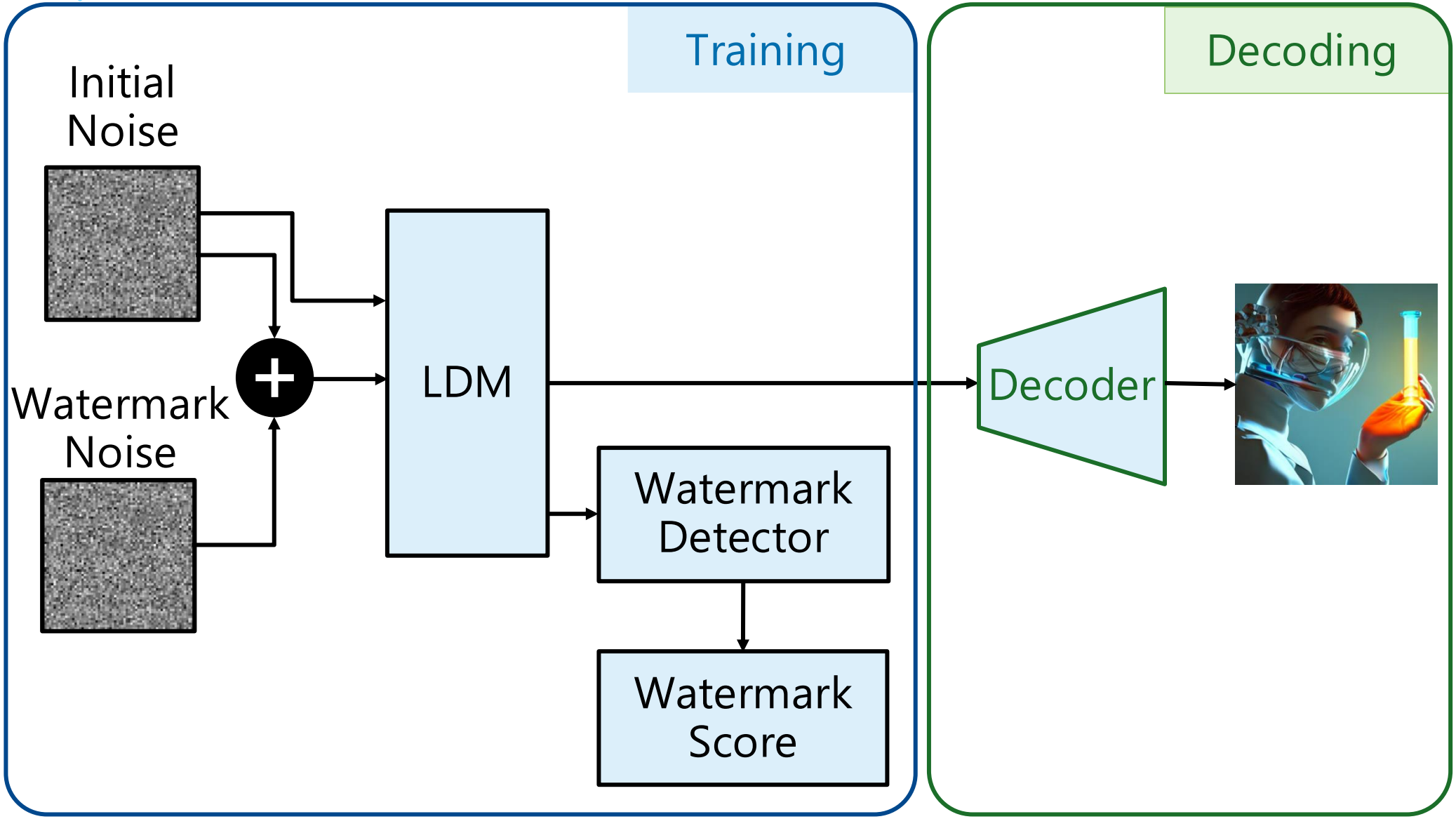
Hard to predict!



Detector

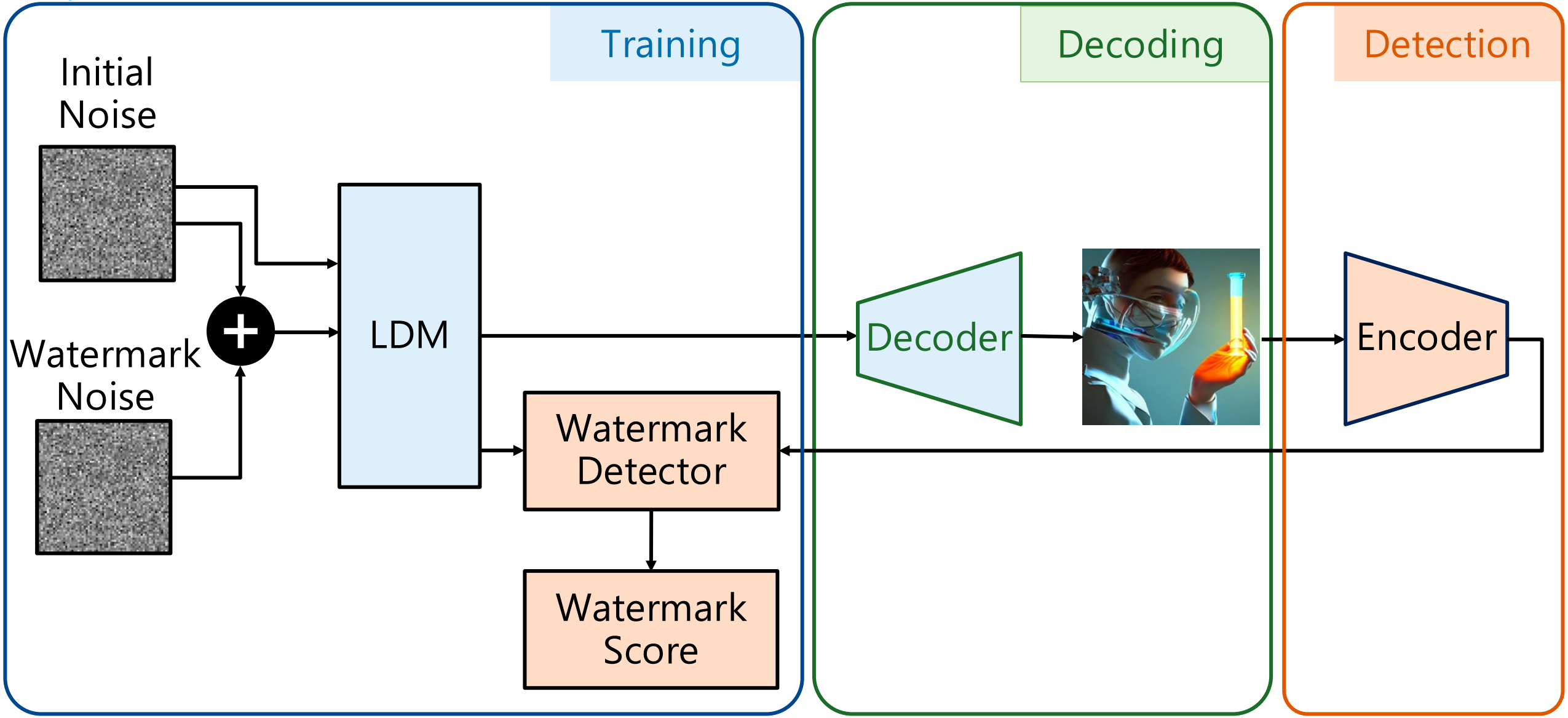
Generated

SERUM: Simple, Efficient, Robust, Unifying Mark

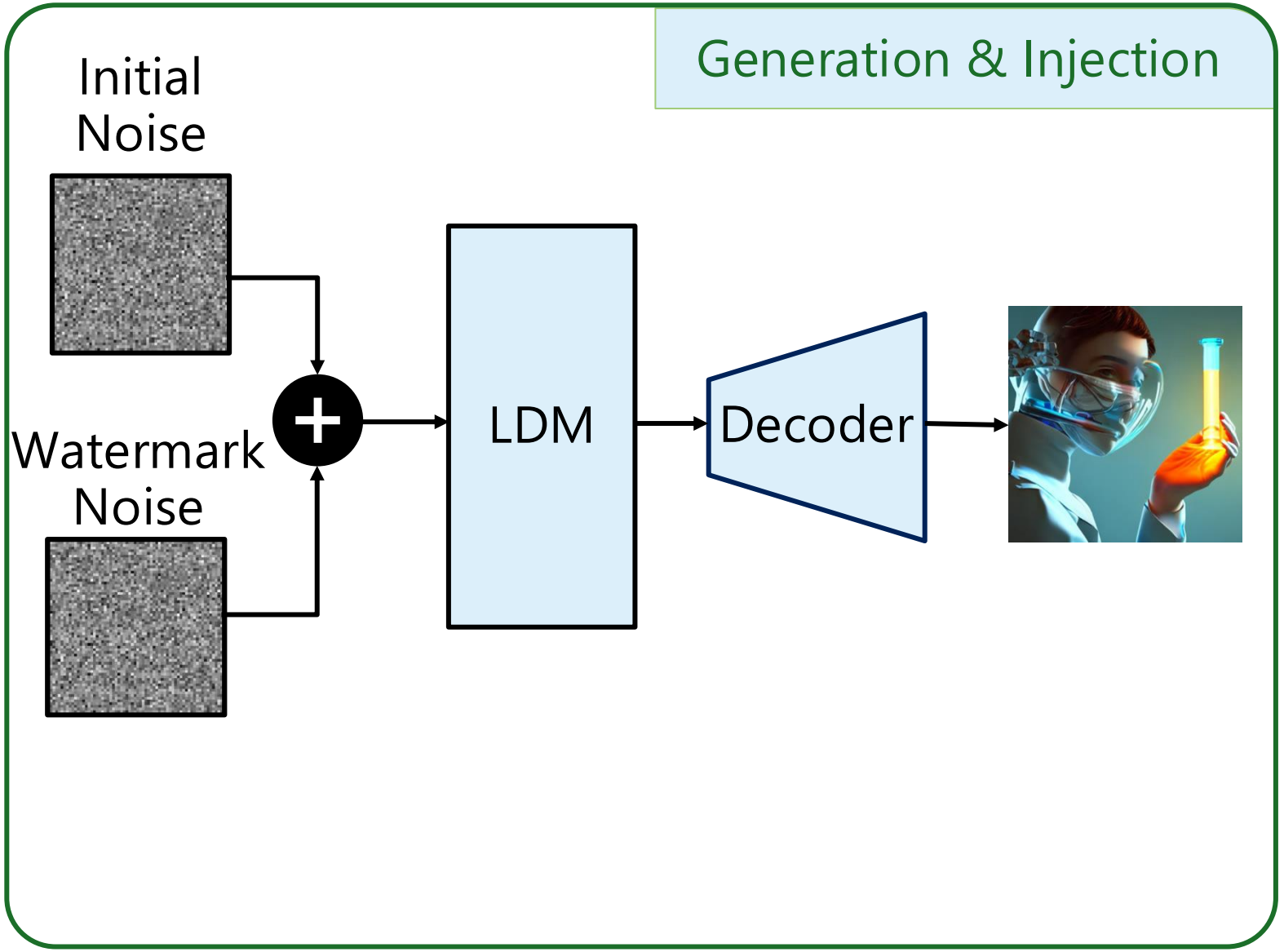


Kociszewski,..., Boenisch, **Dziedzic** *"SERUM: Simple, Efficient, Robust, & Unifying Marking for Diffusion Models" ICLR'26*

SERUM: Simple, Efficient, Robust, Unifying Mark



SERUM Watermark: Generation & Injection



Watermark Noise Generation:

$W = \mathcal{N}(0, I)$ (fixed per user)

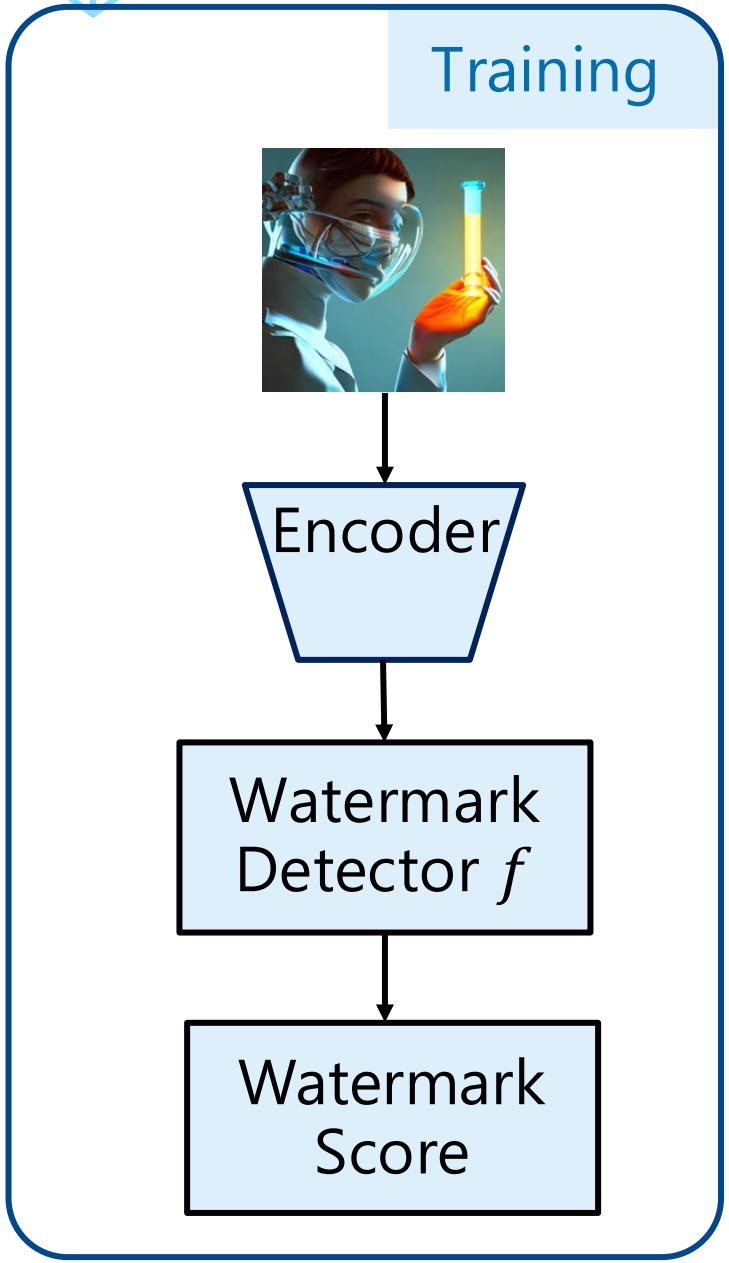
Watermark Injection:

$$\eta' = \sqrt{1-\alpha}\eta + \sqrt{\alpha}W$$

η –Initial Gaussian Noise

α –controls the balance between watermark's detectability and image diversity

SERUM: Lightweight Watermark Detector



Loss for detecting watermarked images:

$$\mathcal{L}_w = \underbrace{-\log f\,(x^*)}_{\text{marked clean}} \underbrace{- logf\left(\mathcal{T}\left(x^*_{:,:m}\right)\right)}_{\text{marked transformed}} \underbrace{- logf(x_t^*)}_{\text{marked precomputed}}$$

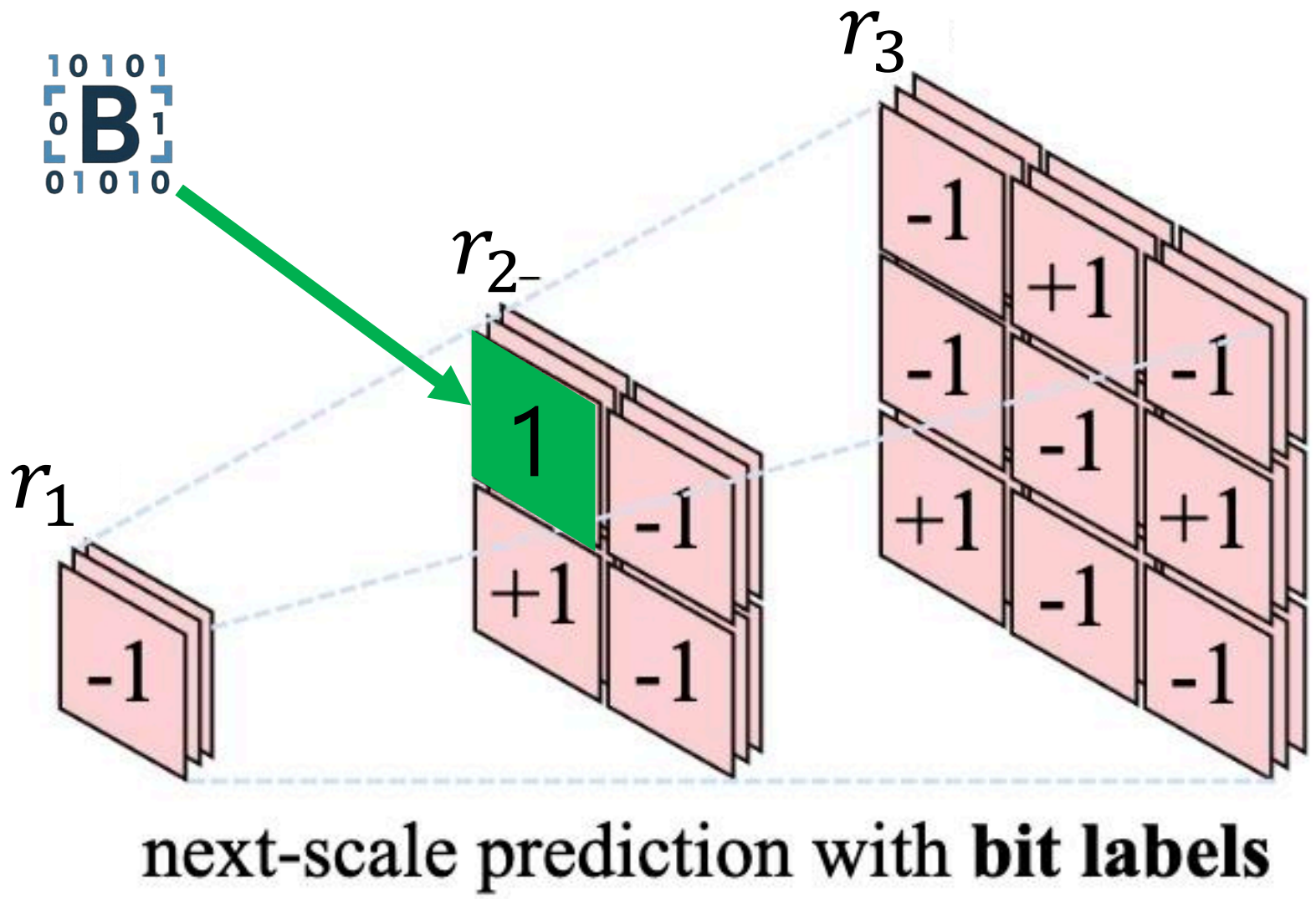
Loss for detecting clean images (no watermark added):

$$\mathcal{L}_n = \underbrace{-\log(1 - f\,(x)}_{\text{clean}} \underbrace{- logf\left(1 - \mathcal{T}\left(x_{:,:m}\right)\right)}_{\text{transformed}} \underbrace{- \log(1 - f(x_t)}_{\text{precomputed}}$$

Loss for training the watermark detector: $\mathcal{L} = \mathcal{L}_w + \mathcal{L}_n$

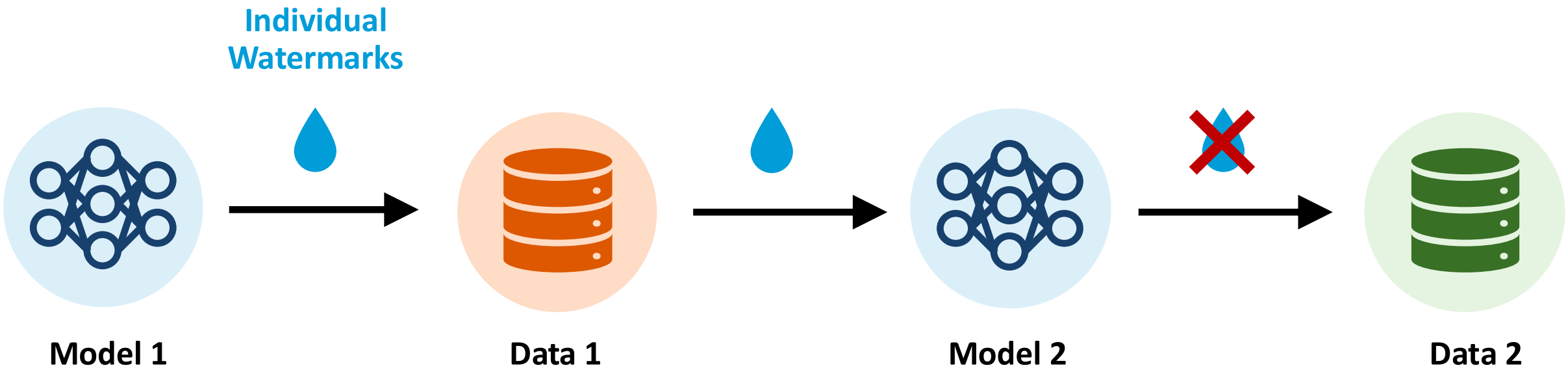
BitMark for Bit-Wise Image Generation





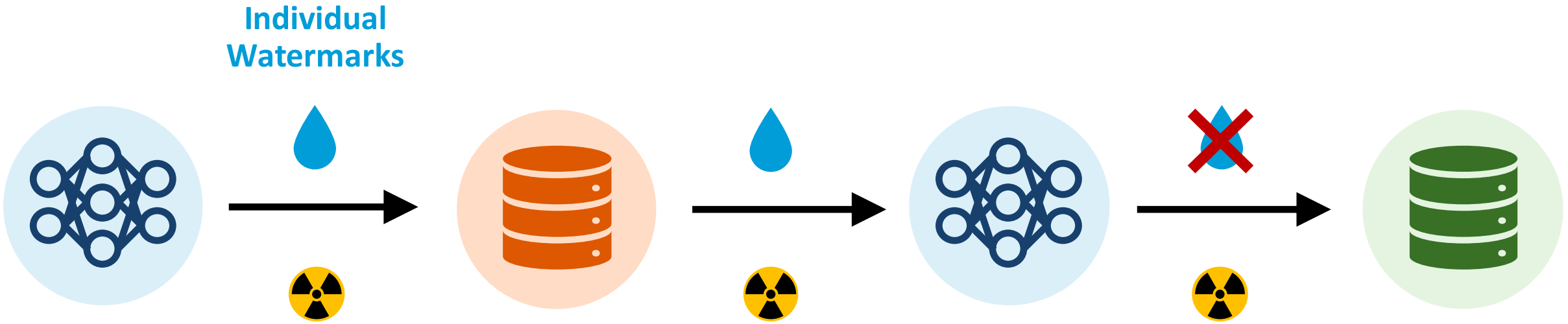
Individual Marks Ignored During Training





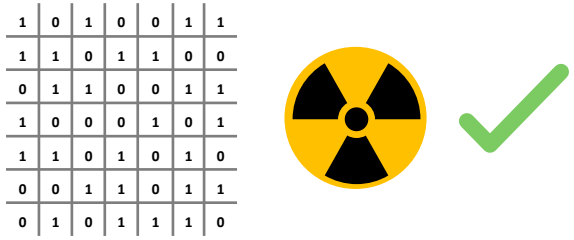
Persistent and Radioactive Marks



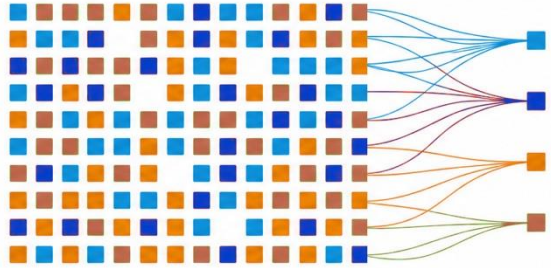


Kerner, ..., Dziedzic. NeurIPS 2025

Radioactive Learnable Marks

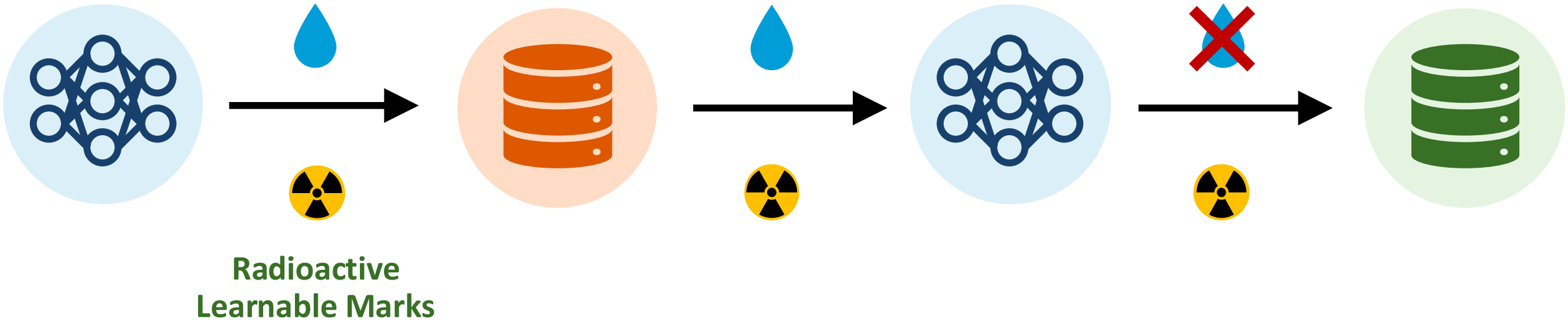


Radioactive Marks

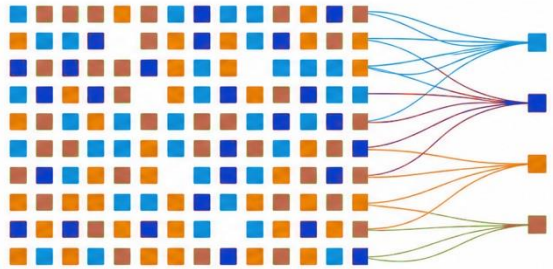


Detect Generated Data & Its Reuse

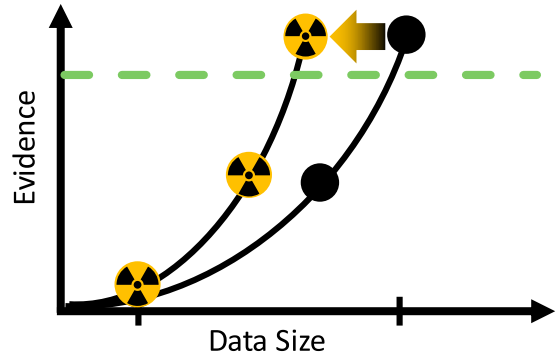




Radioactive Marks



Integrated into a statistical framework



to amplify the signals for data identification

Agent Auditing: Identity Verification and Provenance Tracing



	Passive fingerprint	Active watermark
Identity verification *Is it what it claims to be?* LLM agents	**AGENTPROV (ours)** natural tool choices	**AGENTWM** secret bias in tool calls
Provenance tracing *What was it trained on?* control agents	**ORL-AUDITOR** offline RL datasets	**WORLDMARK (ours)** agents in world models

CISPA
HELMHOLTZ CENTER FOR INFORMATION SECURITY

BitMark: Watermarking Bitwise Image Generative Models

Louis Kerner, Michel Meintz, Bihe Zhao, Franziska Boenisch, Adam Dziedzic

Advances in Neural Information Processing Systems 39 (2025)

Image Generative Artificial Intelligence

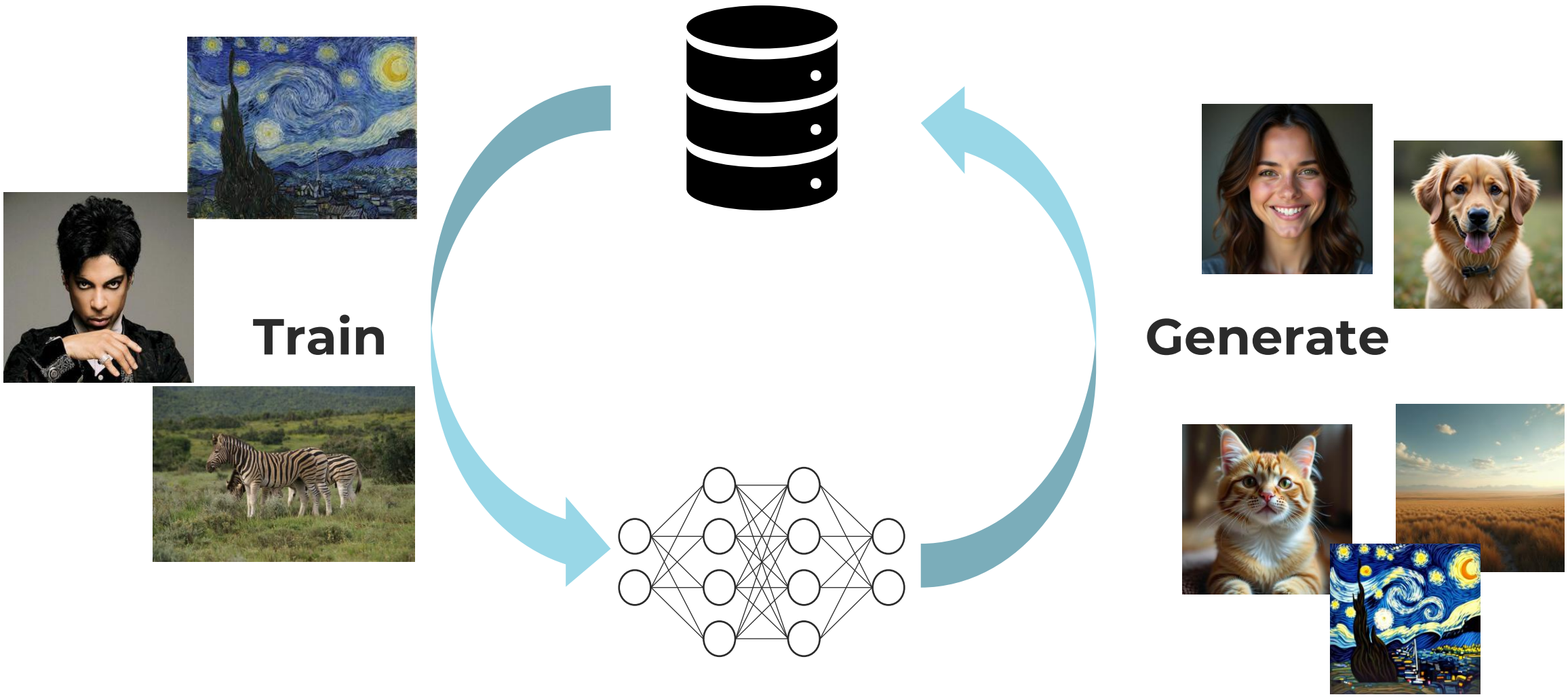


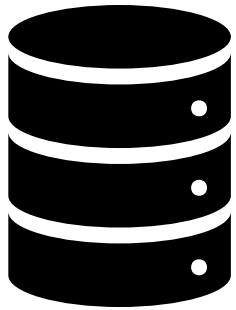
Image Generative Artificial Intelligence

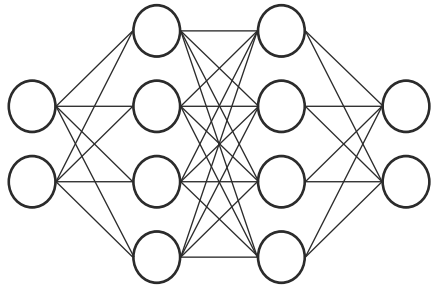






Train









Generate



Harmful Data

Watermarking





02 Aug 2026: Enforcement of Article 50

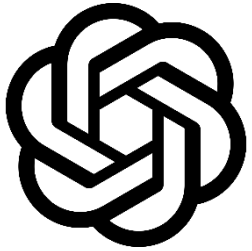
2. Providers of AI systems [...] **generating synthetic audio, image, video or text content**, shall ensure that the **outputs of the AI system are marked in a machine-readable format** and **detectable as artificially generated or manipulated** [...].







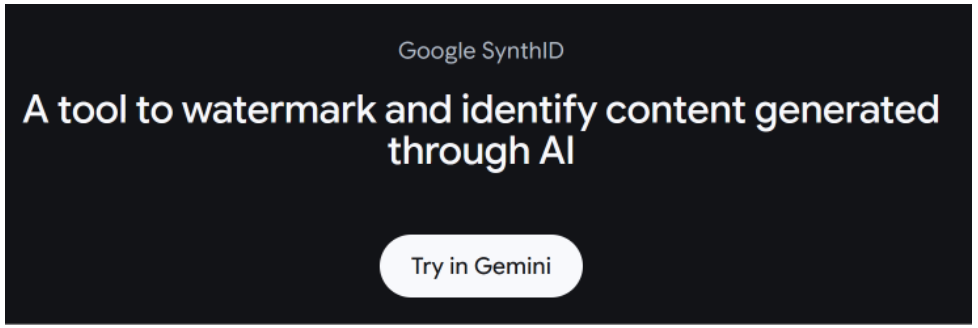
https://www.anthropic.com/news/claude-text-watermark





https://openai.com/index/eu-text-provenance/





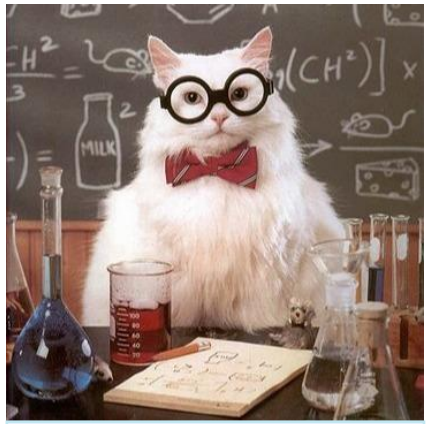
https://deepmind.google/models/synthid/

(Water-)Marking – Application (Images)

Appended to the data

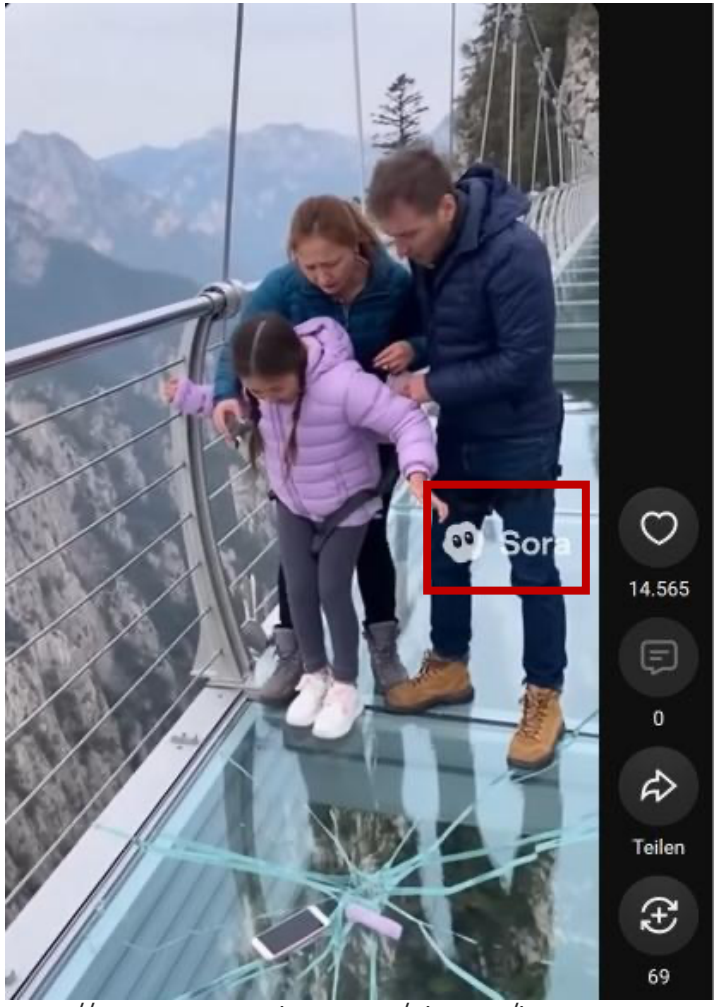
Applied on the data

Metadata



"dateCreated": "2026-10-11T12:00:00Z„
"label": "c2pa.actions",
"data": { "actions": [{ "action": "c2pa.watermarked",
"parameters": {
"watermark_id": "WM-2026-8942", "recipient": "internal-team" } }

visible



https://www.youtube.com/shorts/k43pJvIN6p4

„invisible“

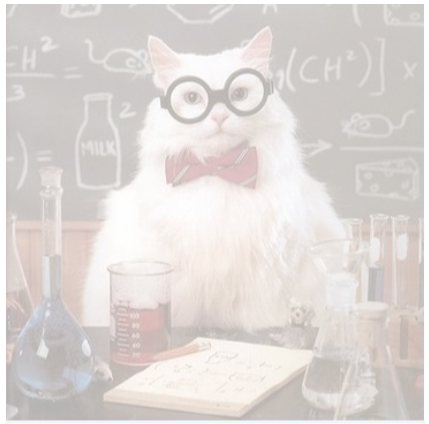


(Water-)Marking – Application (Images)

Appended to the data

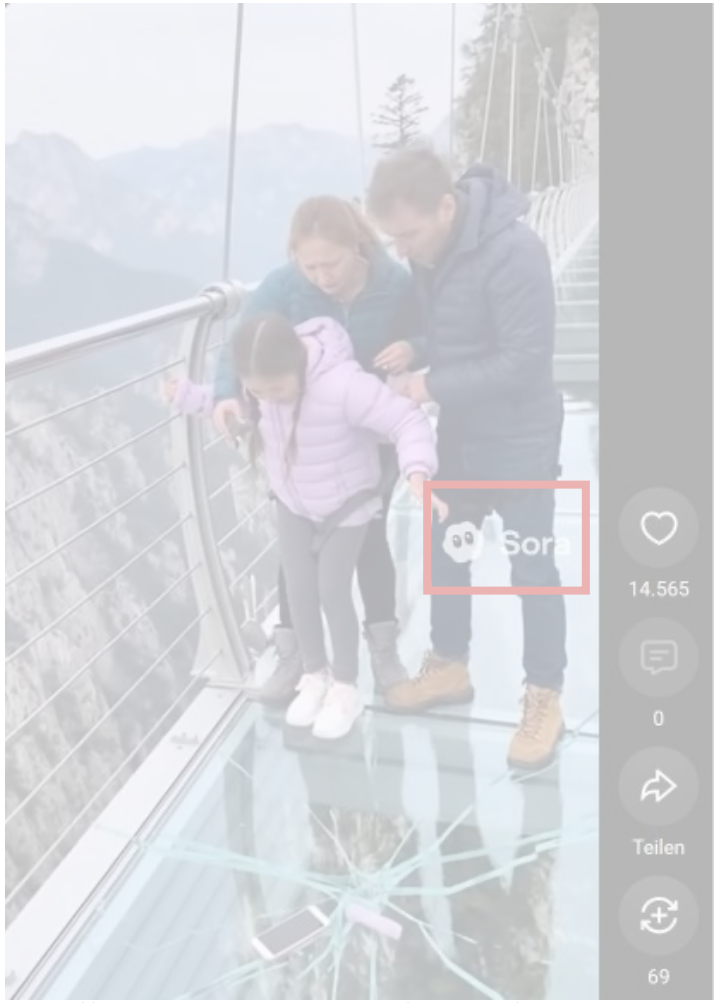
Applied on the data

Metadata



"dateCreated": "2026-10-11T12:00:00Z„
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"data": { "actions": [{ "action": "c2pa.watermarked",
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"watermark_id": "WM-2026-8942", "recipient": "internal-team" } }

visible



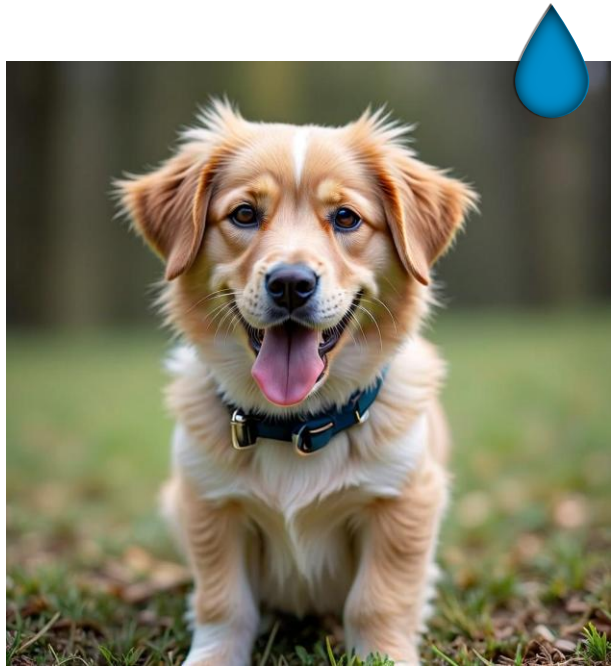
https://www.youtube.com/shorts/k43pJvIN6p4

„invisible“



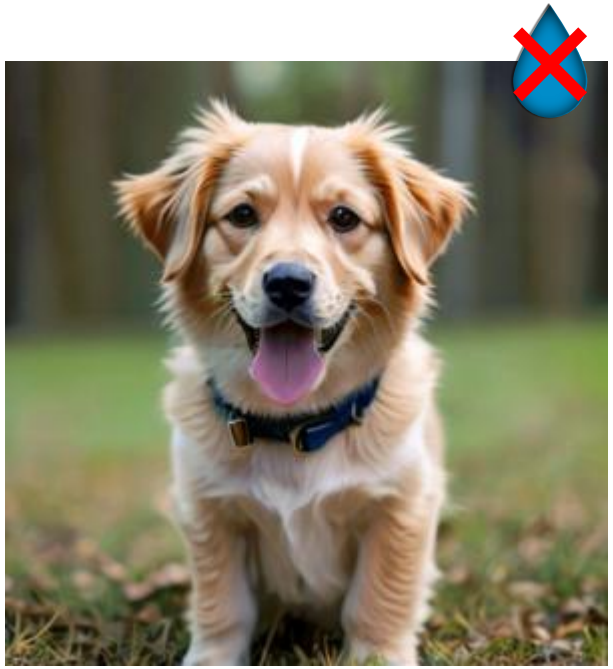
Post-Processing Watermarks





Post-Processing Watermarks

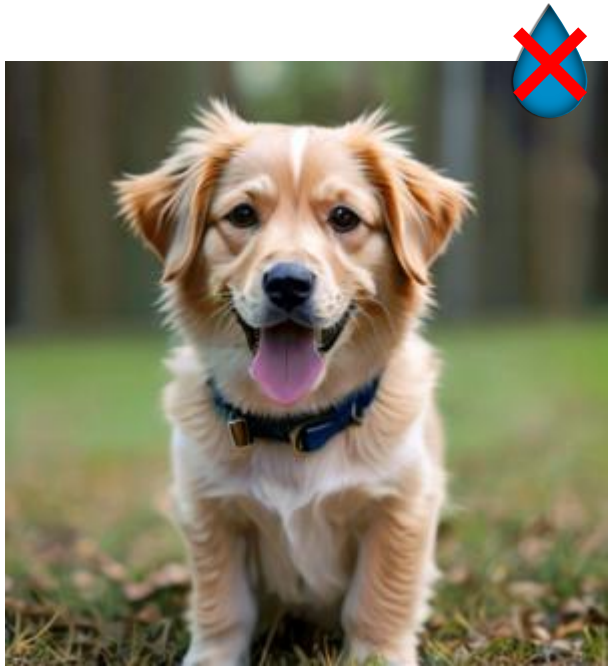


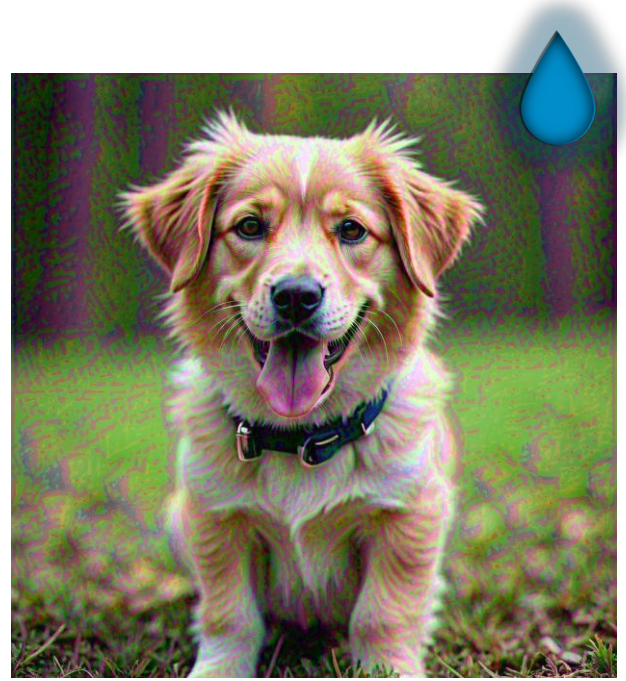


Easy to <u>remove</u> without degrading data quality

Post-Processing Watermarks



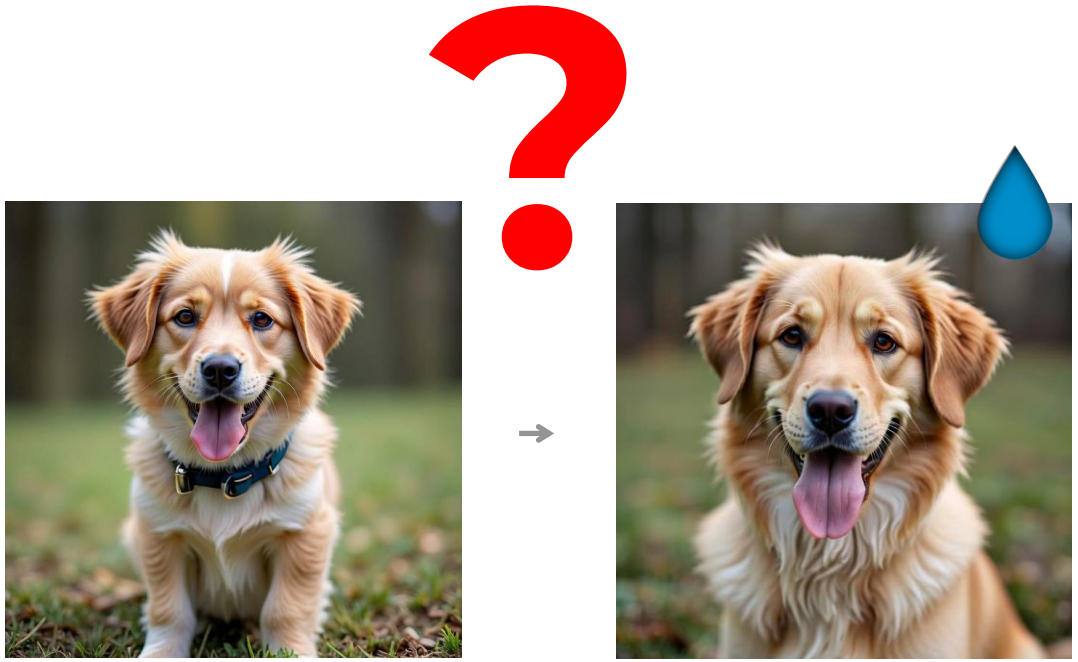




Easy to <u>remove</u> without degrading data quality

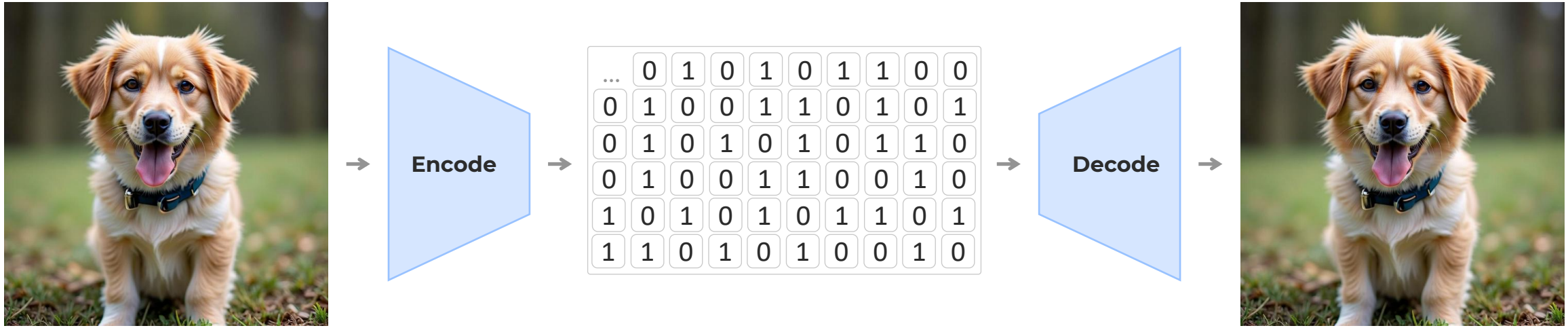
Stronger <u>embedding</u> degrades the data quality.

In-Generation Watermarks



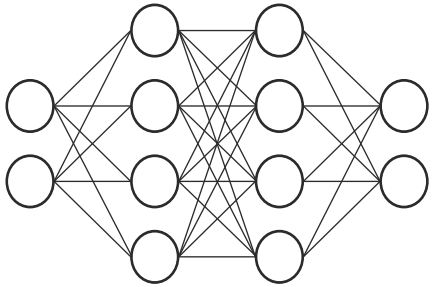
Apply the watermark to the content during generation, coupling the mark with the data‘s contnet.

Bitwise Autoregressive Generation – Infinity



Infinity: Scaling Bitwise AutoRegressive Modeling for High-Resolution Image Synthesis @ CVPR 2025
Jian Han, Jinlai Liu, Yi Jiang, Bin Yan, Yuqi Zhang, Zehuan Yuan, Bingyue Peng, Xiaobing Liu

Bitwise Autoregressive Generation – Infinity



predict

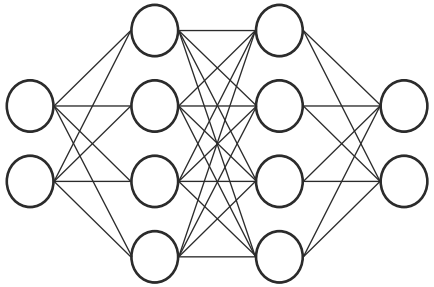
...	0	1	0	1	0	1	1	0	0
0	1	0	0	1	1	0	1	0	1
0	1	0	1	0	1	0	1	1	0
0	1	0	0	1	1	0	0	1	0
1	0	1	0	1	0	1	1	0	1
1	1	0	1	0	1	0	0	1	0

Decode

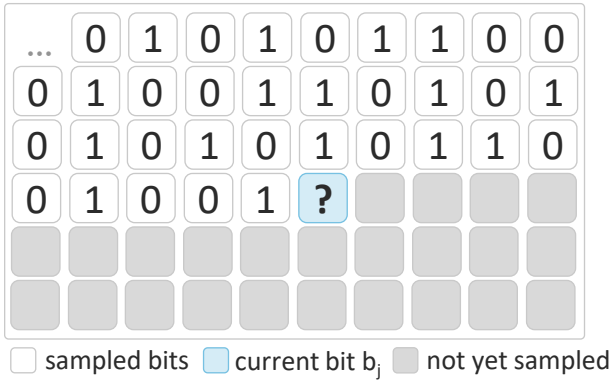


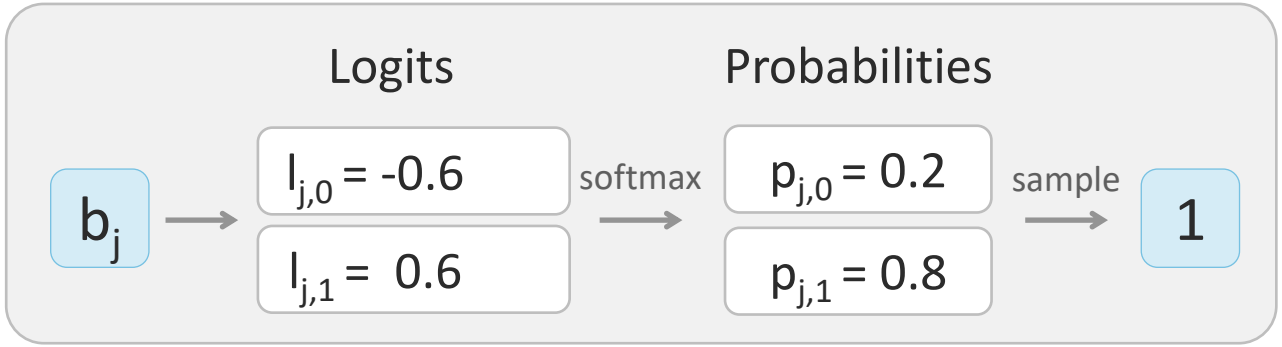
Infinity: Scaling Bitwise AutoRegressive Modeling for High-Resolution Image Synthesis @ CVPR 2025
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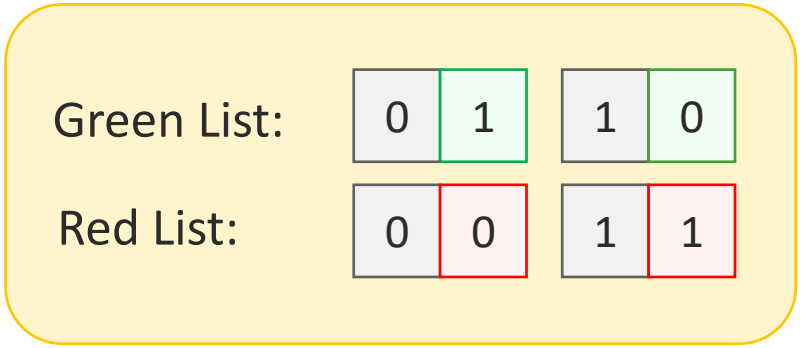
Bitwise Autoregressive Generation – Infinity



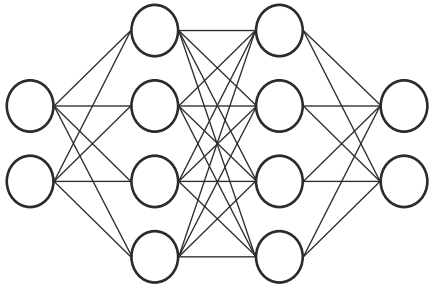
predict



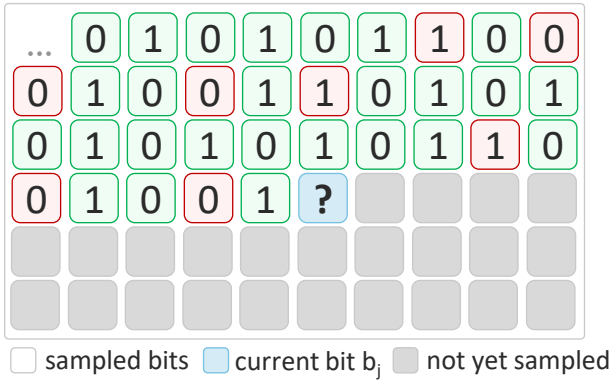


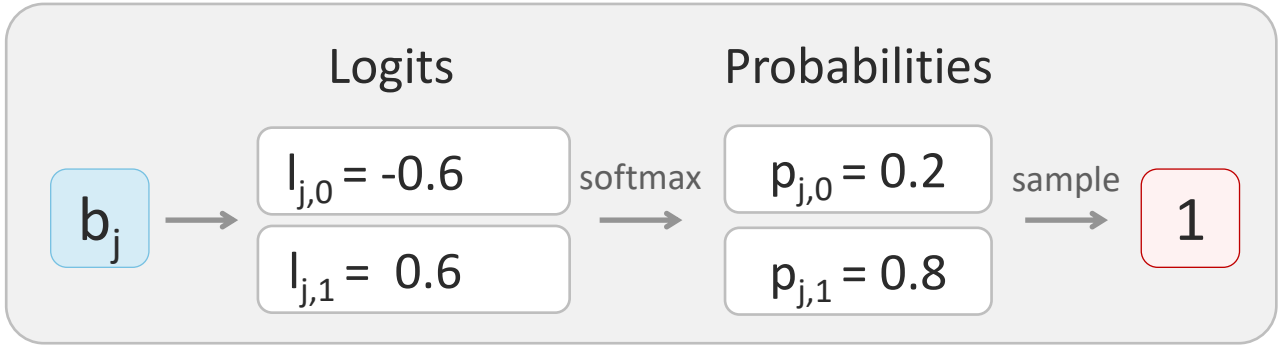


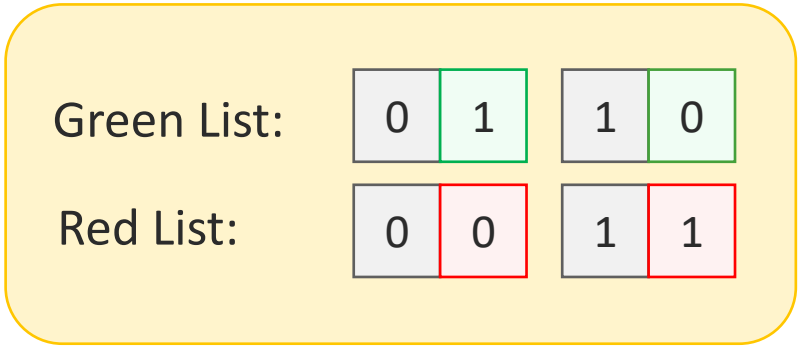
Embedding BitMark



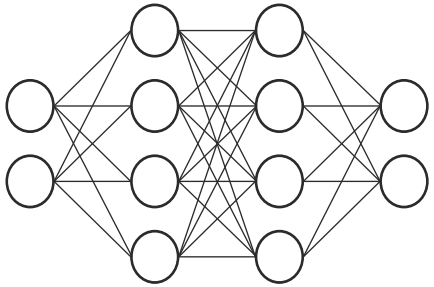
predict



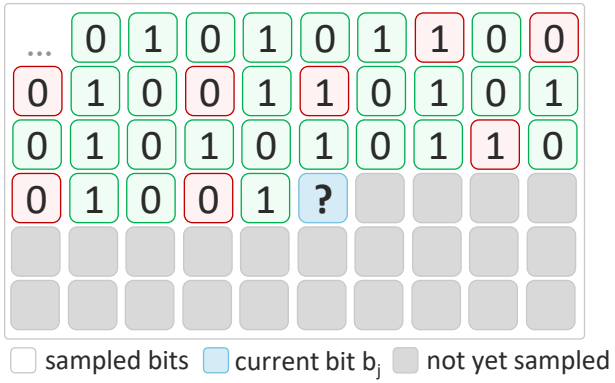




Embedding BitMark



predict



$\delta = 2$

b_j →

Logits

$l_{j,0} = -0.6 + \delta$

$l_{j,1} = 0.6$

softmax →

Probabilities

$p_{j,0} = 0.2$

$p_{j,1} = 0.8$

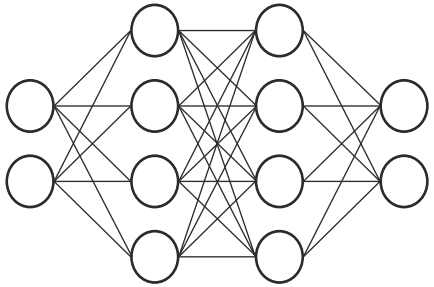
sample →

1

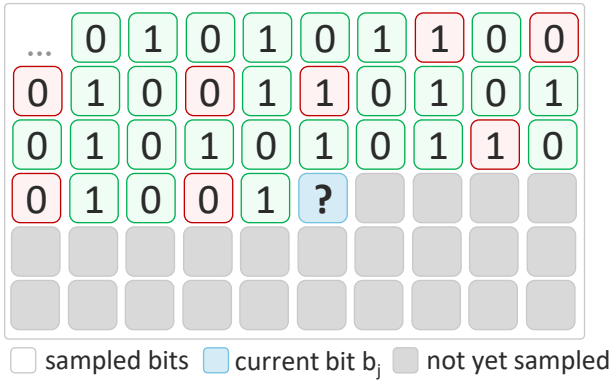
Green List: 0 1 | 1 0

Red List: 0 0 | 1 1

Embedding BitMark



predict

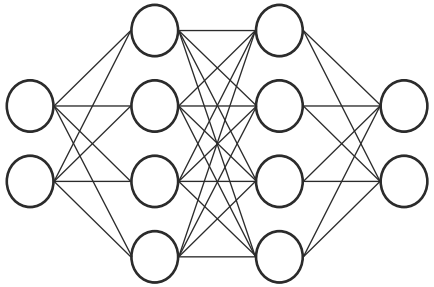


$\delta = 2$

	Logits		Probabilities		
b_j →	$l_{j,0} = -0.6 + 2$	softmax →	$p_{j,0} = 0.69$	sample →	0
	$l_{j,1} = 0.6$		$p_{j,1} = 0.31$		

Green List:	0 1		1 0	
Red List:	0 0		1 1	

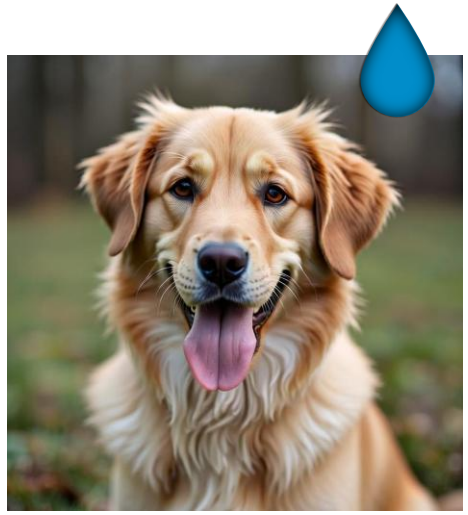
Embedding BitMark



predict

...	0	1	0	1	0	1	1	0	0
0	1	0	0	1	1	0	1	0	1
0	1	0	1	0	1	0	1	1	0
0	1	0	0	1	0	1	1	1	0
1	0	1	0	0	1	1	0	0	1
0	0	1	0	1	0	1	0	1	0

Decode



Green List:	0	1	1	0
Red List:	0	0	1	1

Detecting BitMark

H_0 : This image is generated with no knowledge of the green list.

Green List: 0 1 1 0

Red List: 0 0 1 1

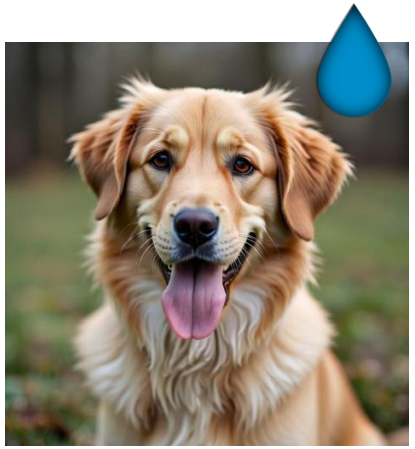


→ Encode →

1	0	0	0	1	0	1	1	0	0
0	0	0	0	1	1	0	0	0	1
0	1	1	1	0	1	1	1	1	0
0	1	0	0	1	1	0	0	1	0
1	0	0	0	1	0	1	1	1	1
1	1	0	1	1	1	0	0	1	0

$$z = (C - \gamma T)/\sqrt{T\gamma(1-\gamma)}$$

z = 0.51



→ Encode →

1	0	1	0	1	0	1	1	0	0
0	1	0	0	1	1	0	1	0	1
0	1	0	1	0	1	0	1	1	0
0	1	0	0	1	0	1	1	1	0
1	0	1	0	0	1	1	0	0	1
0	0	1	0	1	0	1	0	1	0

z = 3.88

Results – Quality

$\delta = 2$





$\delta = 5$

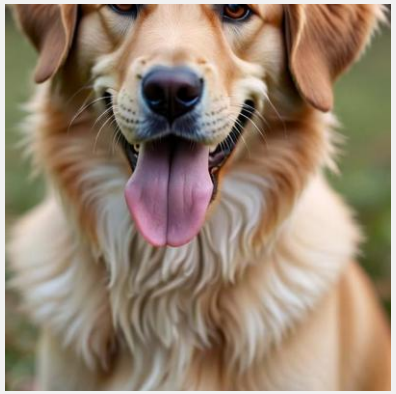




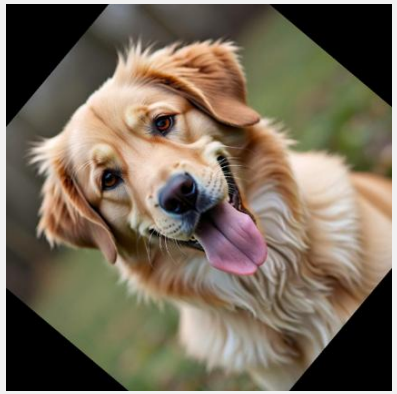
Results – Robustness



Orignal



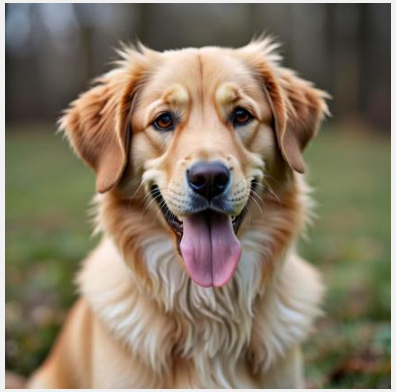
Crop 0.5x



Rotation (0-180°)



Color (Hue 0.3)



Horizontal Flip



CtrlRegen

Results – Robustness

Infinity-2B generated images.

Watermark method	Original	Color	Crop	Rotate	Horizontal	CtrlRegen
StegaStamp						
RivaGAN						
TrustMark						
BitMark (δ = 2) Ours						

Robustness (TPR@1%FPR ↑)

*Using a rotation estimation tool

Results – Robustness

Infinity-2B generated images.

Watermark method	Original	Color	Crop	Rotate	Horizontal	CtrlRegen
StegaStamp	100.0	98.7	1.0	96.7	33.8	44.2
RivaGAN	99.7	99.4	99.4	32.1	0.0	1.6
TrustMark	99.9	2.2	1.4	2.7	99.8	1.1
BitMark ($\delta = 2$) Ours	**100.0**	**99.6**	**98.8**	***98.3**	**100.0**	**91.6**

Robustness (TPR@1%FPR ↑)

*Using a rotation estimation tool

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Thank you!
Questions?

CISPA HELMHOLTZ CENTER FOR INFORMATION SECURITY

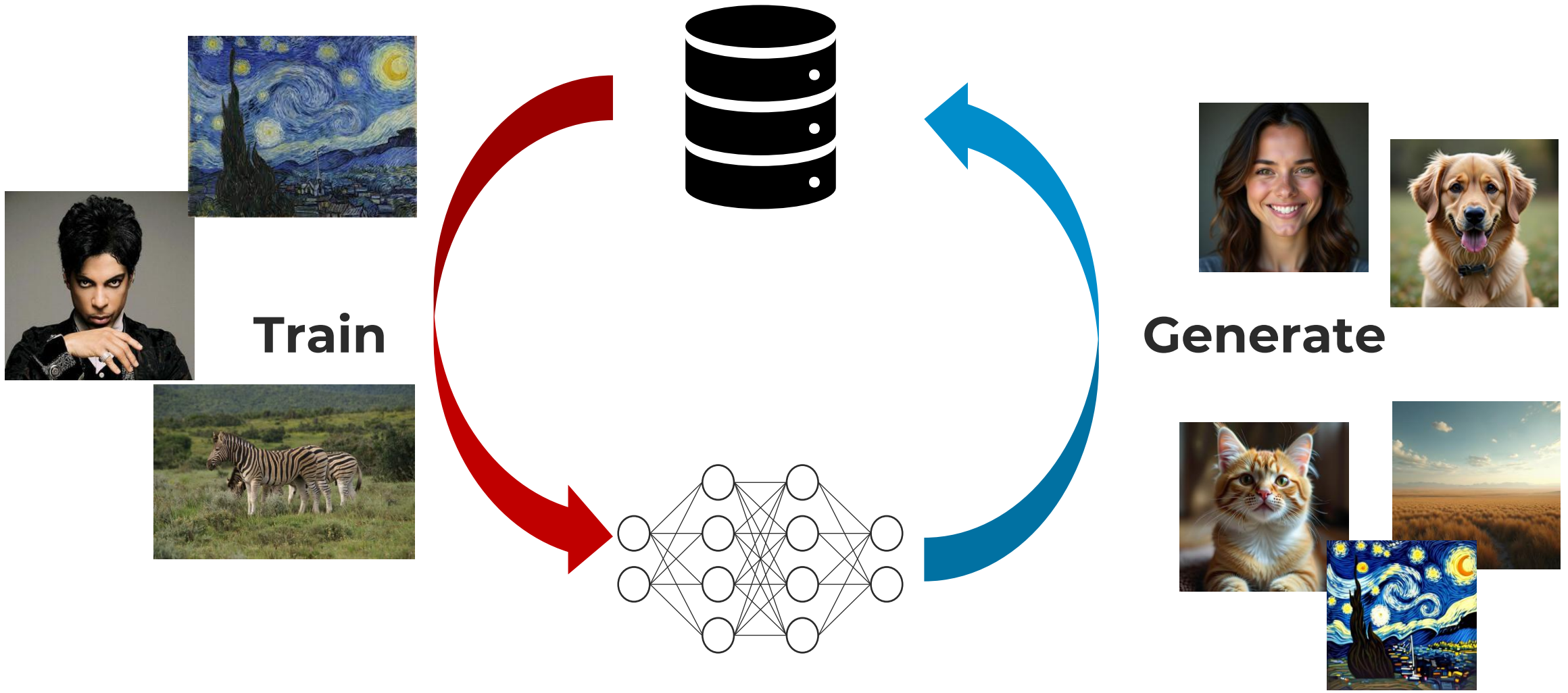
SprintML

RADIUM: RadioActive Decay of Image-Underlaid Marks

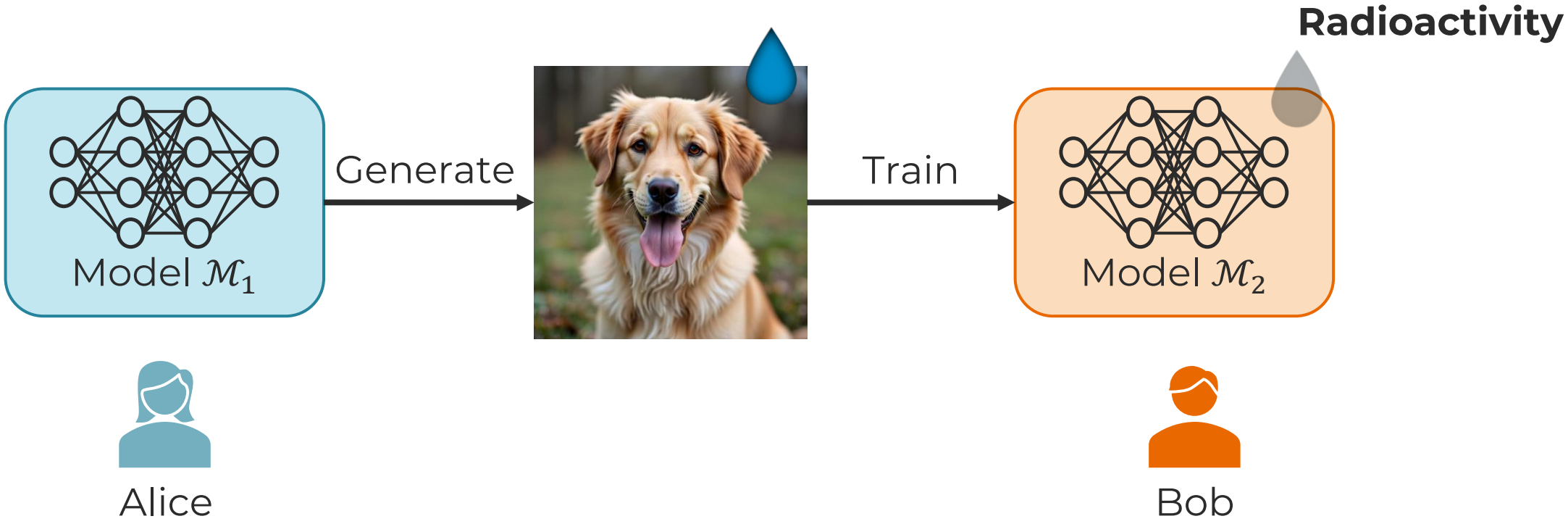
Michel Meintz[*], *Louis Kerner*[*] , *Maitri Vignesh Shah*, *Simon Hector*, *Franziska Boenisch*, *Adam Dziedzic*

CISPA Helmholtz Center for Information Security | ML in PL

Image Generative Artificial Intelligence



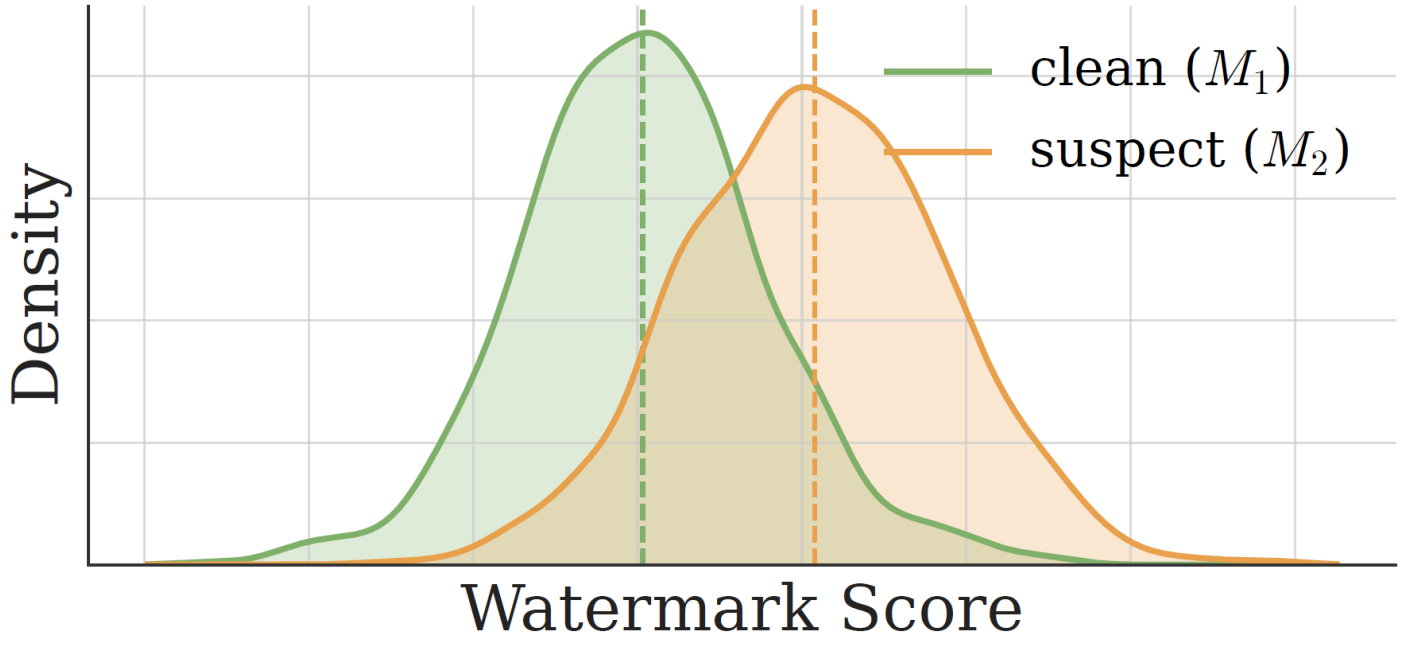
Radioactivity in Model Derivatives



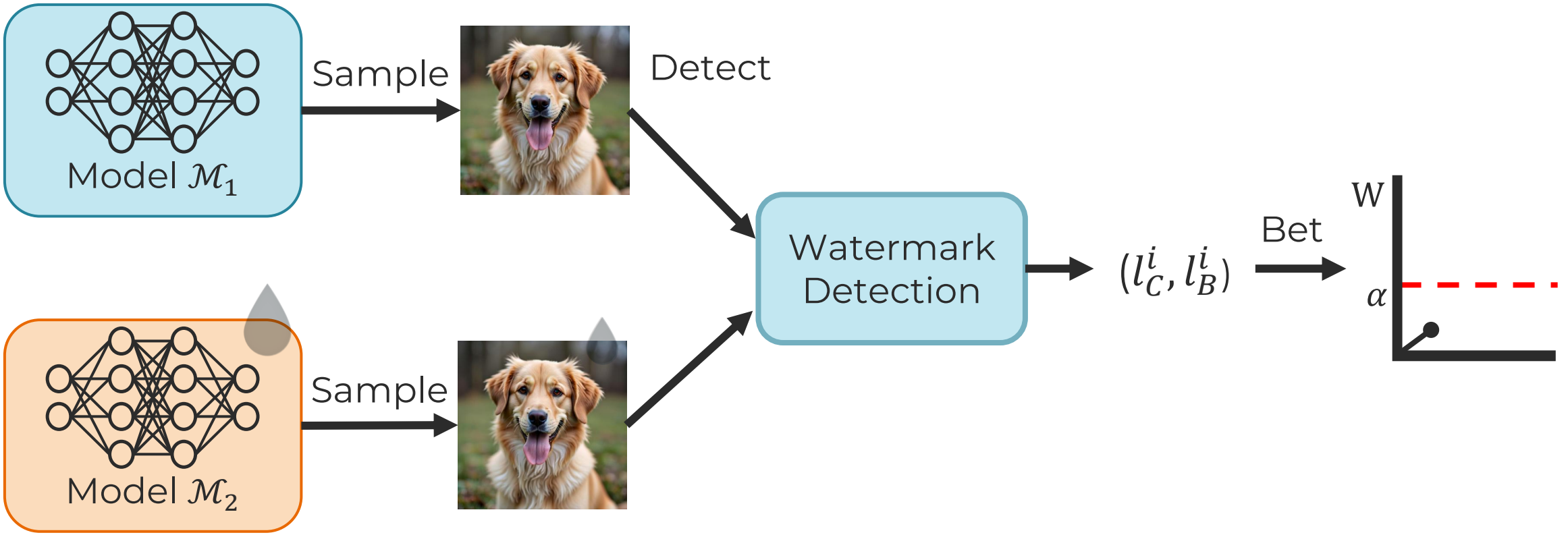
- Bob uses data generated by Alice to train his $\mathcal{M}_2$
- Alice wants to find out if $\mathcal{M}_2$ is trained on her data and thus a derivative of $\mathcal{M}_1$
- Alice has black-box access to $\mathcal{M}_2$ and white-box access to $\mathcal{M}_1$

One Image Does Not Carry Enough Information

- The per-image watermark signal is weak
- Dataset-Inference insight: aggregate the signal across many samples
- **And** we want to use the second model $\mathcal{M}_2$

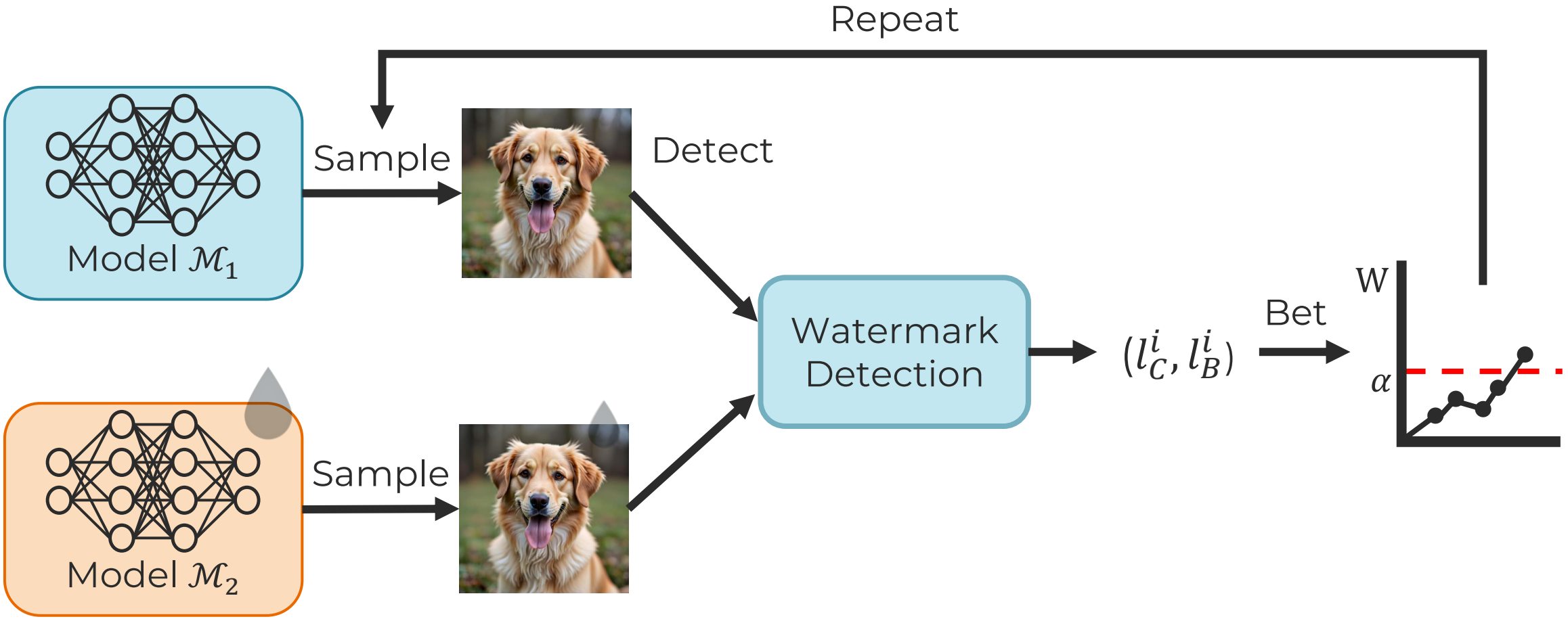


Hypothesis Testing by Betting



- Testing by betting enables us to iteratively sample images until we reach the significance threshold α

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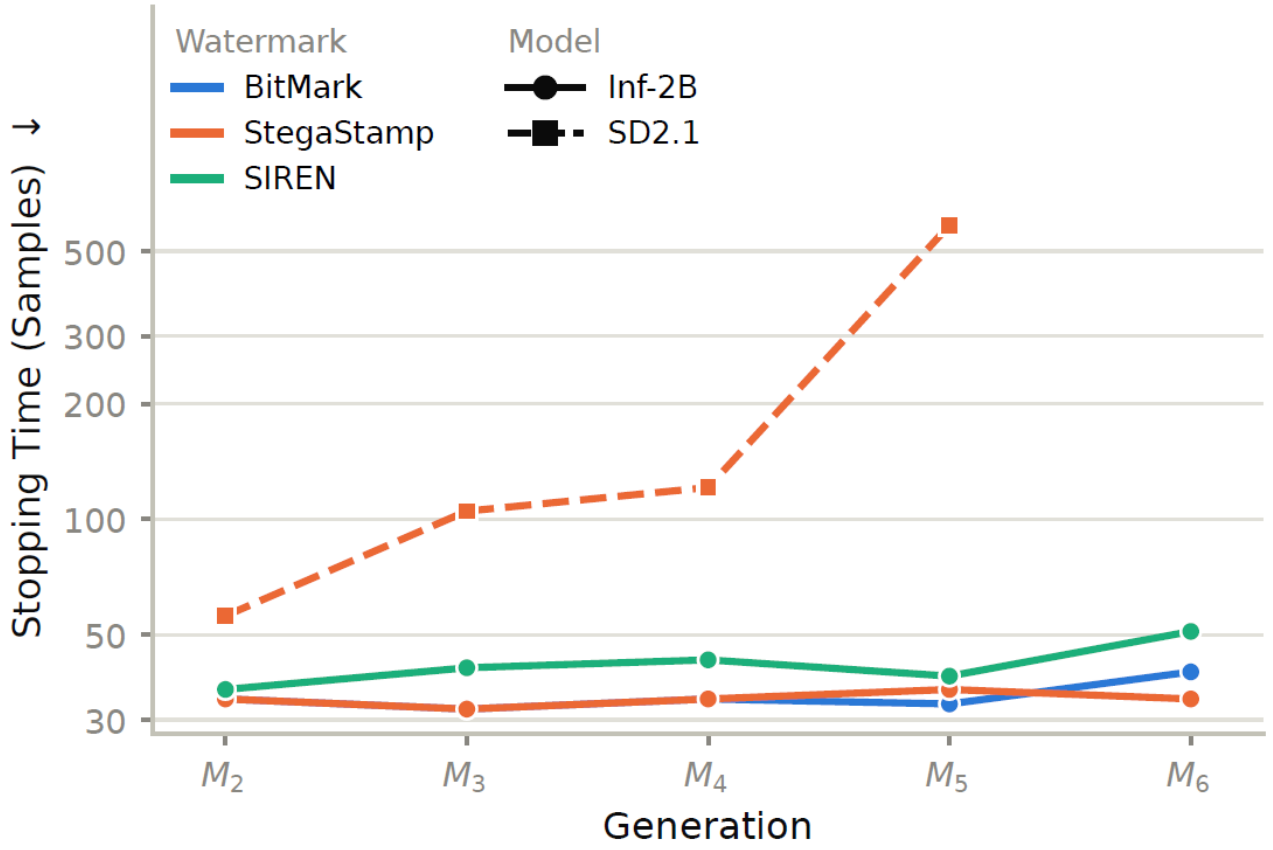
Radioactivity is Disparate Across Watermarks

Watermark	1%	5%	10%	25%	50%	100%
Infinity-2B (autoregressive)						
BitMark	90	90	48	35	41	34
RivaGAN	-	-	581	226	86	45
StegaStamp	-	-	-	678	155	34
SIREN	-	-	-	395	46	36
TrustMark	-	-	-	-	-	170
Stable Diffusion 2.1 (diffusion)						
StegaStamp	-	-	-	-	-	56
RivaGAN	-	-	-	-	-	813
SIREN	-	-	-	-	-	94
TrustMark	-	-	-	-	-	-
TreeRing	-	-	-	-	-	-
Stable Sig.	-	-	-	-	-	-

Samples required until detection with $\alpha = 10^{-5}$; grey = not detected

- **5 / 7** watermarks are radioactive
- Radioactivity is **model architecture dependent**. The autoregressive model learns marks easier.
- **Only BitMark** survives at low data **fractions (1%)**
- Having **access to the training prompts** used by Bob barely helps.

Decay Over Multiple Training Iterations



- $\mathcal{M}_{i+1}$ is trained on 100% of the outputs of $\mathcal{M}_i$
- **Watermarks in Infinity-2B survive** with very limited decay.
- Watermarks in **SD2.1 degrade quick**

Radioactivity Across Architectures is More Difficult

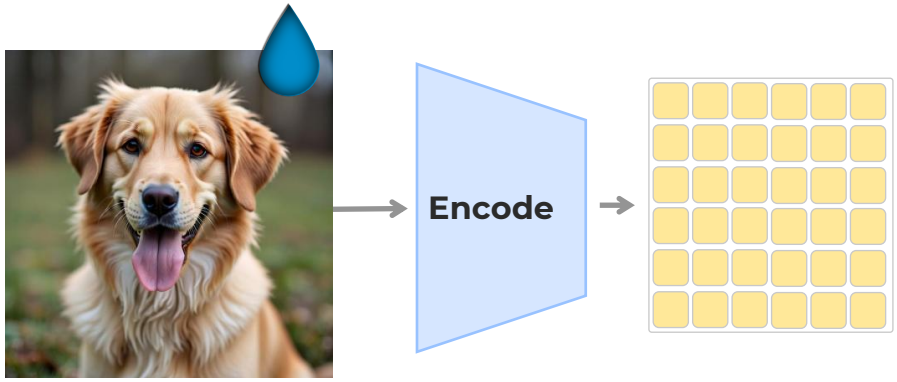
Watermark	$\mathcal{M}_1 \rightarrow \mathcal{M}_2$	25%	50%	100%
BitMark	Inf → SD2.1	-	134	59
BitMark	Inf → SD1.4	-	164	62
SIREN	Inf → SD2.1	-	-	159
SIREN	SD2.1 → SD1.4	-	-	93
StegaStamp	Inf → SD2.1	-	147	79
StegaStamp	Inf → SD1.4	-	-	48
TrustMark	Inf → SD2.1	-	-	112
RivaGAN	any	-	-	-
TrustMark	others	-	-	-

Samples required until detection with $\alpha = 10^{-5}$; grey = not detected

Different architecture for $\mathcal{M}_1$ and $\mathcal{M}_2$:

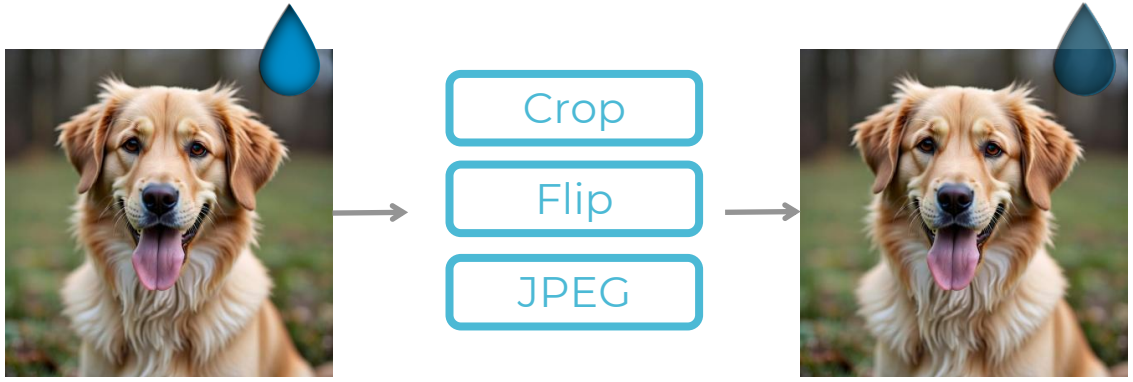
- Detection requires more samples
- Training requires more watermarked data
- Some marks can not be detected at all

What Makes a Watermark Radioactive?



1. Surviving the latent encoding

2. Embedding a similar signal

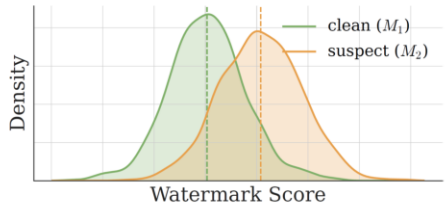


3. Robustness to training augmentations

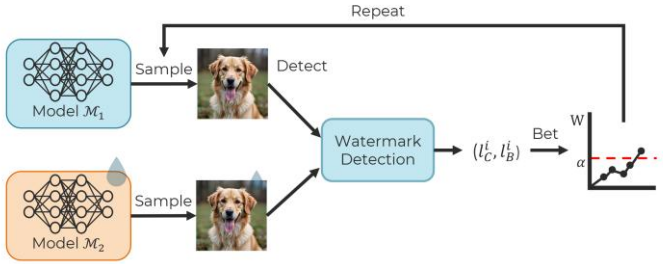
Take-Aways

One Image Does Not Carry Enough Information

- The per-image watermark signal is weak
- Dataset-Inference insight: aggregate the signal across many samples
- **And** we want to use the second model $\mathcal{M}_2$



Hypothesis Testing by Betting



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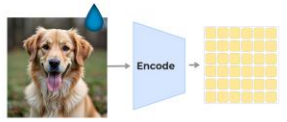
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TrustMark	-	-	-	-	-	-
TreeRing	-	-	-	-	-	-
Stable Sig.	-	-	-	-	-	-

Samples required until detection with $\alpha = 10^{-5}$; grey = not detected

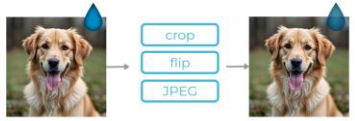
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2. Embedding a similar signal



3. Robustness to training augmentations

Agent Auditing: Identity Verification and Provenance Tracing

Bihe Zhao

CISPA Helmholtz Center for Information Security

ML in PL 2026

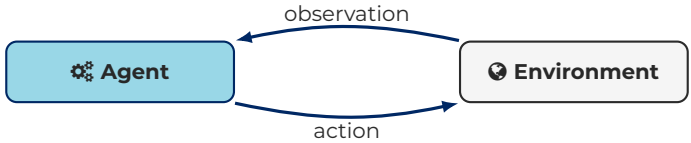
Outline

1. Why audit agents?
2. Identity verification
3. Provenance tracing
4. Summary

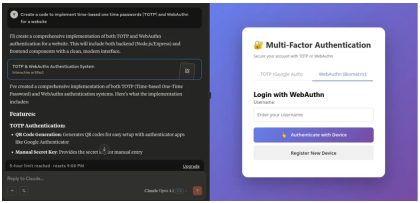


1. Why audit agents?

What is an agent?

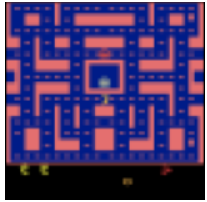


LLM agents act by calling tools



Control agents act by moving





- Both are often trained with reinforcement learning (RL).

Images: Claude Opus 4.1 screenshot, public domain, Wikimedia Commons. Spot robot, photo by Jonte, CC BY-SA 4.0, Wikimedia Commons. Ms. Pac-Man frame from our experiments.

Why audit agents?



Trust

Users pay for one model and may get another.



Safety

A flawed dataset or simulator spreads to every agent trained on it.



Regulation

The EU AI Act asks high-risk AI to keep logs for traceability.

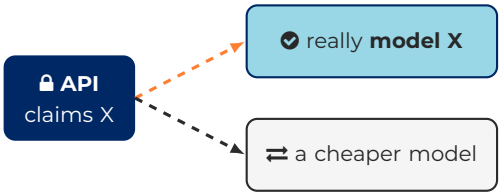
Case from Zhang et al., 2026. Photos: BalticServers.com (CC BY-SA 3.0), Grendelkhan (CC BY-SA 4.0), Wikimedia Commons. EU flag, public domain. EU AI Act, Article 12.

Two audit goals

Identity verification

LLM agents

Is it the model it claims to be?



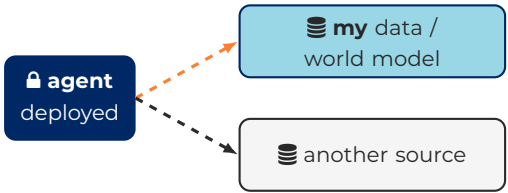
which model is behind it?

Why: get the model you pay for.

Provenance tracing

control agents

What was it trained on?



trained on mine, or not?

Why: licences and recalls.

- Either way, the auditor usually sees only the actions.

A taxonomy for agent auditing

	Passive fingerprint	**Active** watermark
Identity verification *Is it what it claims to be?* LLM agents	**AGENTPROV (ours)** natural tool choices	**AGENTWM** secret bias in tool calls
Provenance tracing *What was it trained on?* control agents	**ORL-AUDITOR** offline RL datasets	**WORLDMARK (ours)** agents in world models



2. Identity verification

for LLM agents

Identity verification, passive

AGENTPROV: is the API serving the model it claims?

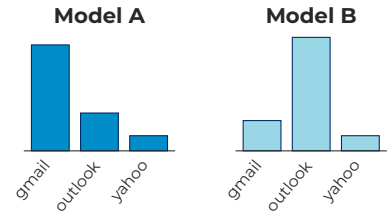
	Passive fingerprint	**Active** watermark
Identity verification *Is it what it claims to be?* LLM agents	**AGENTPROV (ours)** natural tool choices	**AGENTWM** secret bias in tool calls
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AgentProv: tool choices as a fingerprint

our work

- Text audits fail: tool calls return no text.
- Idea: compare which tool a model picks.

Probe: “Send an email to Alice”
Tools offered: `gmail`, `outlook`, `yahoo`
The auditor reads which tool is called



- 20 probes, squared MMD test.

Xun Wang, Bihe Zhao, Michael Backes, Franziska Boenisch, Adam Dziedzic. AgentProv: Auditing Agentic LLM API Providers via Tool-use Policy Probes.

AgentProv: results

our work

	MET text	RUT text	AgentProv tool calls
Different models caught (630 pairs)	100%	100%	**100**%
Same model falsely rejected (36 models)	0/36	2/36	**0**/**36**
False accusations under hidden system prompts	67%	53%	**7**%

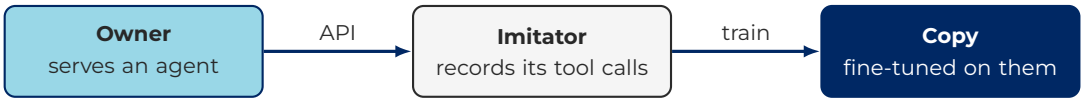
- Same power as text-based tests, far fewer false accusations.
- One audit is 1,000 queries, under $1 per endpoint.

Identity verification, active

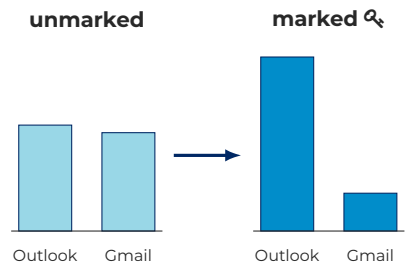
AGENTWM: is this suspect a copy of my agent?

	Passive fingerprint	**Active** watermark
Identity verification *Is it what it claims to be?* LLM agents	AGENTPROV (ours) natural tool choices	**AGENTWM** secret bias in tool calls
Provenance tracing *What was it trained on?* control agents	ORL-AUDITOR offline RL datasets	WORLDMARK (ours) agents in world models

AgentWM: watermarking tool-call paths



- Equivalent tools: Outlook or Gmail.
- Embed: add a secret bias.
- Verify: does the suspect share it?



Liwen Wang, Zongjie Li, Yuchong Xie, Shuai Wang, Dongdong She, Wei Wang, Juergen Rahmel. On Protecting Agentic Systems' Intellectual Property via Watermarking. arXiv:2602.08401, 2026.



3. Provenance tracing

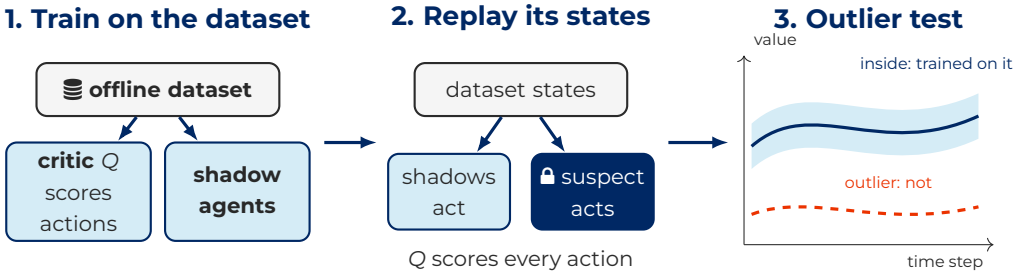
for control agents

Provenance tracing, passive

ORL-AUDITOR: was this agent trained on my offline RL dataset?

	Passive fingerprint	**Active** watermark
Identity verification *Is it what it claims to be?* LLM agents	AGENTPROV (ours) natural tool choices	AGENTWM secret bias in tool calls
Provenance tracing *What was it trained on?* control agents	**ORL-AUDITOR** offline RL datasets	WORLDMARK (ours) agents in world models

ORL-Auditor: how it works



- Agents trained on the same data pick actions of similar value. The band of shadow agents shows what “trained on it” looks like.

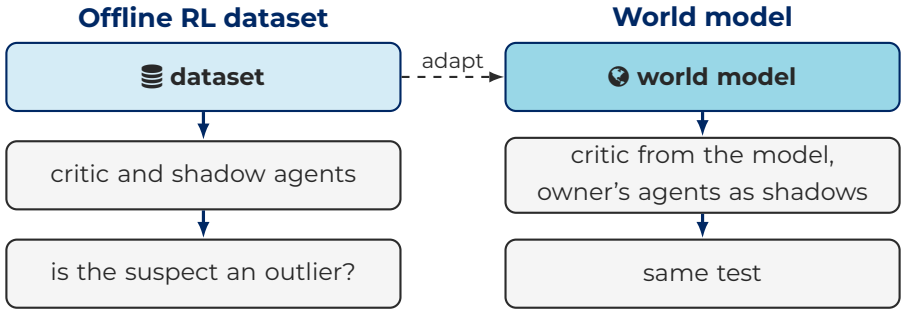
Linkang Du, Min Chen, Mingyang Sun, Shouling Ji, Peng Cheng, Jiming Chen, Zhikun Zhang. ORL-Auditor: Dataset Auditing in Offline Deep Reinforcement Learning. NDSS 2024.

Provenance tracing, active

WorldMark: was this agent trained inside my world model?

	Passive fingerprint	**Active** watermark
Identity verification *Is it what it claims to be?* LLM agents	AgentProv (ours) natural tool choices	AgentWM secret bias in tool calls
Provenance tracing *What was it trained on?* control agents	ORL-Auditor offline RL datasets	**WorldMark (ours)** agents in world models

ORL-Auditor on world models



- On world models it also accuses outsiders. So we need an active approach.

Adaptation from Bihe Zhao, Xun Wang, Juangui Xu, Franziska Boenisch, Adam Dziedzic. Provenance Tracing for Agents Trained inside World Models.

WorldMark: the idea

our work

1. Key picks anchors



secret frame + keyed action

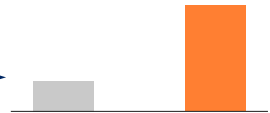
2. Customer trains inside

World model

keyed action
at an anchor:
reward + small bonus

the agent learns the preference

3. Owner audits



keyed-action match rate

- Show the suspect the anchor frames. A much higher match rate means it was trained in our world model.

Bihe Zhao, Xun Wang, Juangui Xu, Franziska Boenisch, Adam Dziedzic. Provenance Tracing for Agents Trained inside World Models.

WorldMark: results

our work

200/200	0/200	>97%
marked agents detected	false accusations	of the clean agents' score

- Four world models: Mamba, transformer, diffusion, recurrent.
- Only actions are read, at $\alpha = 0.01$.

4. Summary

The taxonomy

	Passive	Active
Identity verification LLM agents	**AGENTPROV** tool choices reveal it	**AGENTWM** bias survives imitation
Provenance tracing control agents	**ORL-AUDITOR** values reveal the dataset	**WORLDMARK** agents inherit the key

- Passive: no preparation, works on what is already released.
- Active: planted in advance, gives key-specific evidence.
- All four audit an agent from its actions alone.

References

- Xun Wang, Bihe Zhao, Michael Backes, Franziska Boenisch, Adam Dziedzic. AgentProv: Auditing Agentic LLM API Providers via Tool-use Policy Probes.
- Liwen Wang, Zongjie Li, Yuchong Xie, Shuai Wang, Dongdong She, Wei Wang, Juergen Rahmel. On Protecting Agentic Systems' Intellectual Property via Watermarking. arXiv:2602.08401, 2026.
- Linkang Du, Min Chen, Mingyang Sun, Shouling Ji, Peng Cheng, Jiming Chen, Zhikun Zhang. ORL-Auditor: Dataset Auditing in Offline Deep Reinforcement Learning. NDSS 2024.
- Bihe Zhao, Xun Wang, Juangui Xu, Franziska Boenisch, Adam Dziedzic. Provenance Tracing for Agents Trained inside World Models.
- Yage Zhang, Yukun Jiang, Zeyuan Chen, Michael Backes, Xinyue Shen, Yang Zhang. Real Money, Fake Models: Deceptive Model Claims in Shadow APIs. arXiv:2603.01919, 2026.
- Irena Gao, Percy Liang, Carlos Guestrin. Model Equality Testing: Which Model is this API Serving? ICLR 2025.

Thank you!

Questions?